COMPASS_Synoptic_SmartChamberFluxes_2025_CO2

Stephanie J. Wilson

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Contents

0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data

a. i.e. drops fluxes that have the drop column TRUE or FLAG this

b. adjusting times as needed

4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

0.4 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

```
## [1] "CSV file exists and has been read into the code."
```

0.5 Mark Drop column in Metadata for Retaken Fluxes or other reasons to Drop

```
############## Check For Plot ID

##Sometimes there are other fluxes in the .json files that we don't want to calculate
  #change drop column to TRUE if the plot doesn't include a site code in the Plot_ID
metadata <- metadata %>%
  rename(Plot = Plot_ID) %>%
  mutate(
    drop = ifelse(
      !grepl("SWH|GCW|MSM|GWI|GCReW", Plot),
      # Check if Plot_ID does not contain the specified strings
      TRUE,  # Set drop to TRUE if it doesn't contain any of the strings
      drop   # Keep the original value of drop if it contains one of the strings
    ) )

############## Check for retakes

##Sometimes we have to retake a flux in the field due to issues or ebullition
##We need to change the drop column to reflect using only the latest flux
  #In order to we expect an "a" after the label, but sometimes this is not the case so we will fix that
metadata <- metadata %>%
    mutate(Plot = case_when(
    str_detect(Plot, "-retake") ~ str_replace(Plot, "-retake", "a"),
    str_detect(Plot, "-Retake") ~ str_replace(Plot, "-Retake", "a"),
    str_detect(Plot, "-b") ~ str_replace(Plot, "-b", "b"),
    TRUE ~ Plot  # Keeps the original value if none of the conditions match
  ))

#If the flux was taken more than once, flag the others so that we use the latest measurement
# Extract the base plot (e.g., "GCW-TR-1" from "GCW-TR-1", "GCW-TR-1a", etc.)
metadata$base_plot <- str_extract(metadata$Plot,  "^[^-]+-[A-Za-z]+-[A-Za-z]+-[A-Za-z]+")
```

```r
# Extract the replicate number (the number after the last hyphen)
metadata$replicate <- as.numeric(str_extract(metadata$Plot, "(?<=-)[0-9]+(?=[a-zA-Z]?$)"))

# Extract the suffix (if any, the letter after the replicate number)
metadata$suffix <- str_extract(metadata$Plot, "(?<=\\d)[a-zA-Z]$")

# If there's no suffix, set it to an empty string
metadata$suffix[is.na(metadata$suffix)] <- ""

# Sort by base plot, replicate number, and suffix (to keep highest suffix) - #####INSTEAD WE WANT TO CH
metadata <- metadata %>%
  group_by(base_plot, replicate) %>%
  arrange(base_plot, replicate, desc(suffix), .by_group = TRUE) %>%
  mutate(drop = row_number() != 1) %>% #the entry with the latest suffix will keep false all others wil
  #slice_head(n = 1) %>% #if you wanted to remove a the line rather than flag it
  ungroup()


################# Check Observation lengths

#if the observation length is over 5 mins then change drop column to true
metadata <- metadata %>%
  mutate(drop = if_else(Total_Time > 300, TRUE, drop))


################# Remove columns with drop = TRUE
metadata <-  metadata %>%
  filter(drop == "FALSE")
```

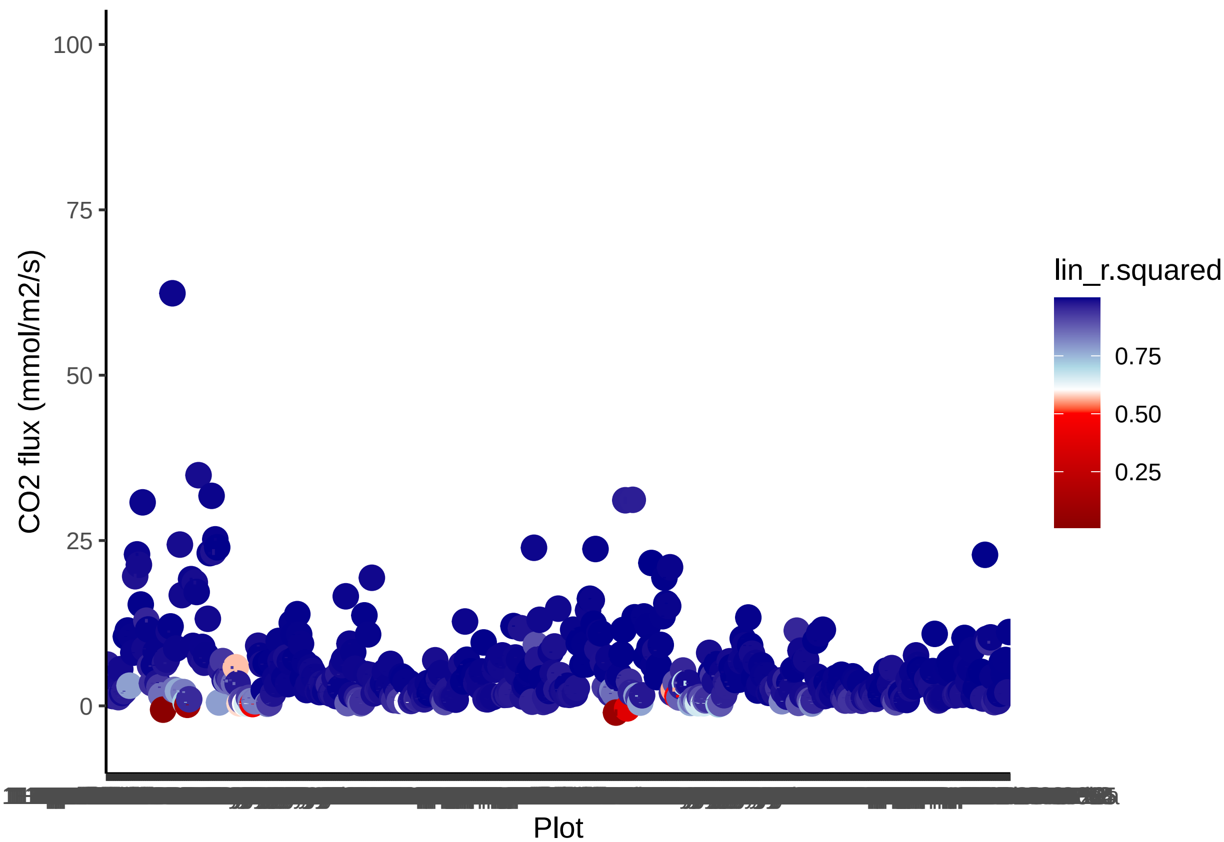
##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.6 Match up the metadata to the rest of the data

0.7 Change the units from ppm to umol

0.8 Calculate fluxes

0.9 Visualize fluxes and sort by r2



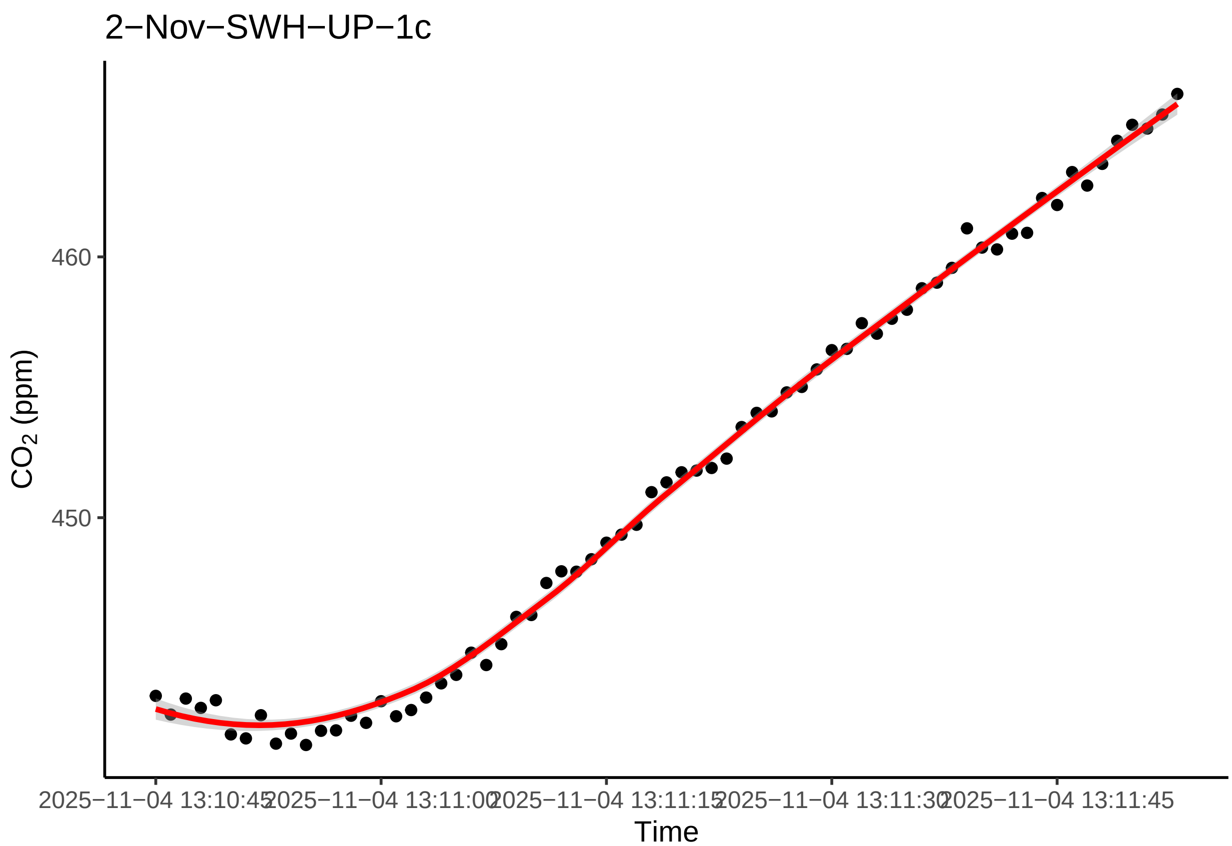
0.10 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

```
tobechecked <- c(
)

colnames <- c("Plot")

#Loop to run five random plots subset above
for(P in tobechecked){
  r1 <- subset(alldat, Plot == P)
  p1 <- ggplot(r1, aes(as.POSIXct(TIMESTAMP), co2)) + geom_point() +
  scale_x_datetime(breaks = "15 sec") +
  theme_classic() +
  ylab(expression(paste( CO [2], " (ppm)"))) +
```

```r
  xlab("Time") + labs(title = P) +
  geom_smooth(color="red")
  print(p1)
  }


#list of the ones that I changed and checked and looked good?
fixedplots <- c("2-Sep-SWH-TR-7", "2-Jul-SWH-TR-3", "1-May-SWH-TR-6a", "1-Jul-SWH-TR-3",
                "1-Jul-SWH-TR-10", "2-Jul-SWH-TR-2", "2-Mar-GWI-TR-5", "2-Mar-GWI-UP-3a",
                "2-May-SWH-TR-6a", "1-Mar-GWI-UP-3a", "1-May-SWH-TR-7a",
                "1-Jul-SWH-TR-9", "1-May-SWH-TR-1", "2-Mar-SWH-UP-1", "2-Mar-SWH-UP-2",
                "2-Mar-SWH-UP-5", "2-Nov-SWH-TR-2", "2-Nov-SWH-TR-4", "2-Nov-SWH-TR-5",
                "1-Jul-SWH-TR-2", "1-Jul-SWH-TR-6", "1-Jul-SWH-TR-7a", "2-May-MSM-UP-2",
                "2-May-MSM-UP-4", "2-Jul-GWI-UP-5", "1-Jul-SWH-TR-3", "2-Jul-SWH-TR-6",
                "2-May-MSM-UP-2", "1-Jul-SWH-TR-7a")
```

0.11 Spot check a few fluxes with R2 > 0.9

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

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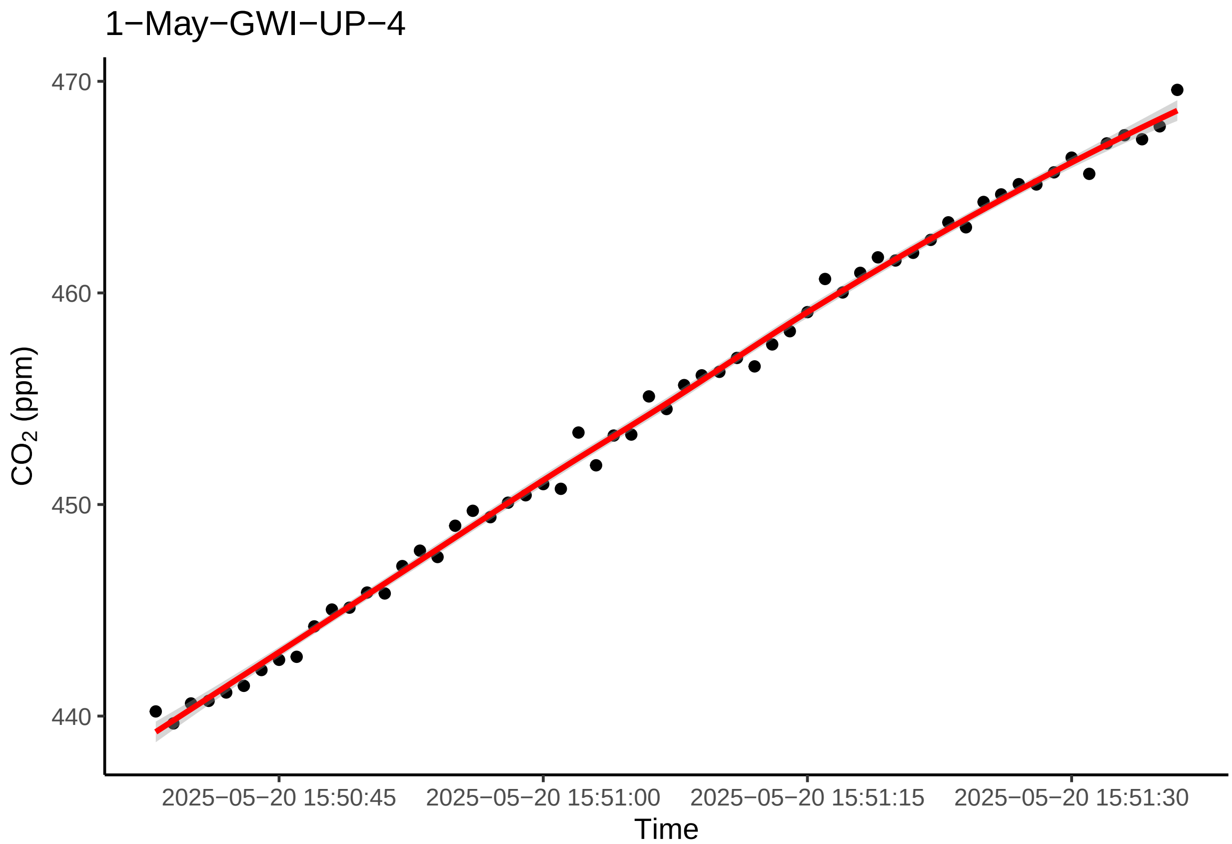


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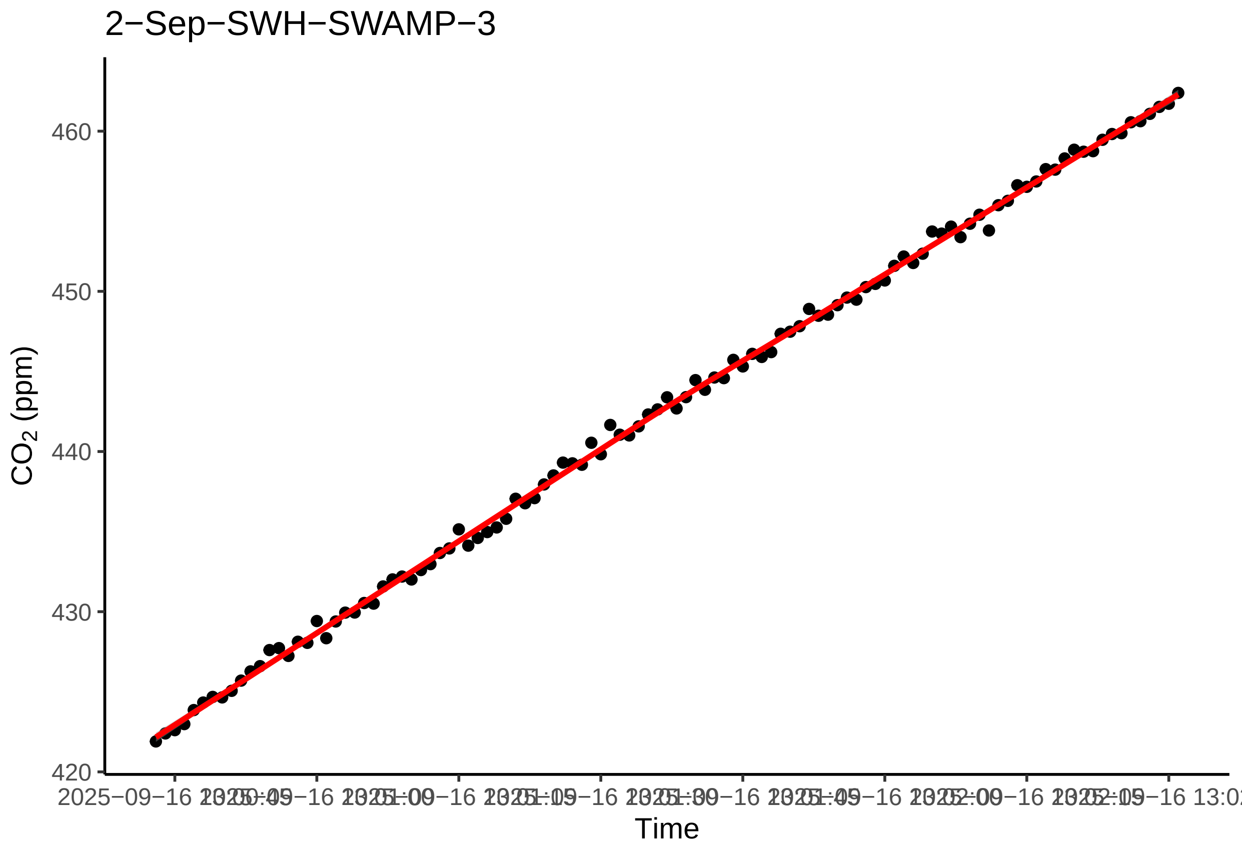


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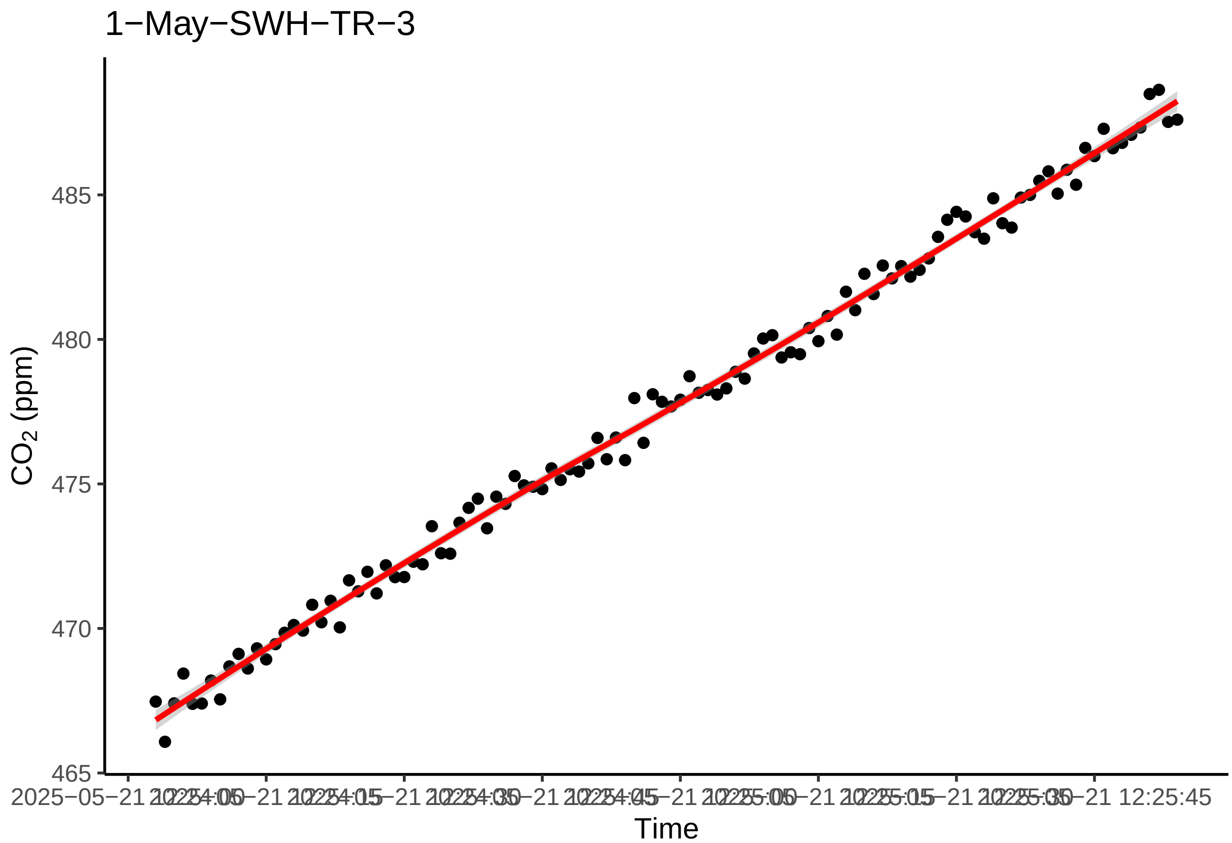


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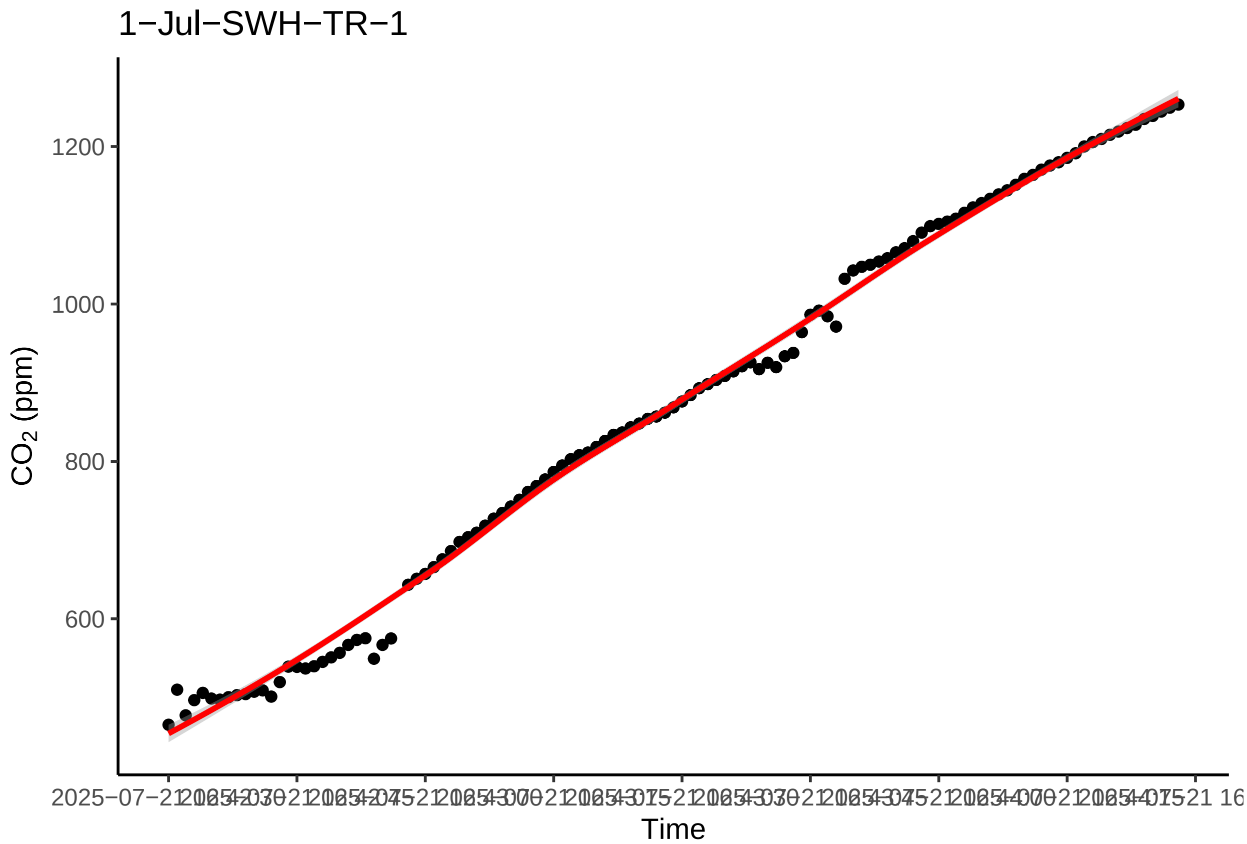

1−Jul−SWH−TR−1
CO2 (ppm)
1200
1000
800
600
Time

0.12 Take a look at the fluxes <0.9 and a flux estimate below 1

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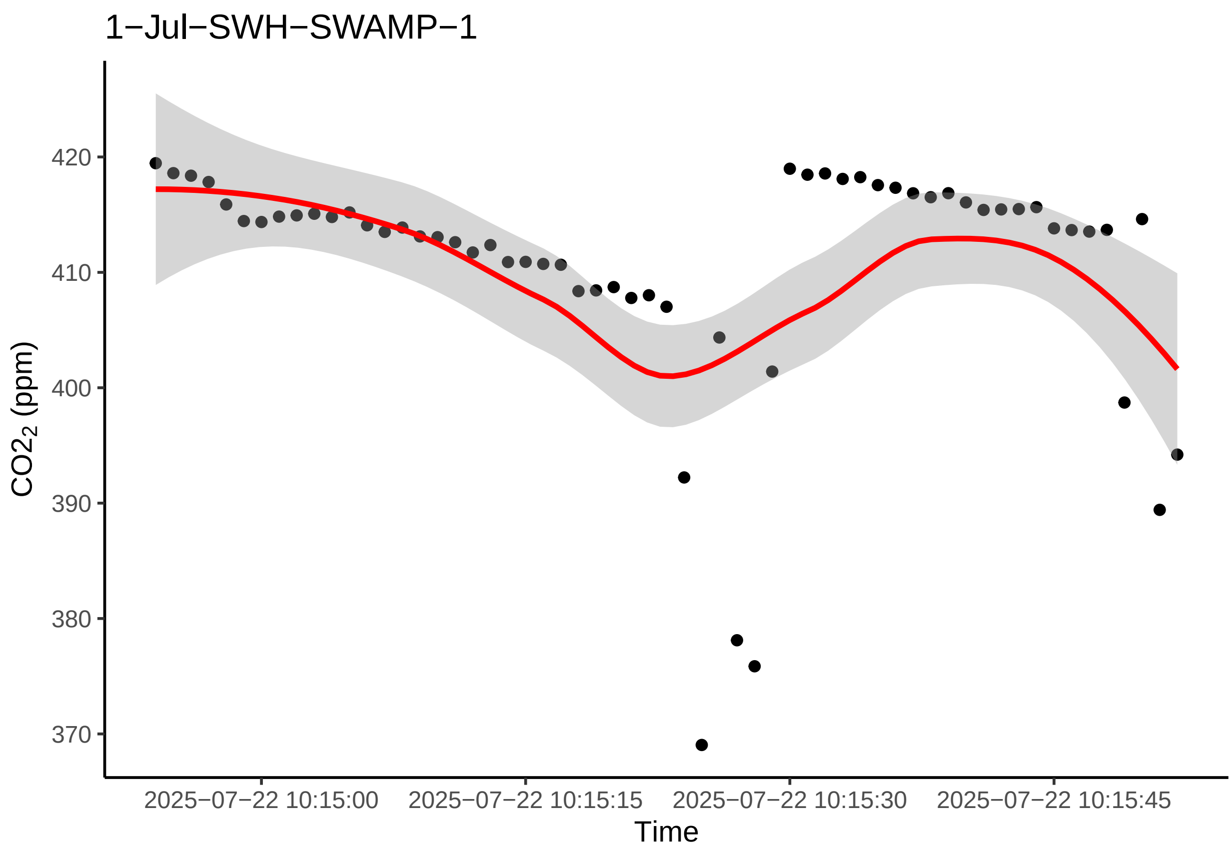


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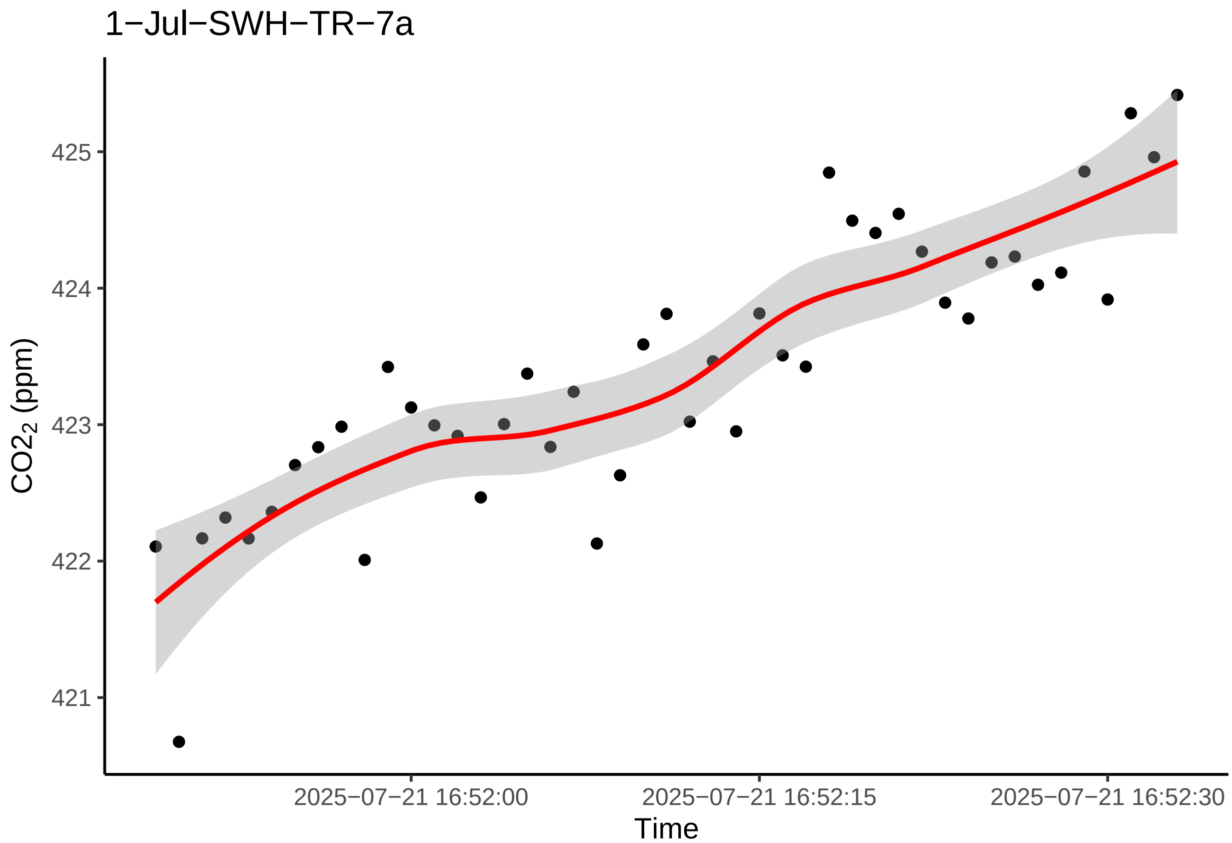


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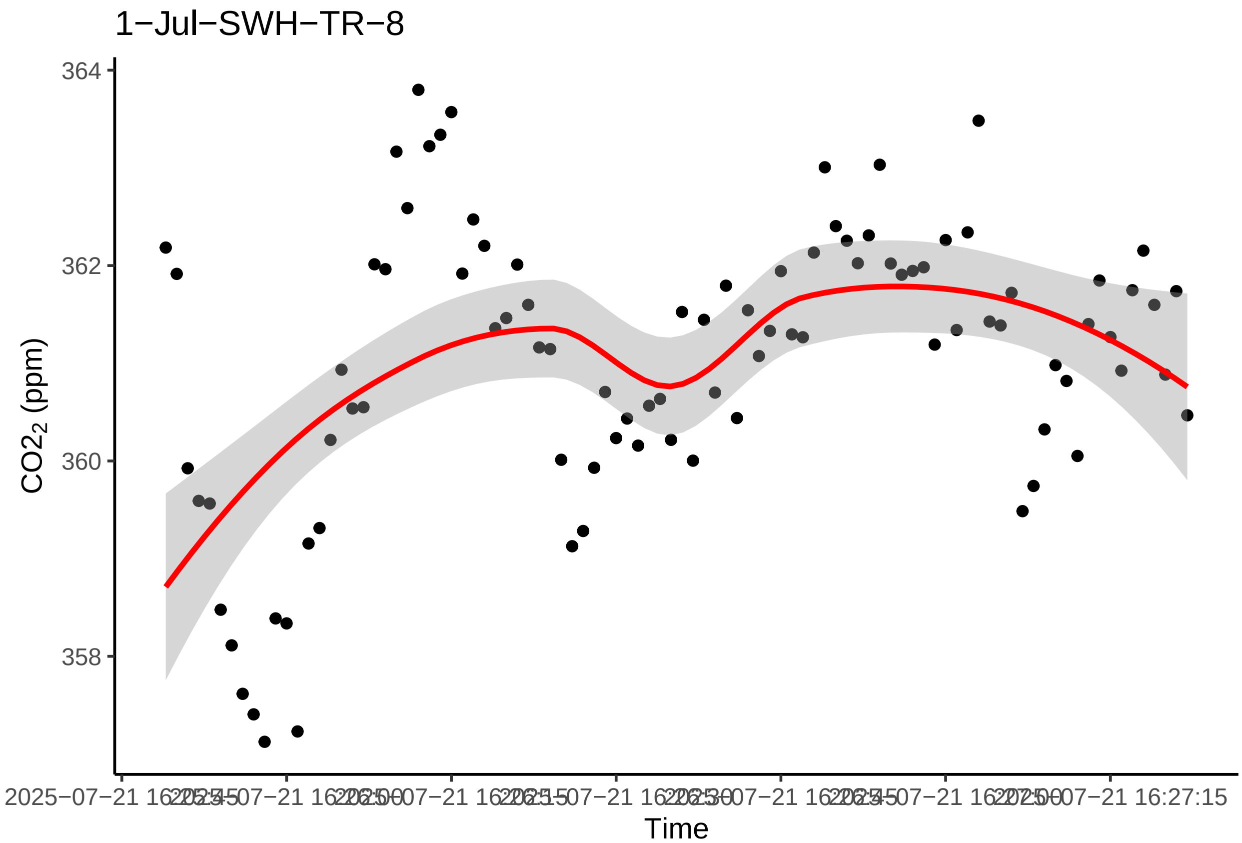


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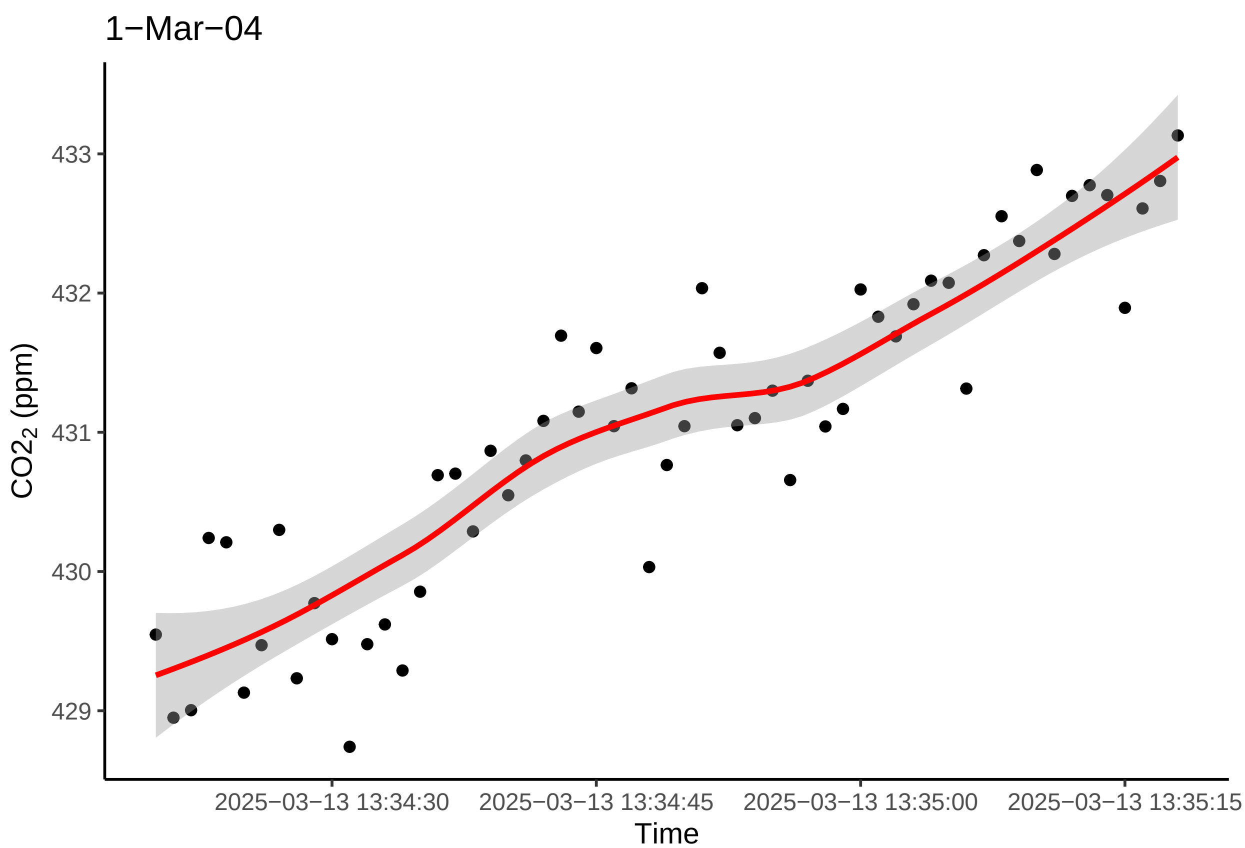


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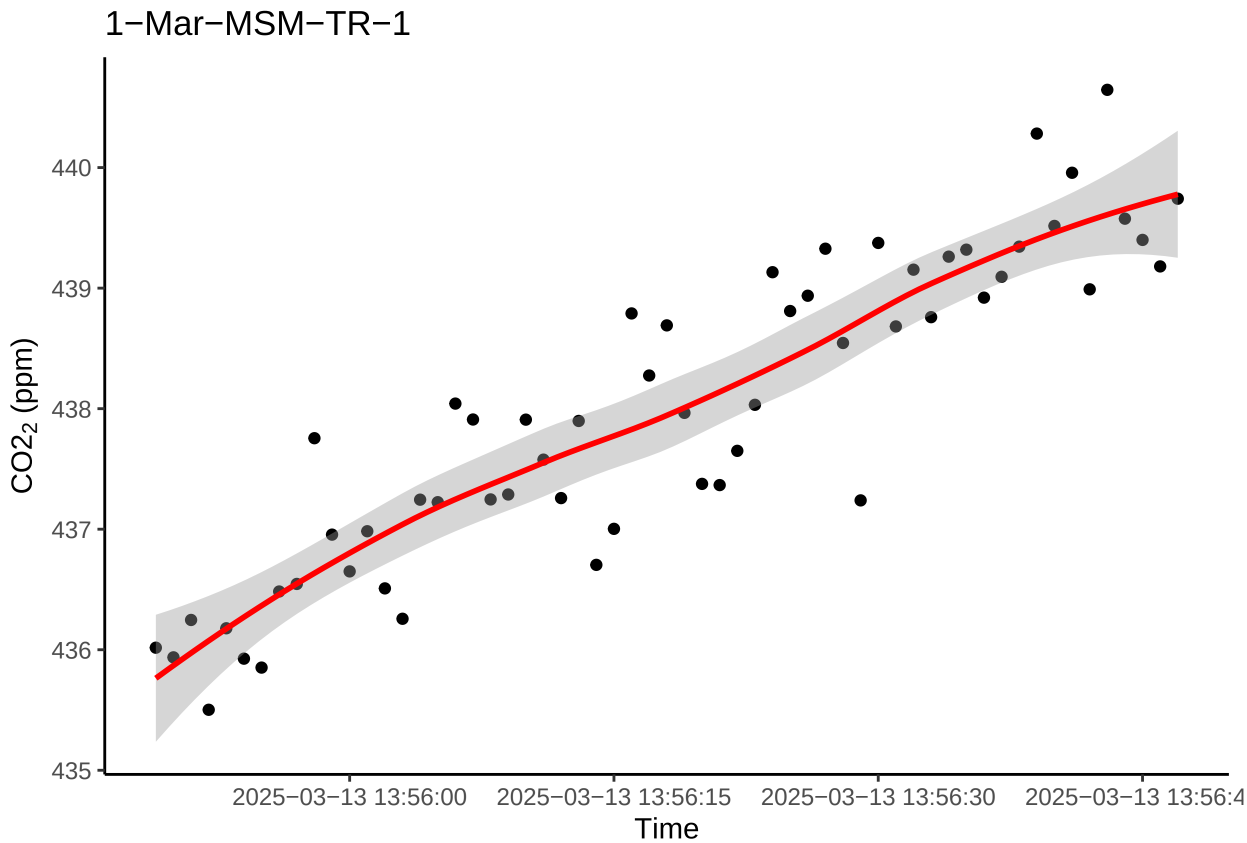


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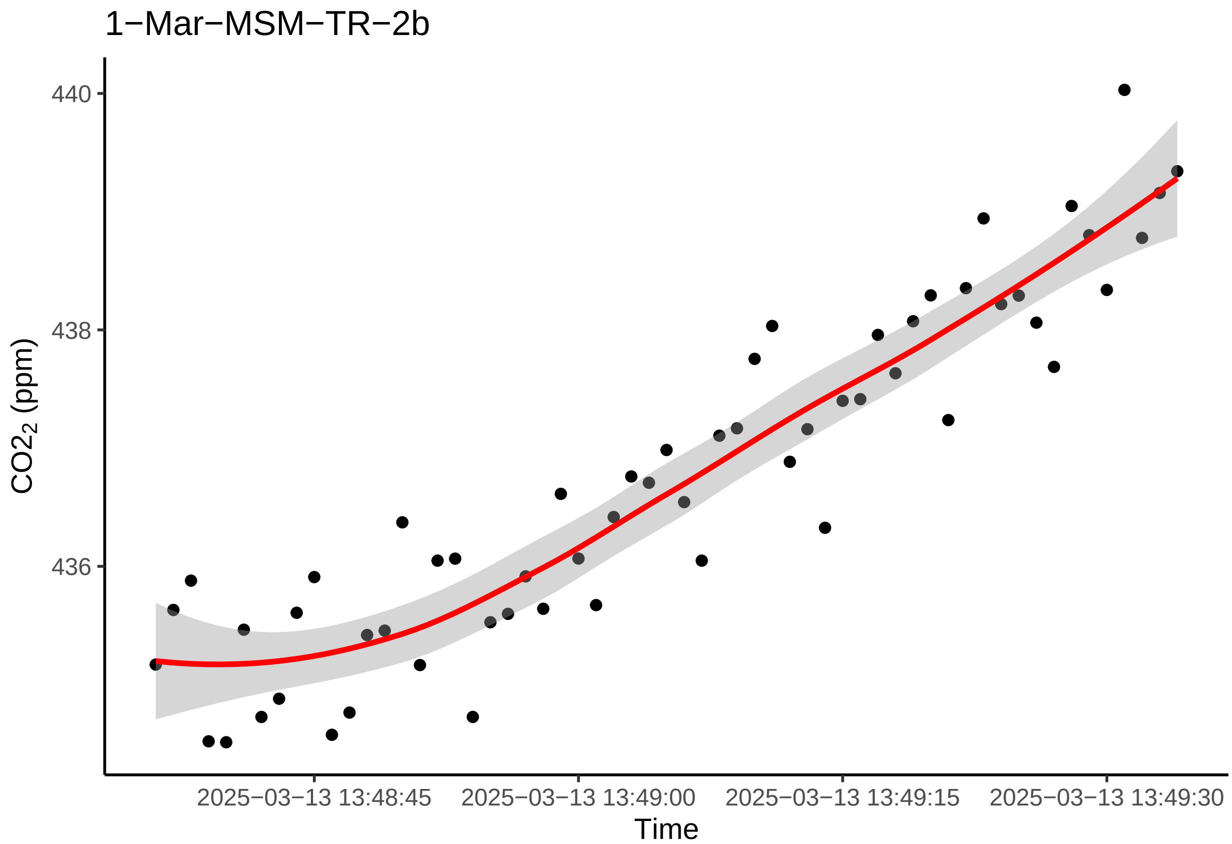


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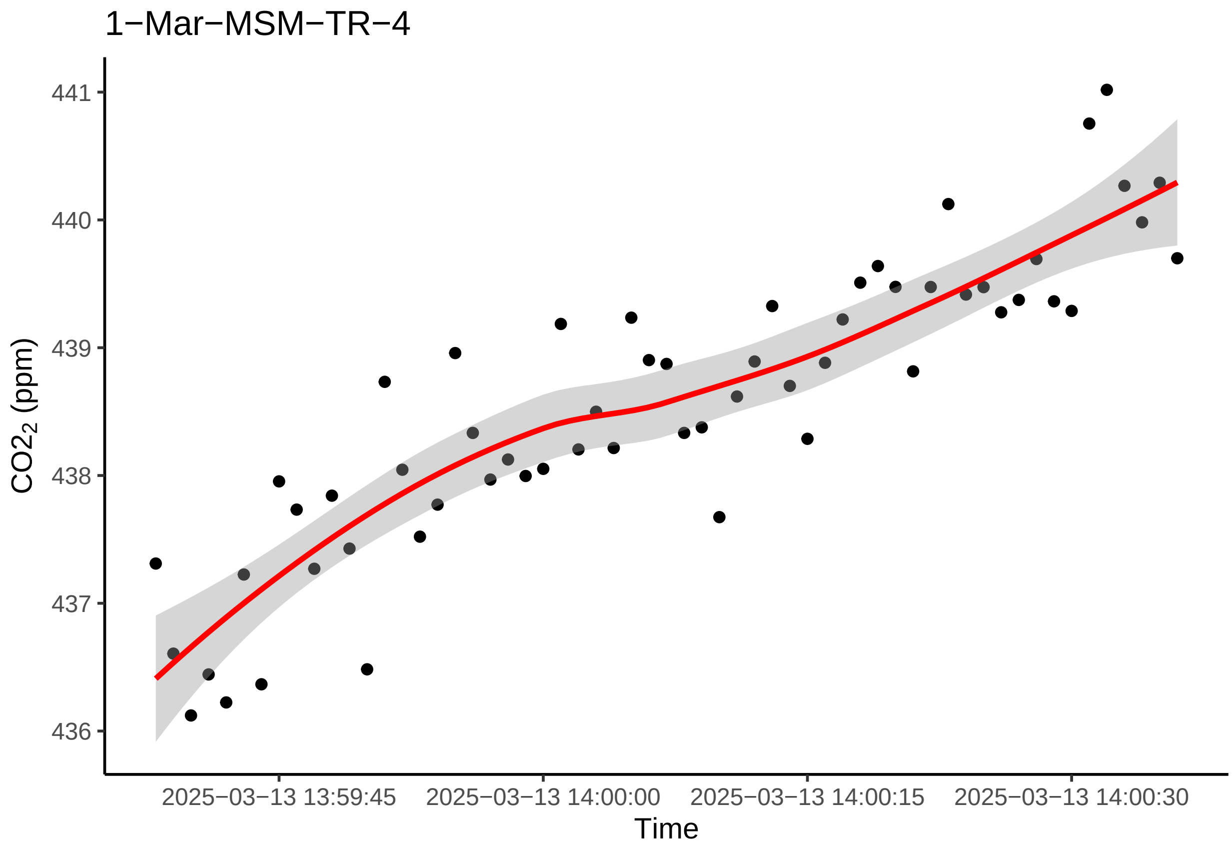


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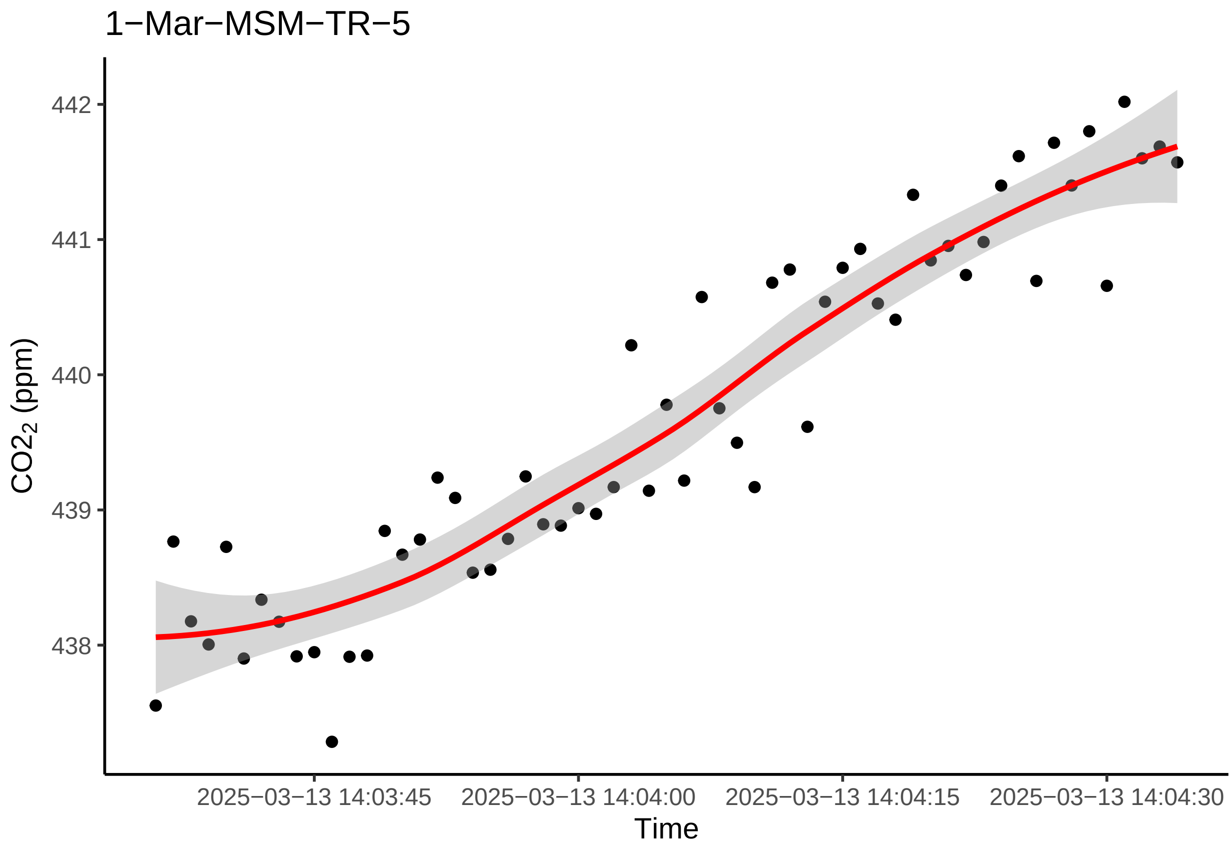


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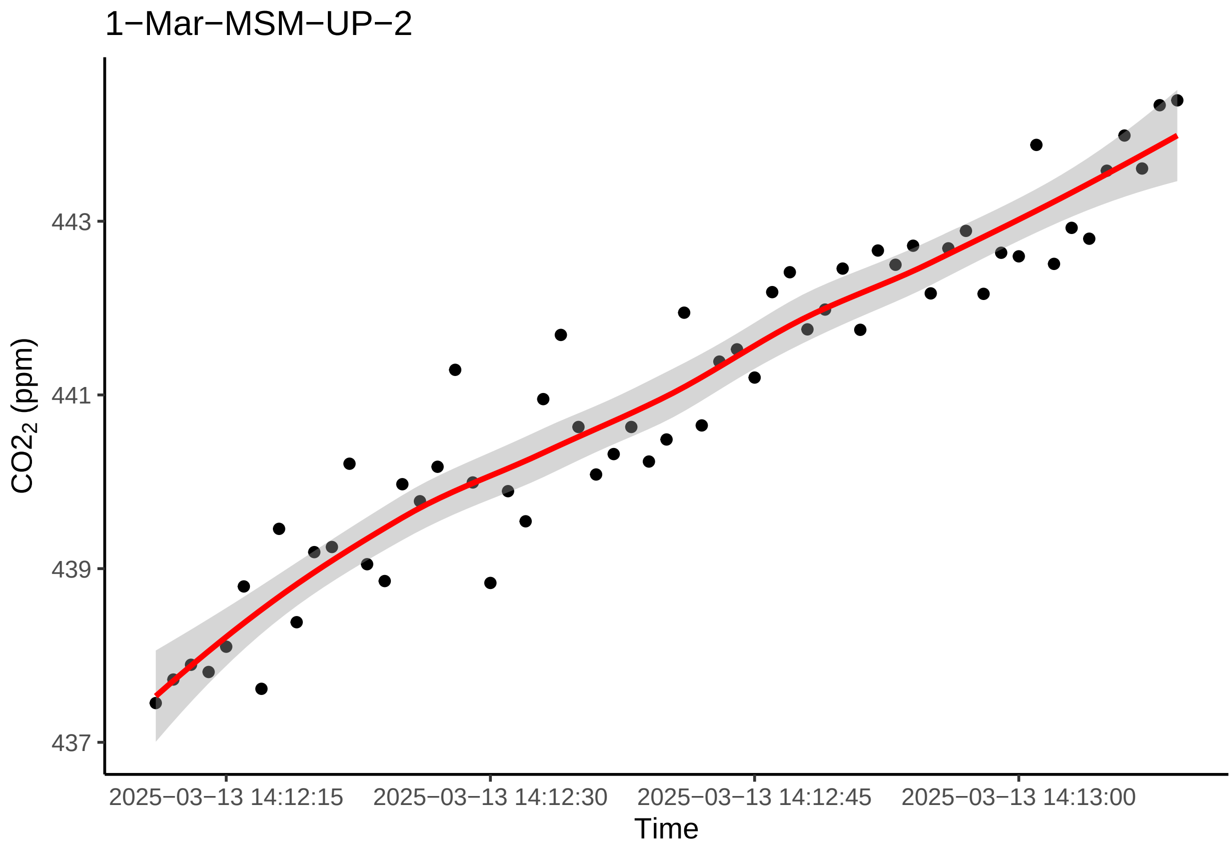


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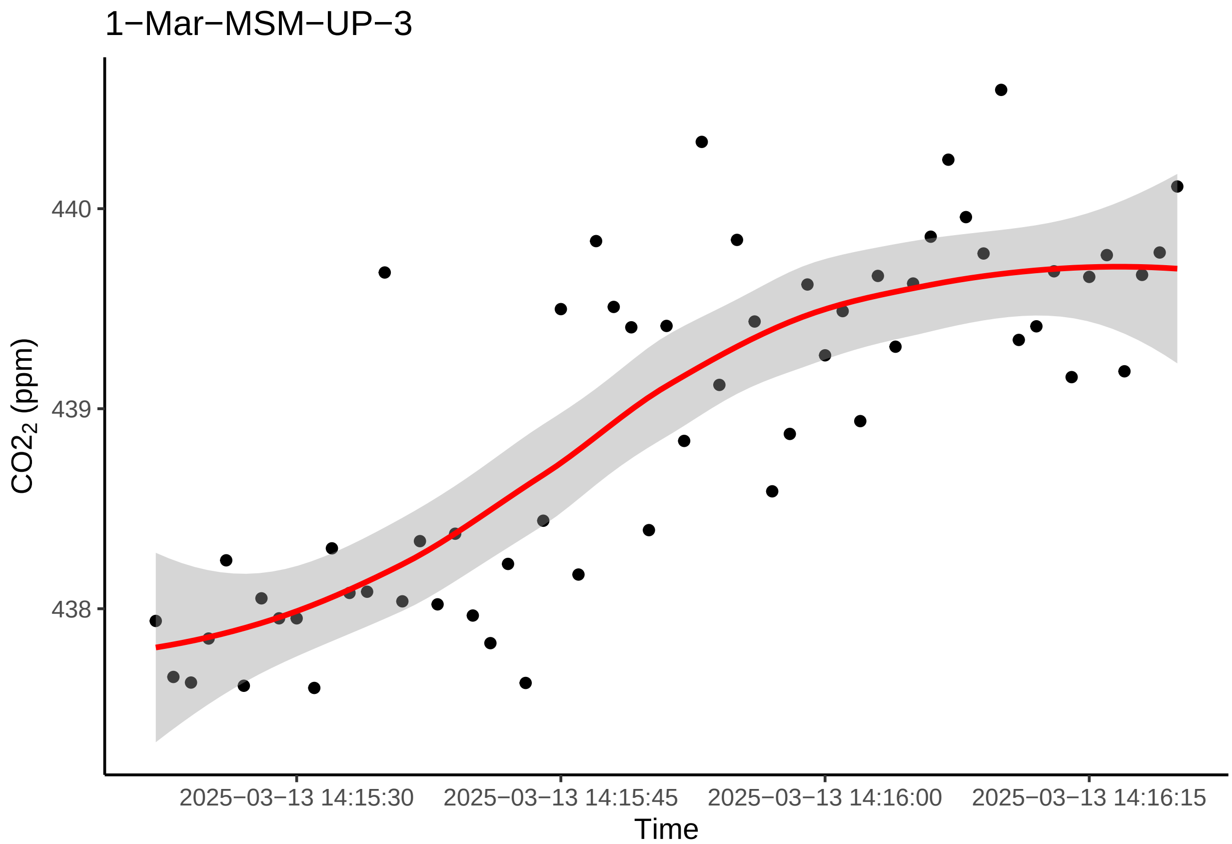


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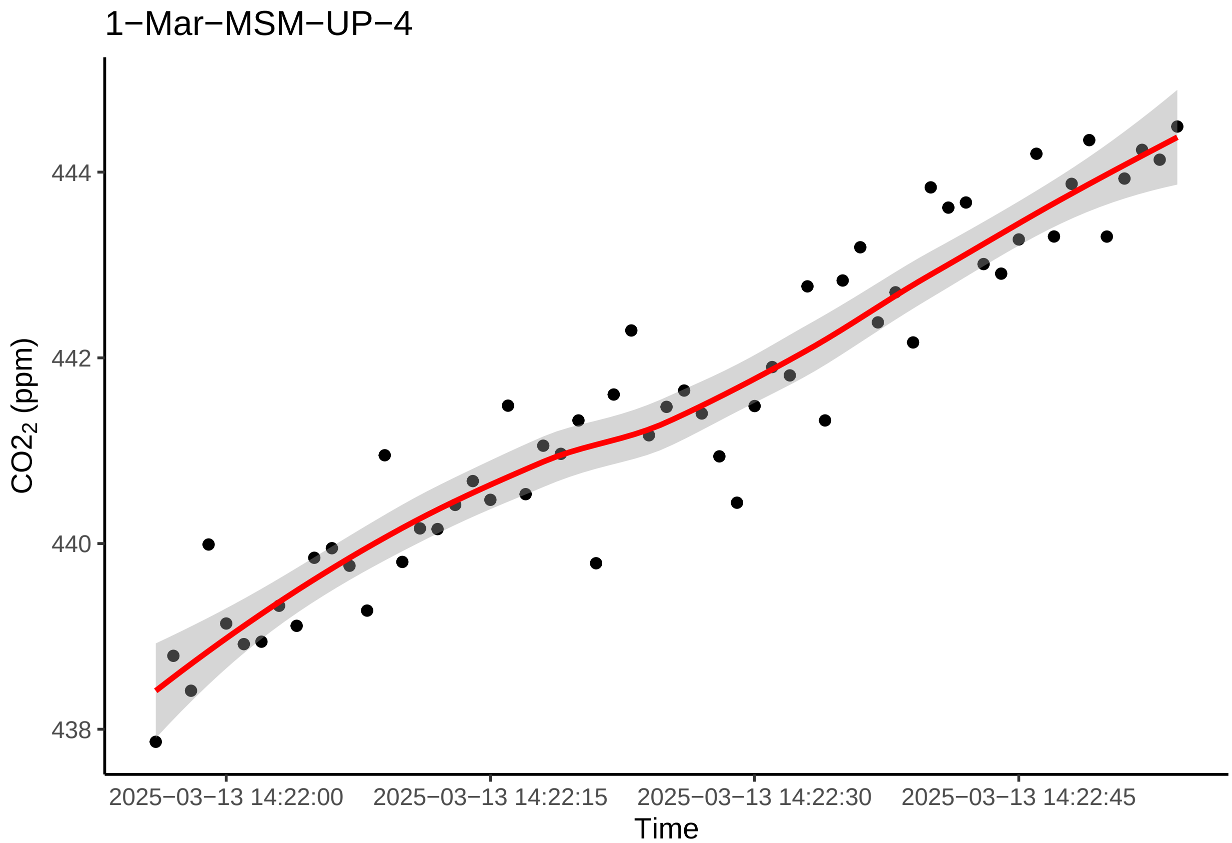


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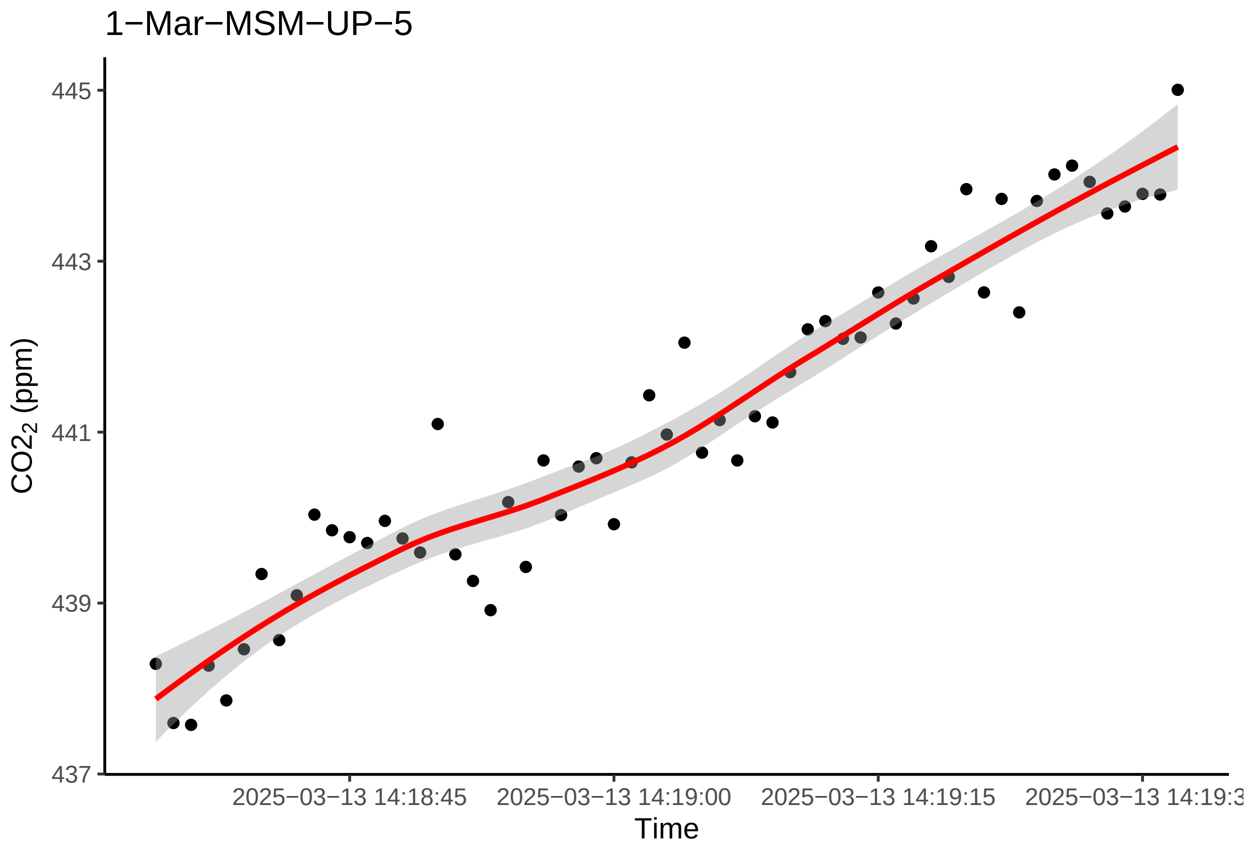


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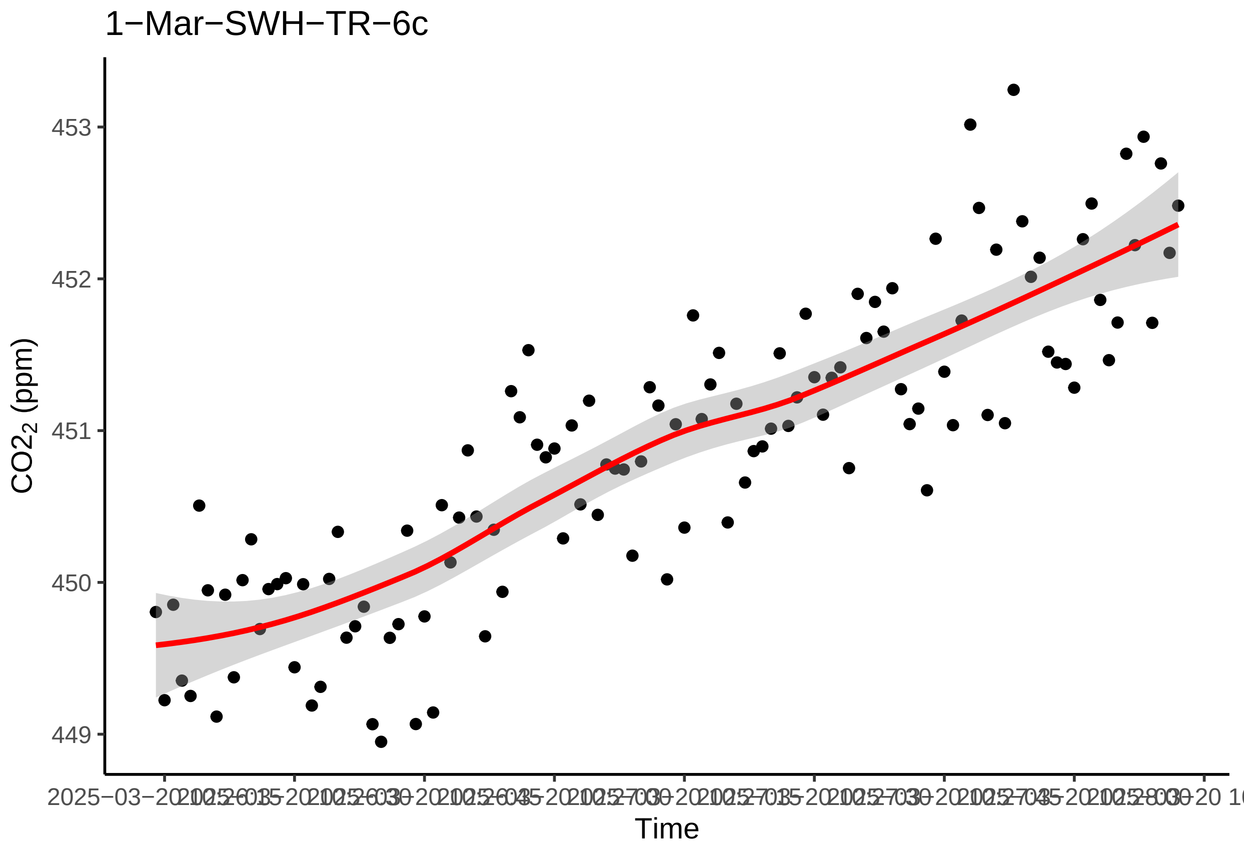


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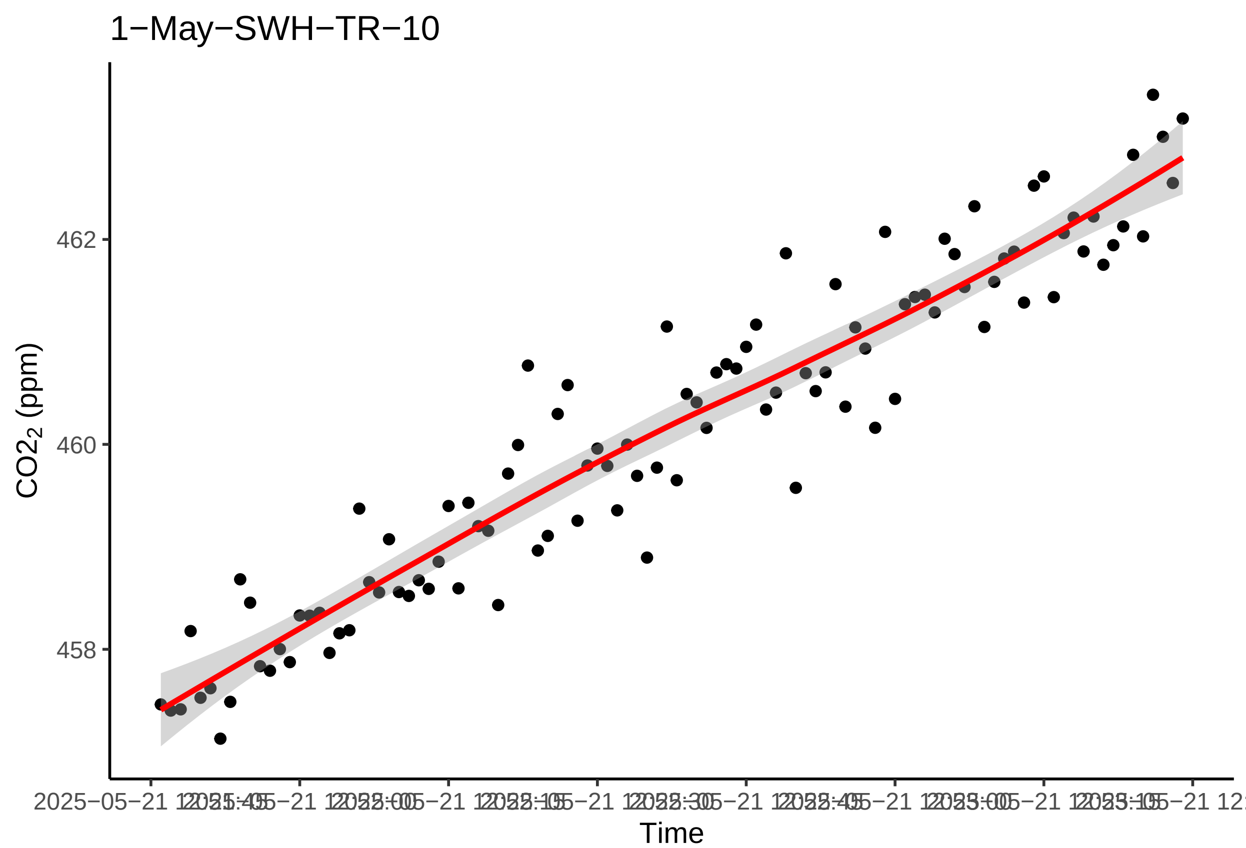


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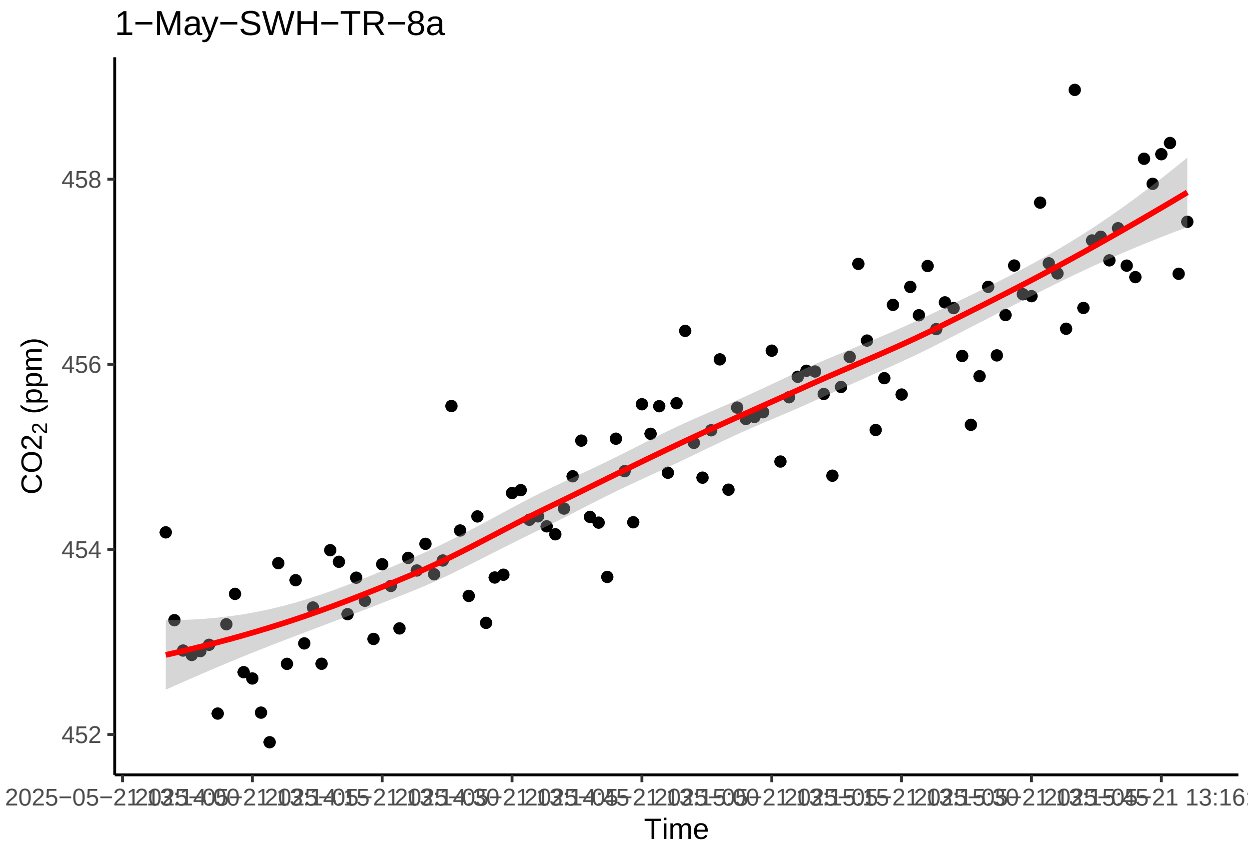


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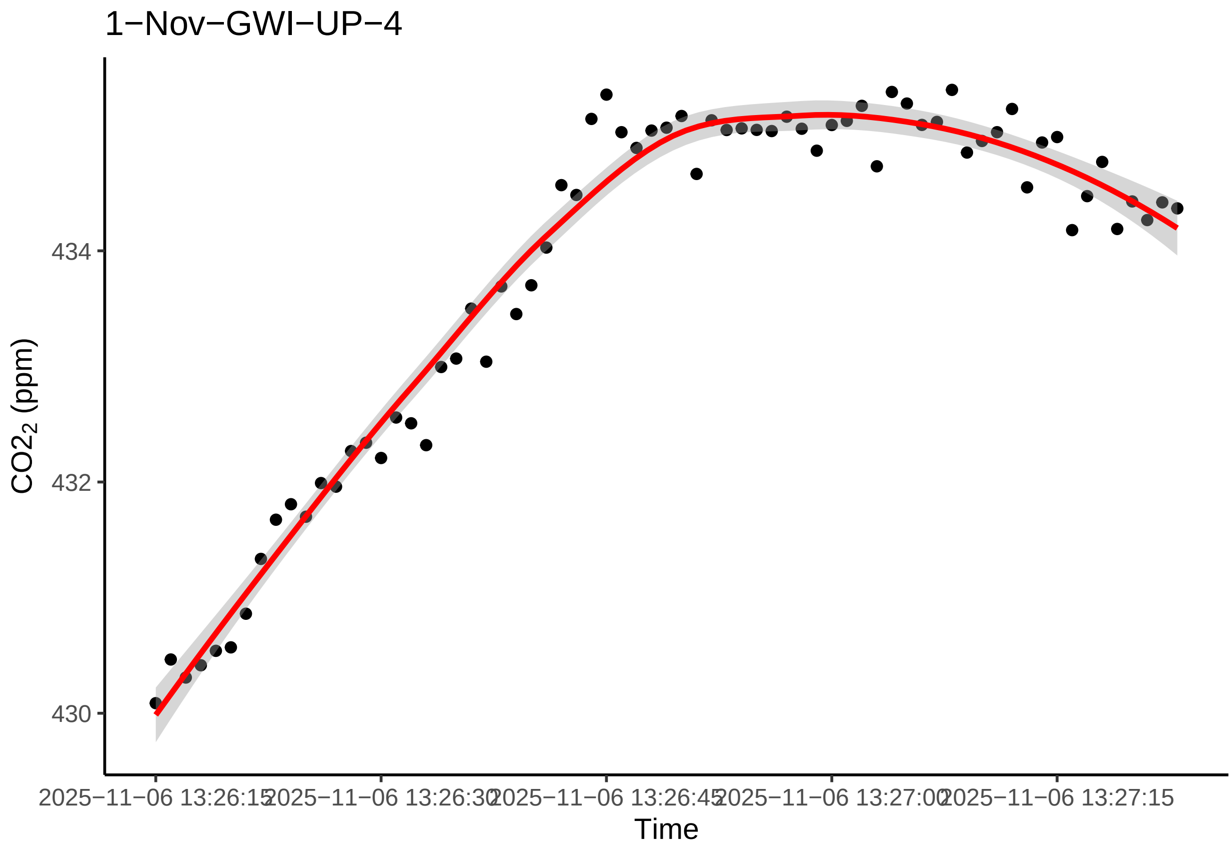


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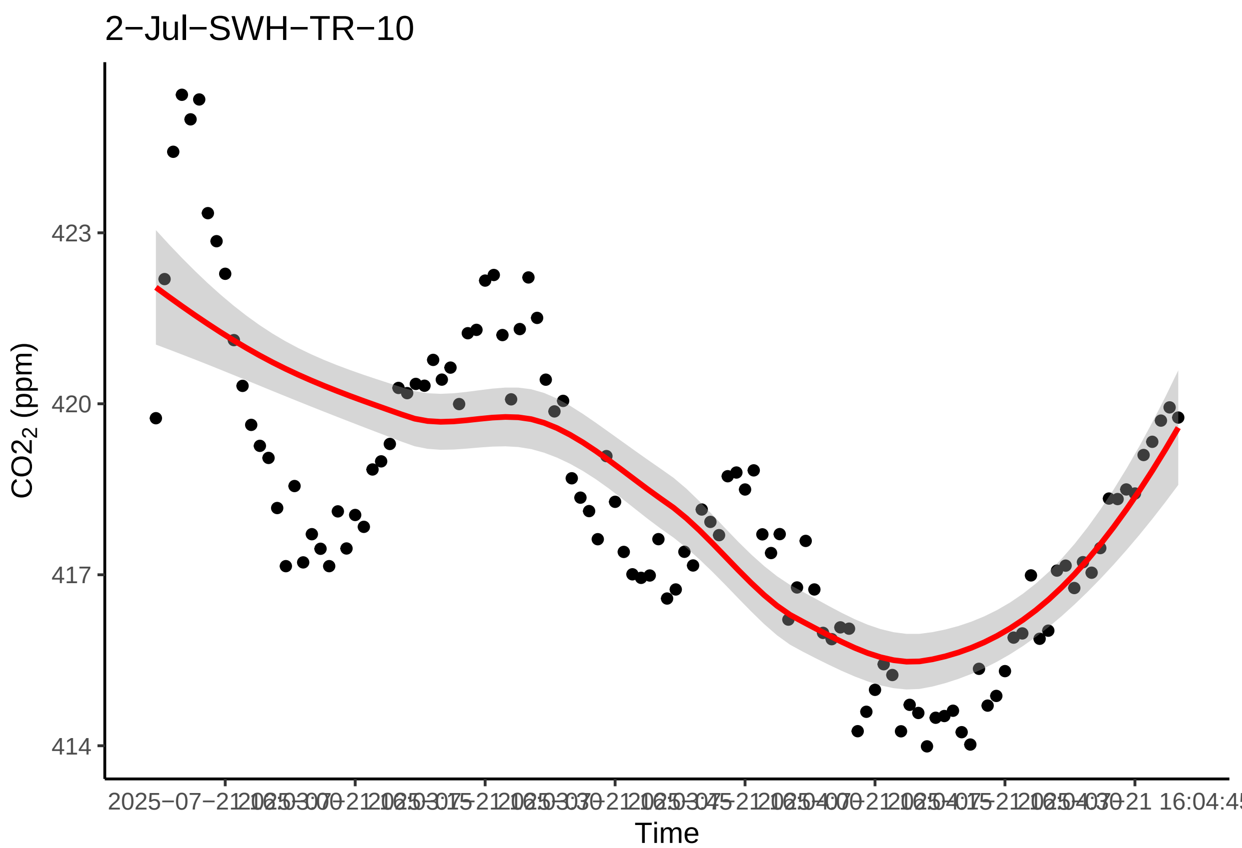


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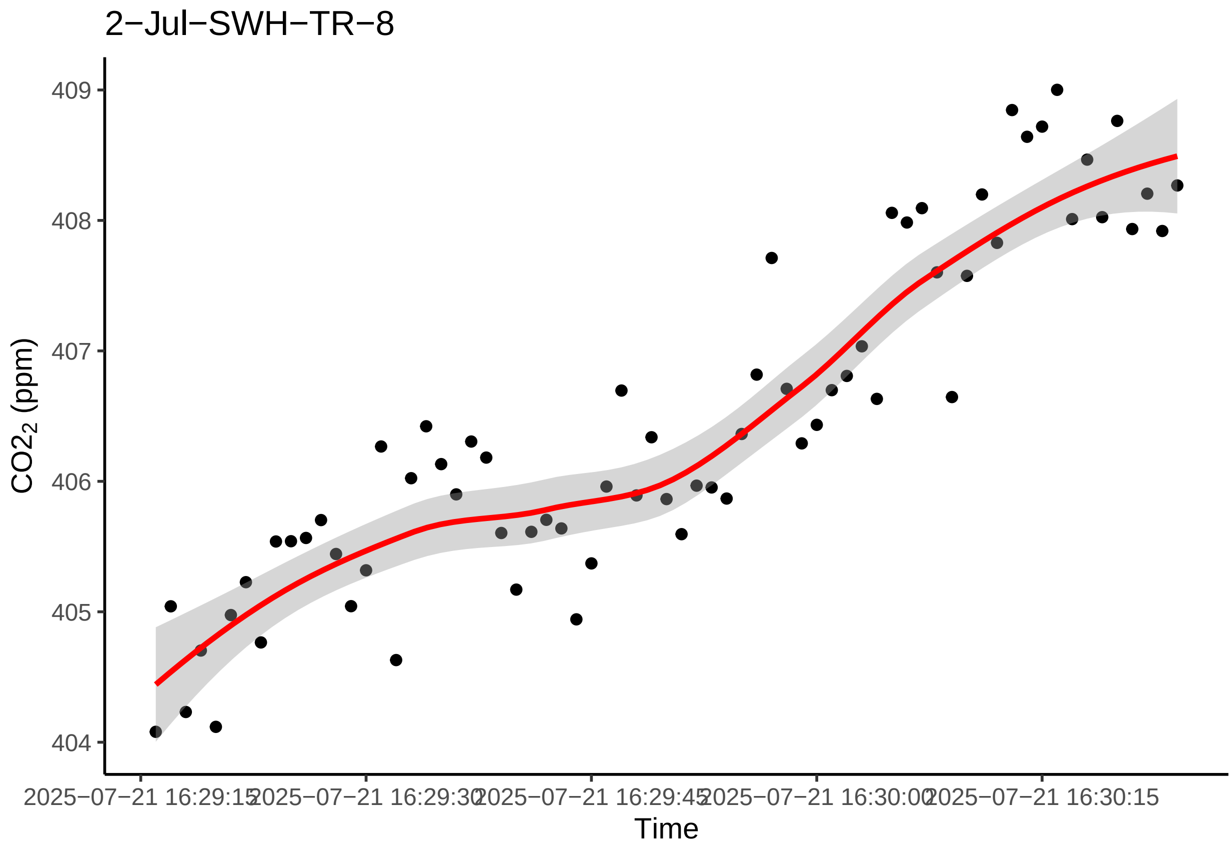


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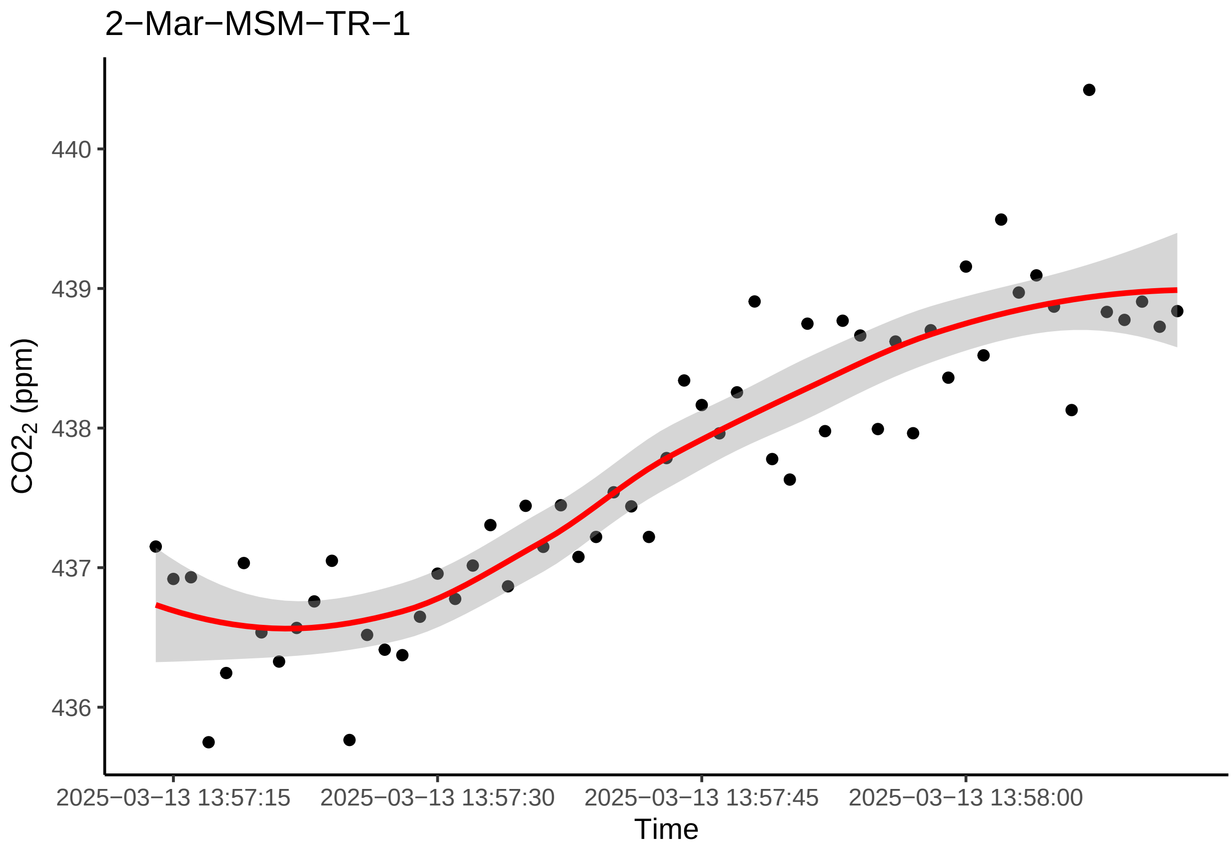


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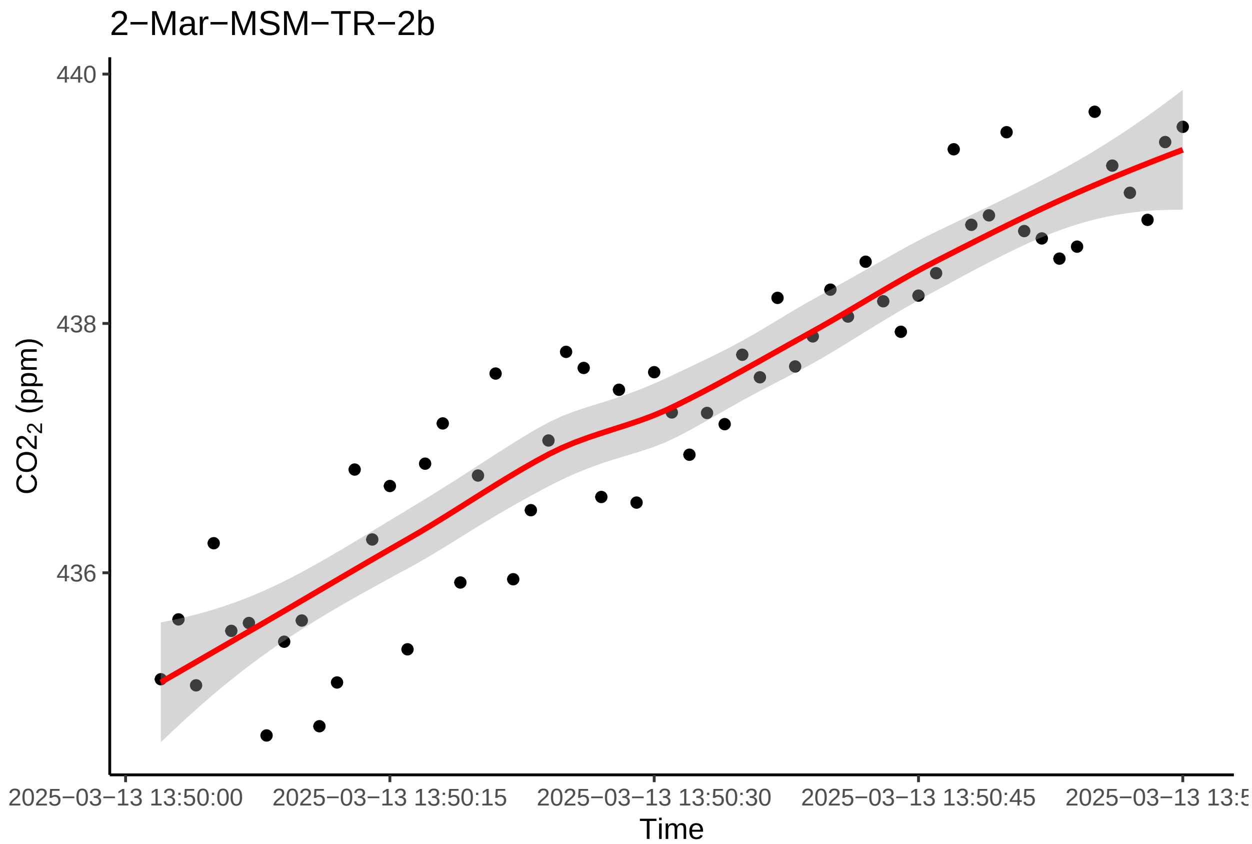


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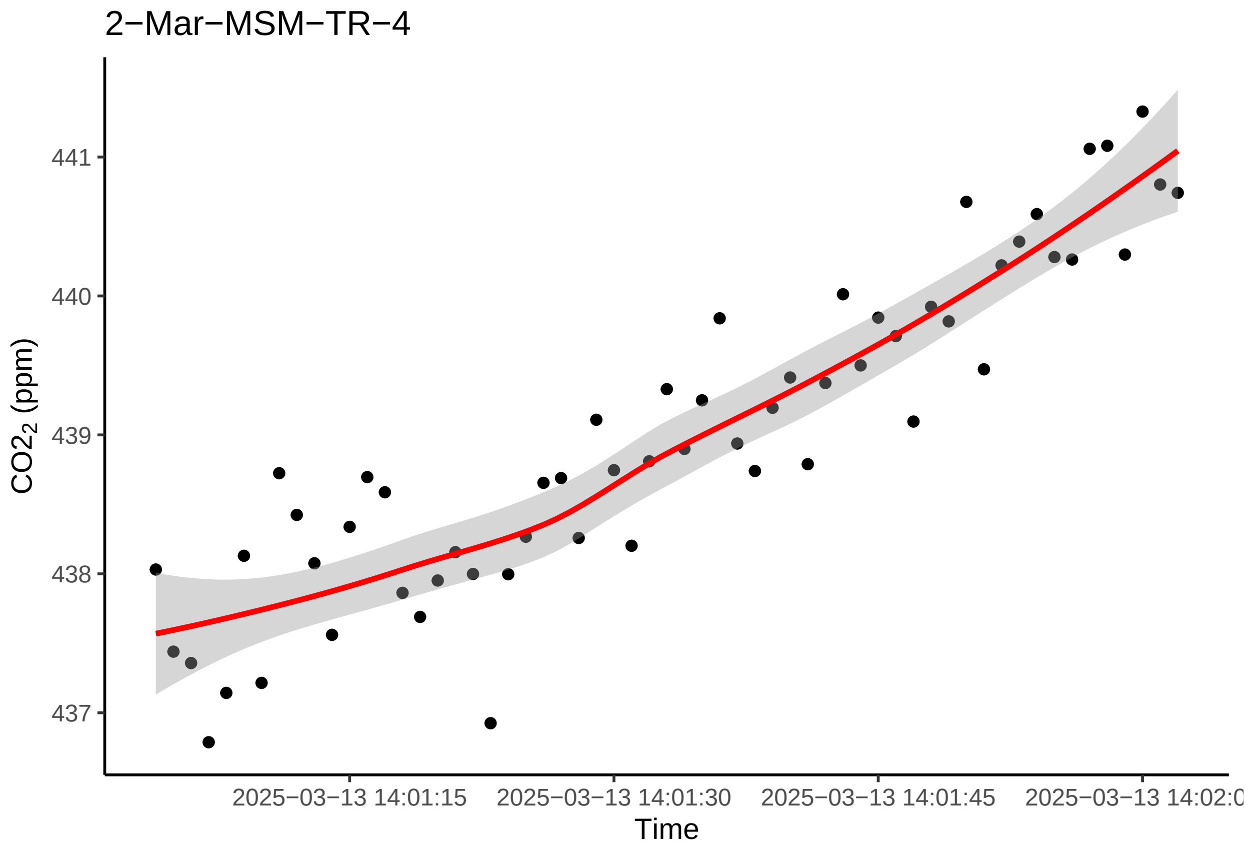


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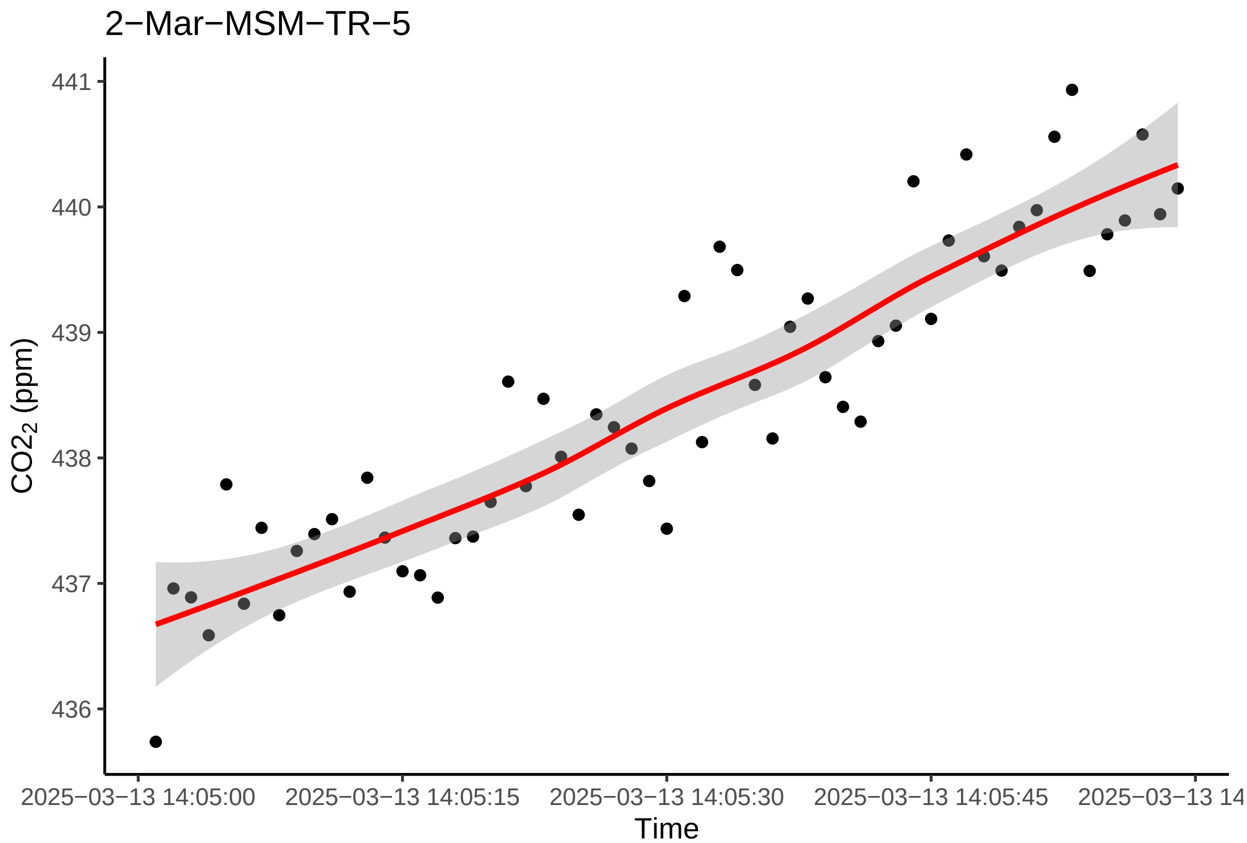


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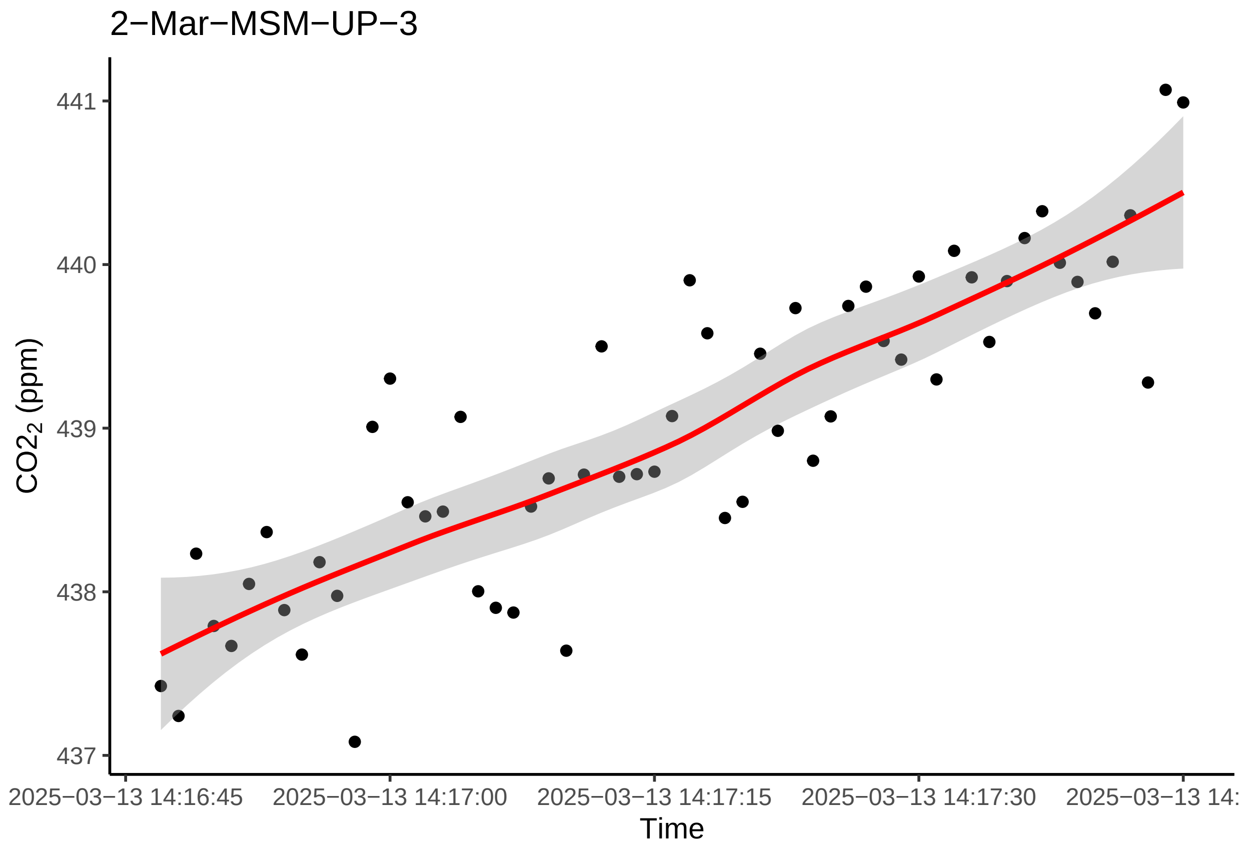


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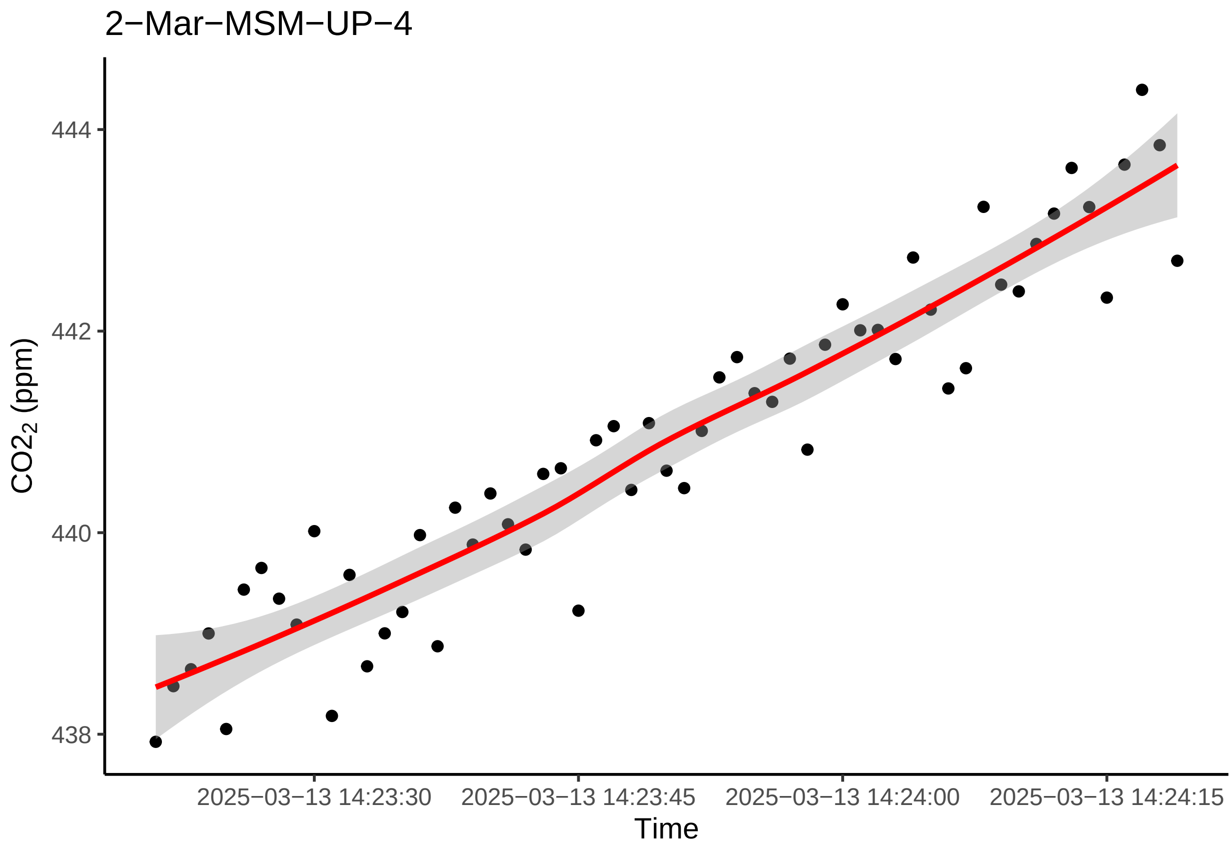


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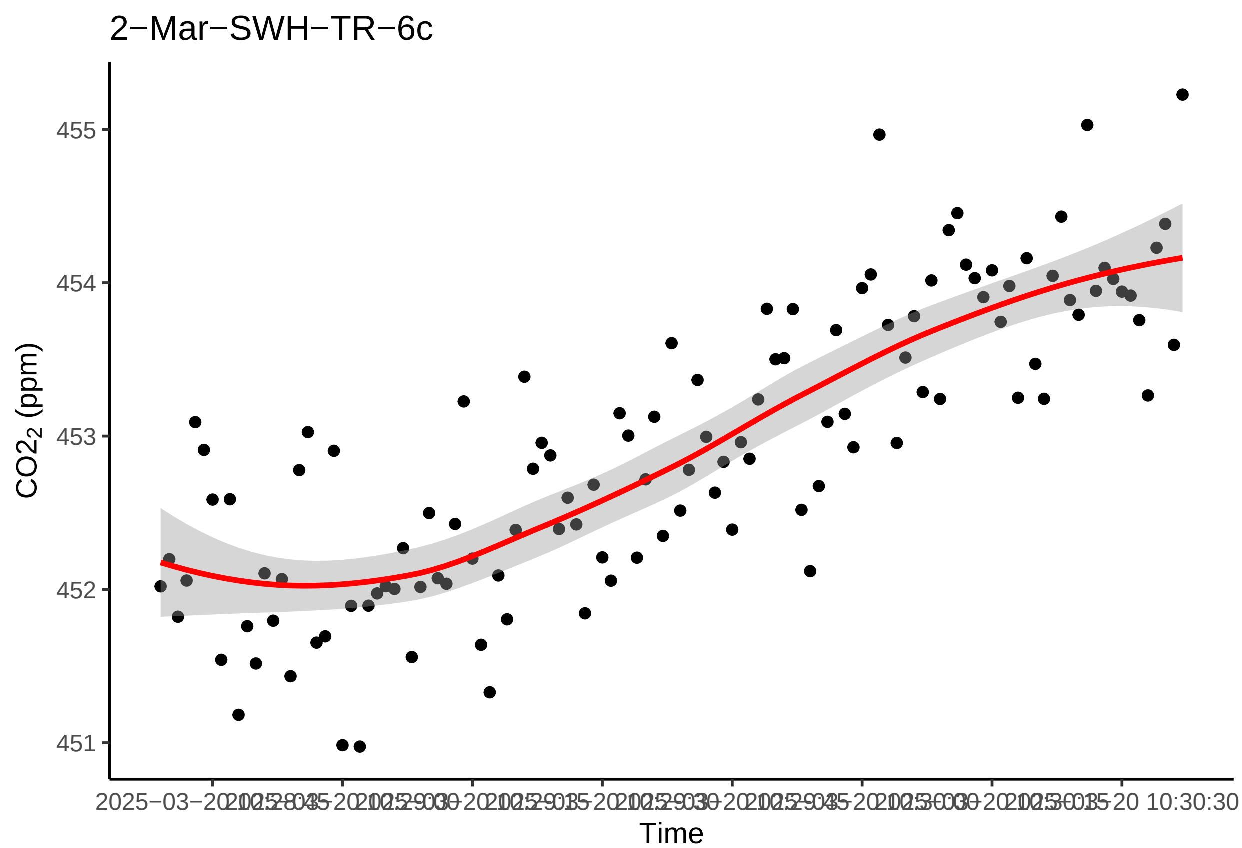


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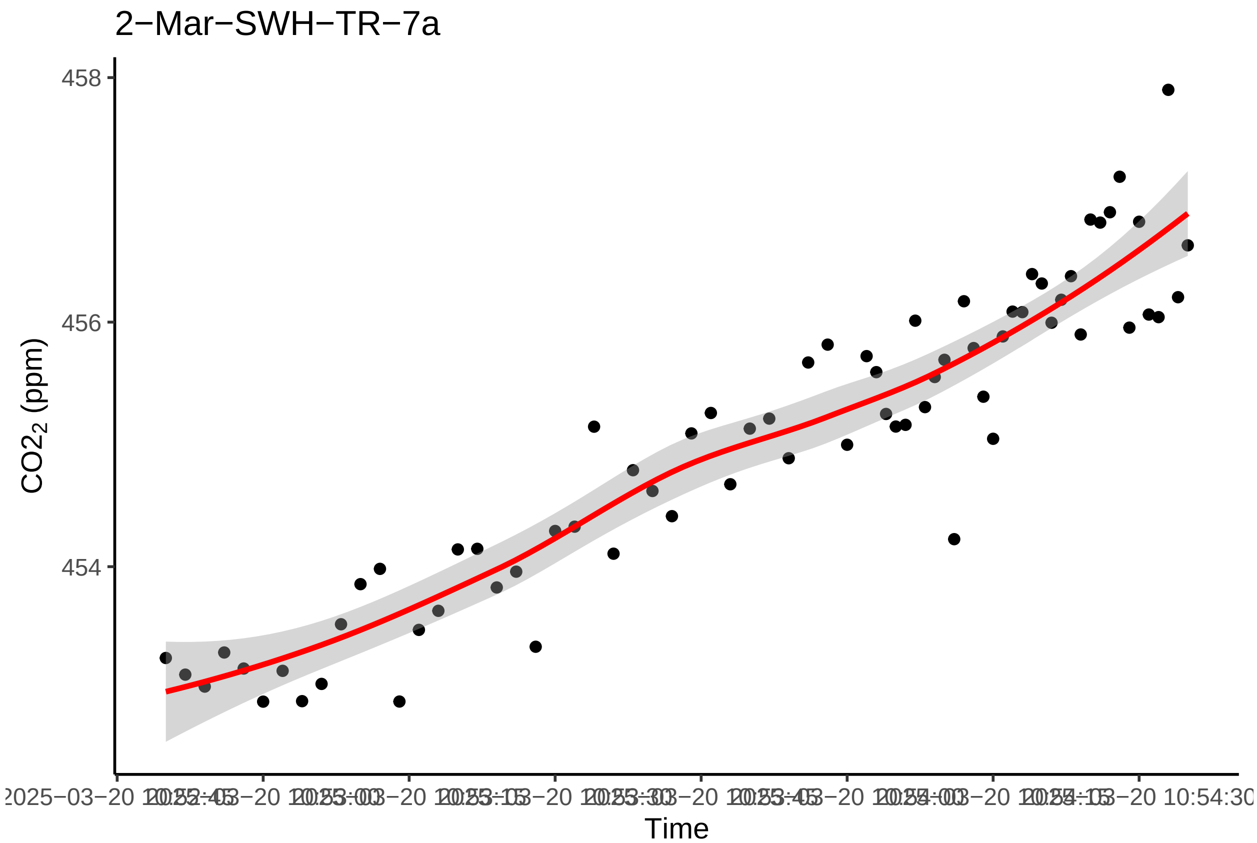


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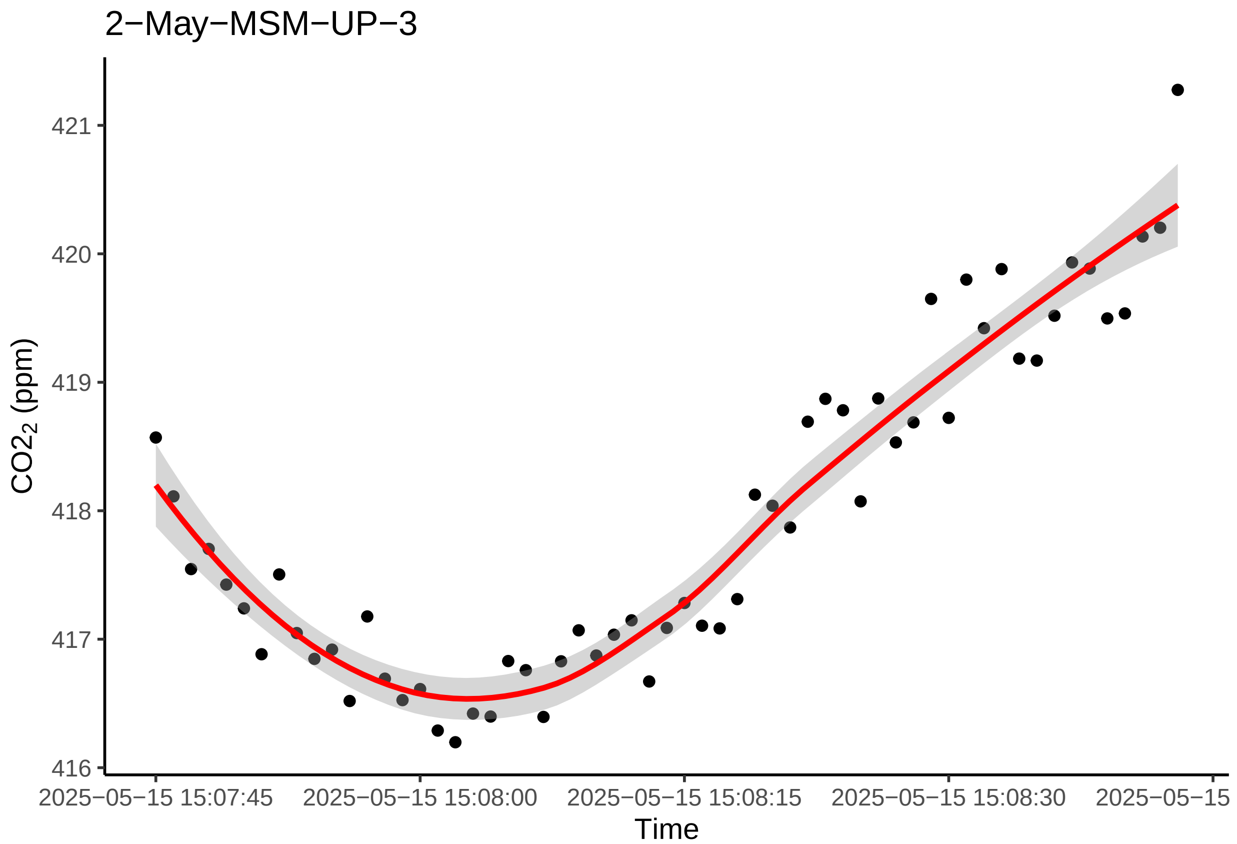


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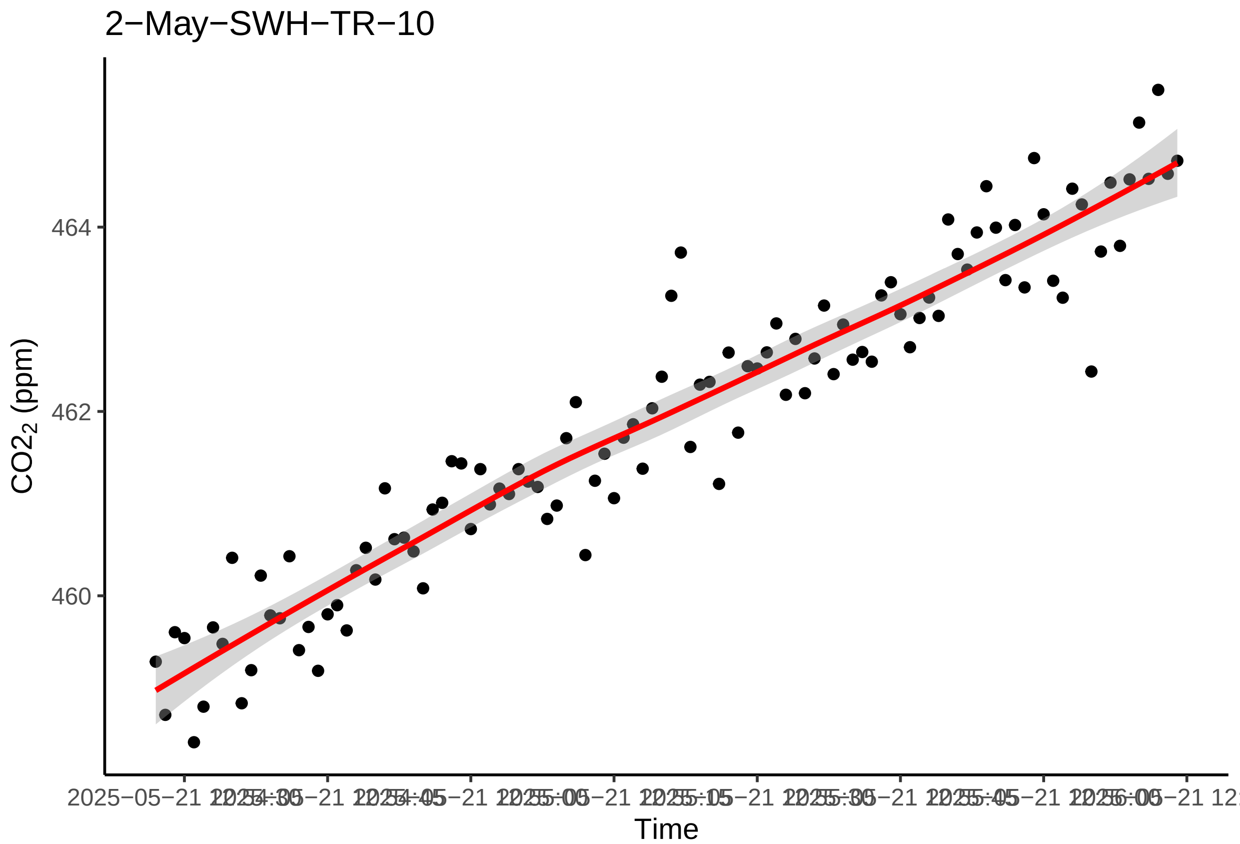


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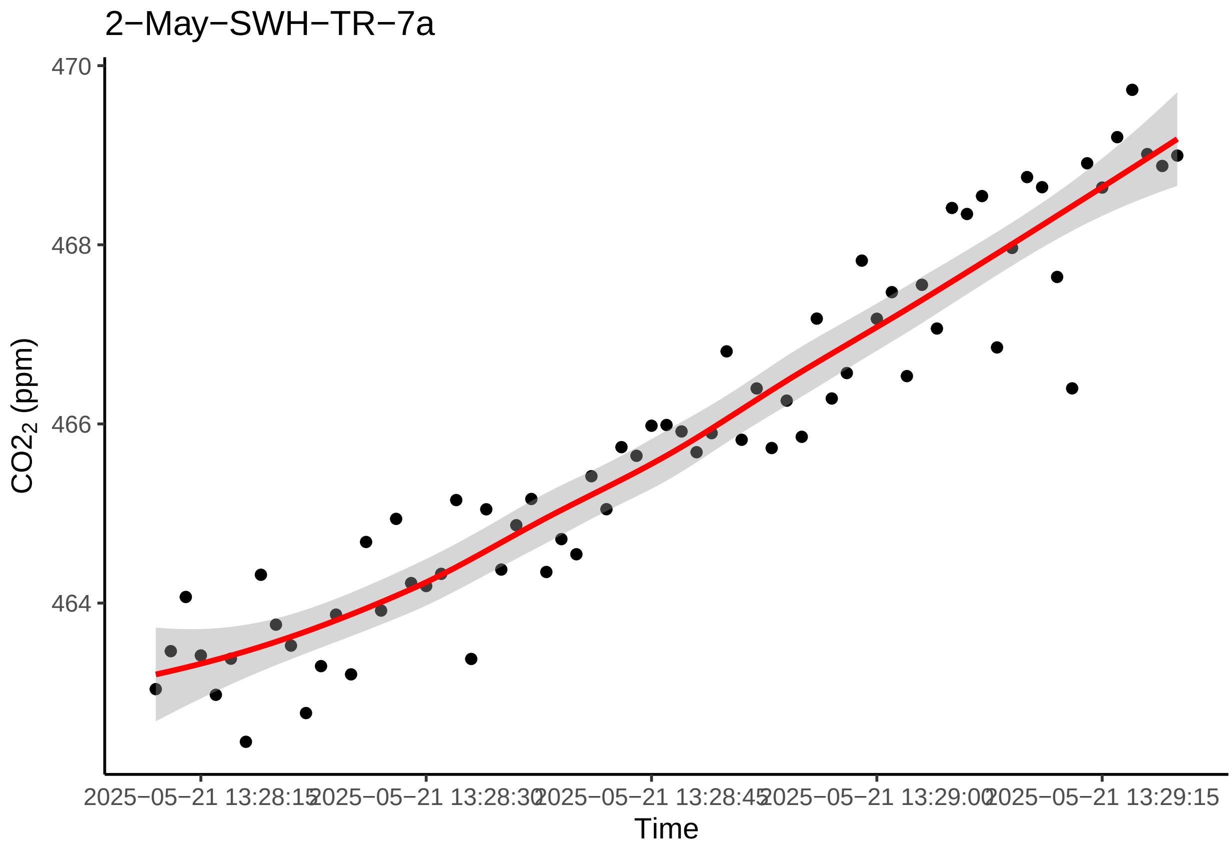


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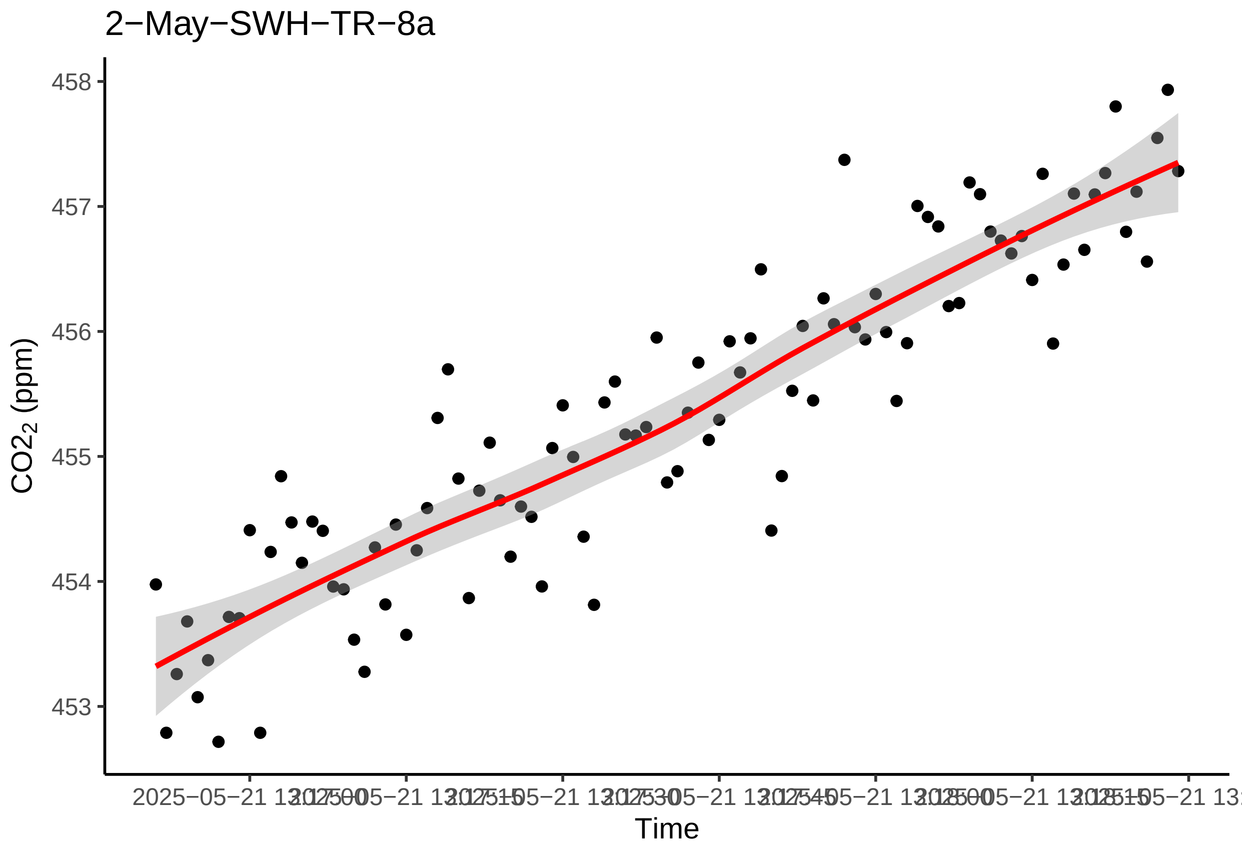


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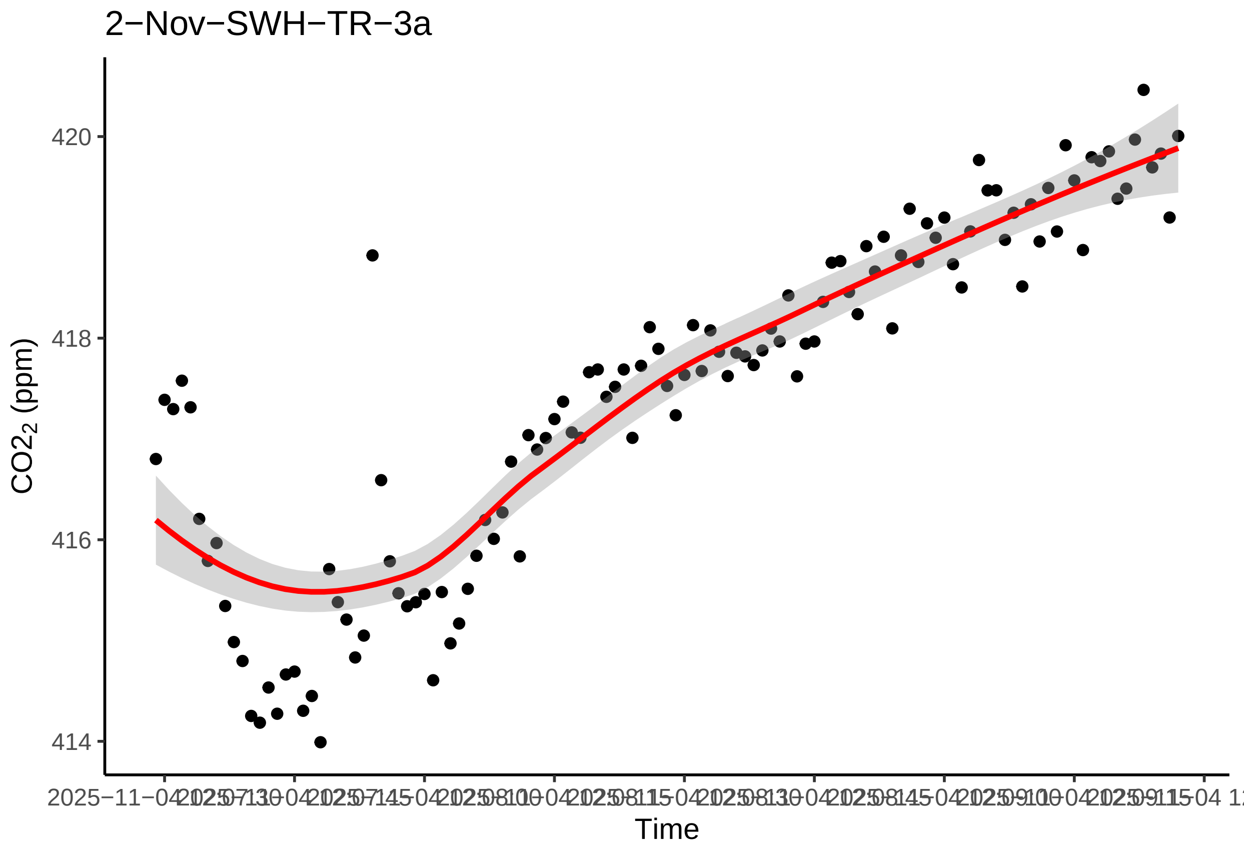
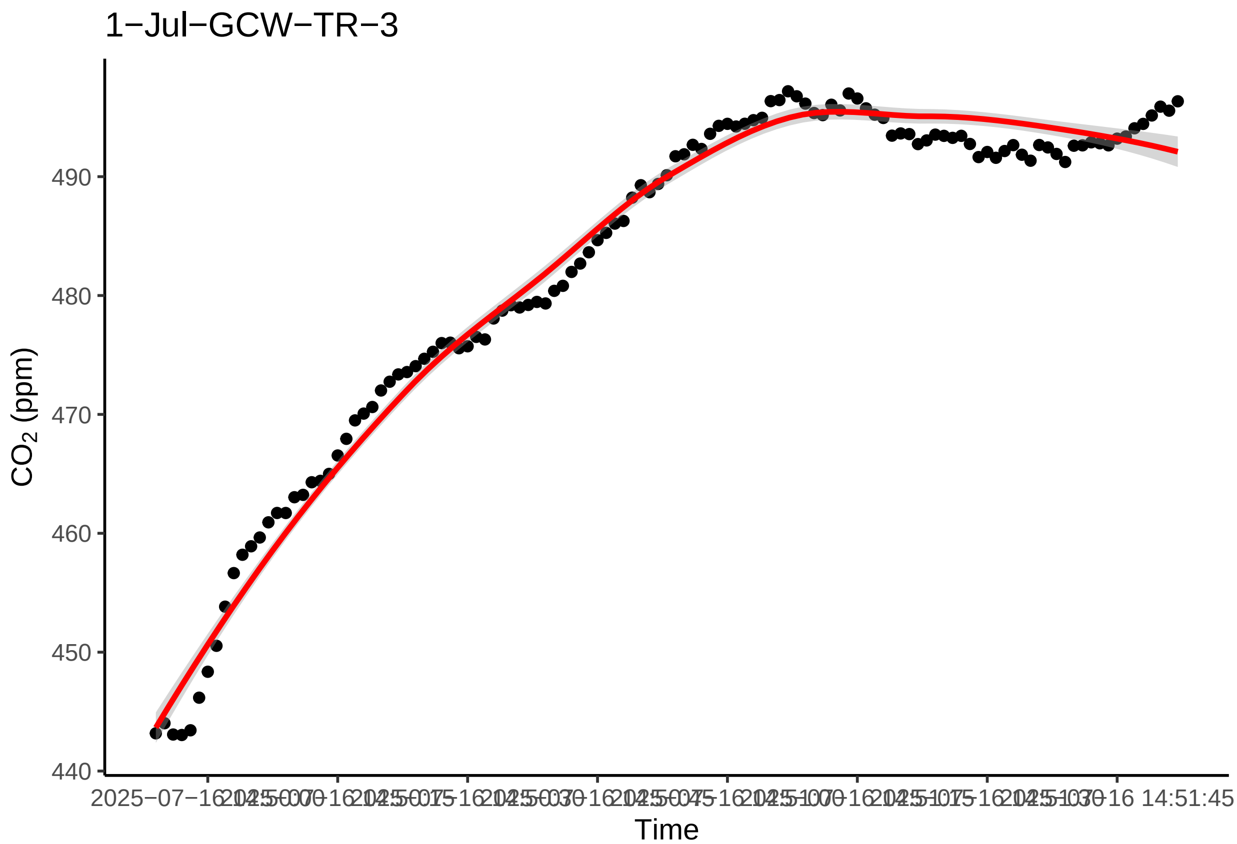

2−Nov−SWH−TR−3a
420
418
416
414
CO2$_2$ (ppm)
Time

0.13 Take a look at the bad fluxes <0.9 and have a high flux estimate (>1)

```
## [1] "See if we want to remove these fluxes or if we want to try and recalculate these ones"
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

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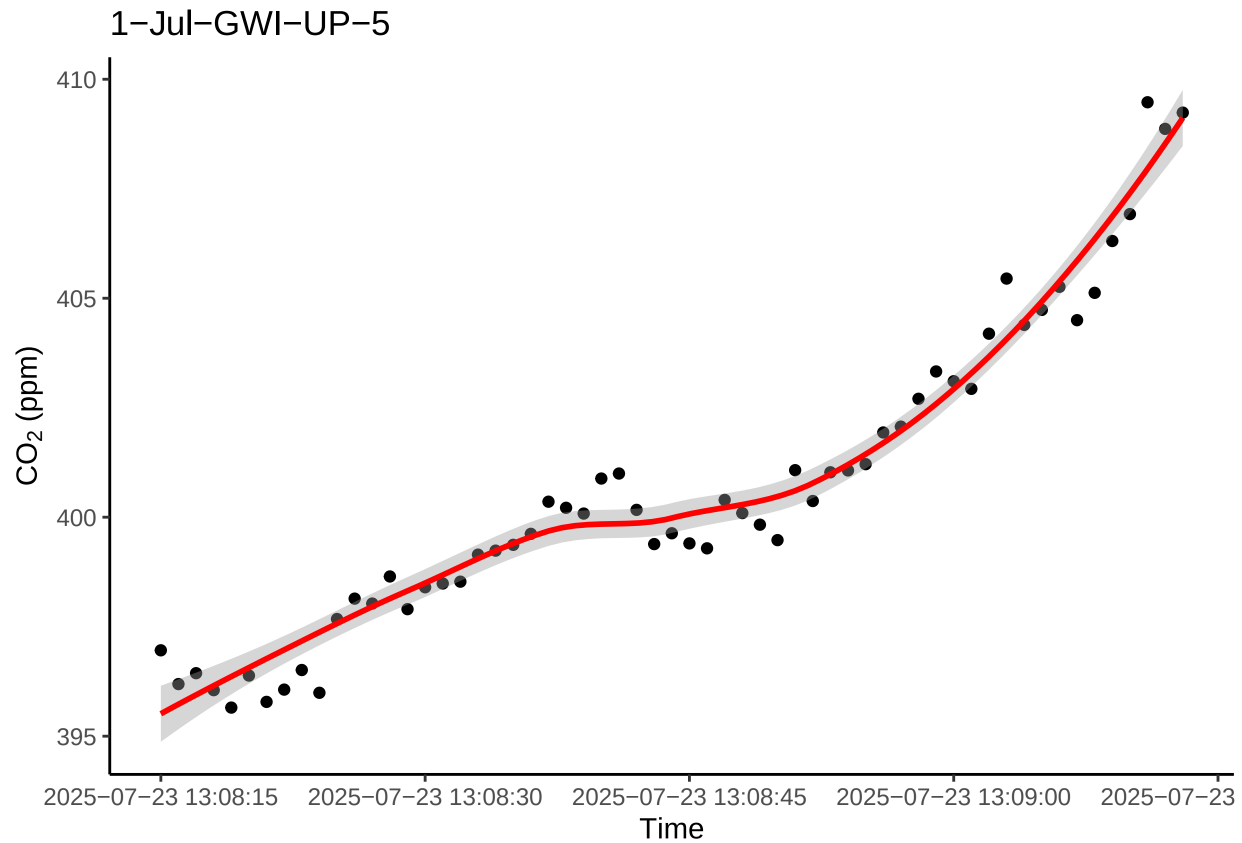


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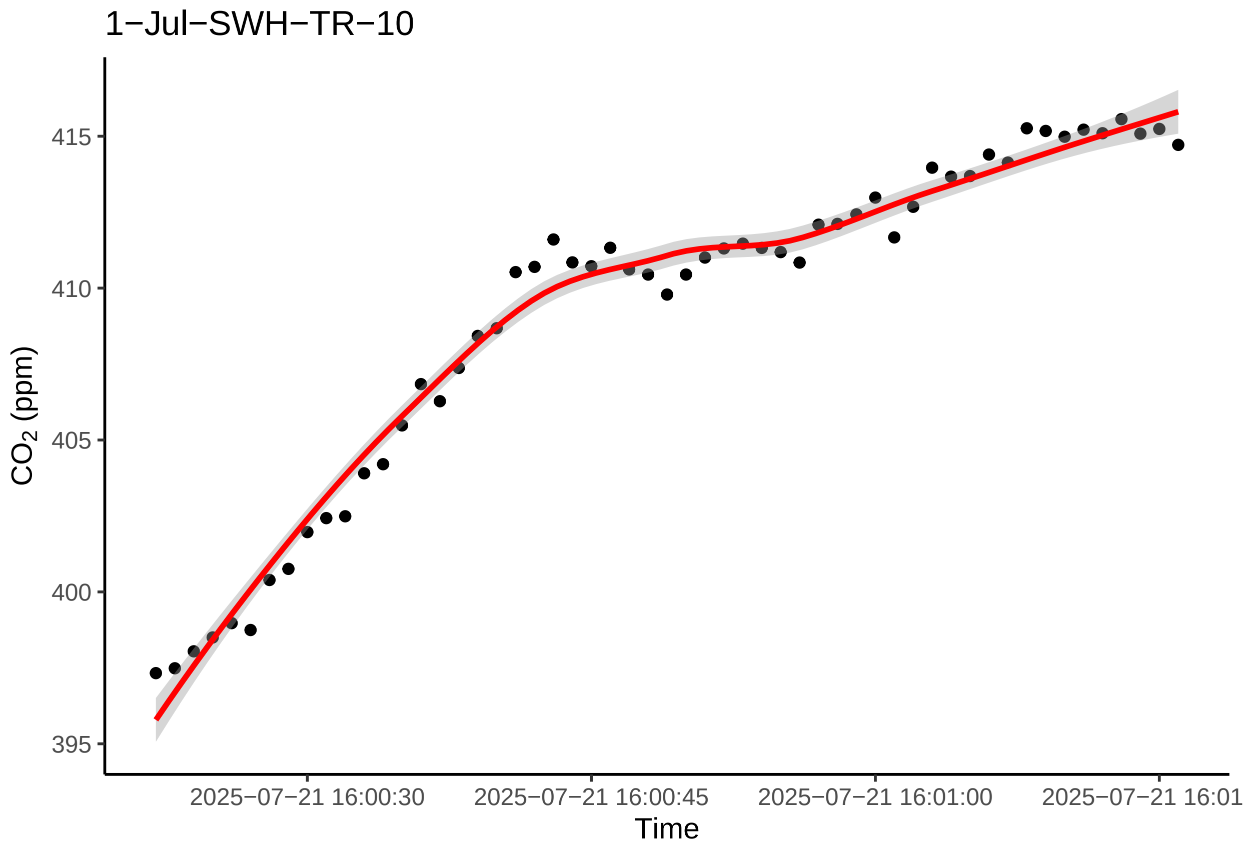


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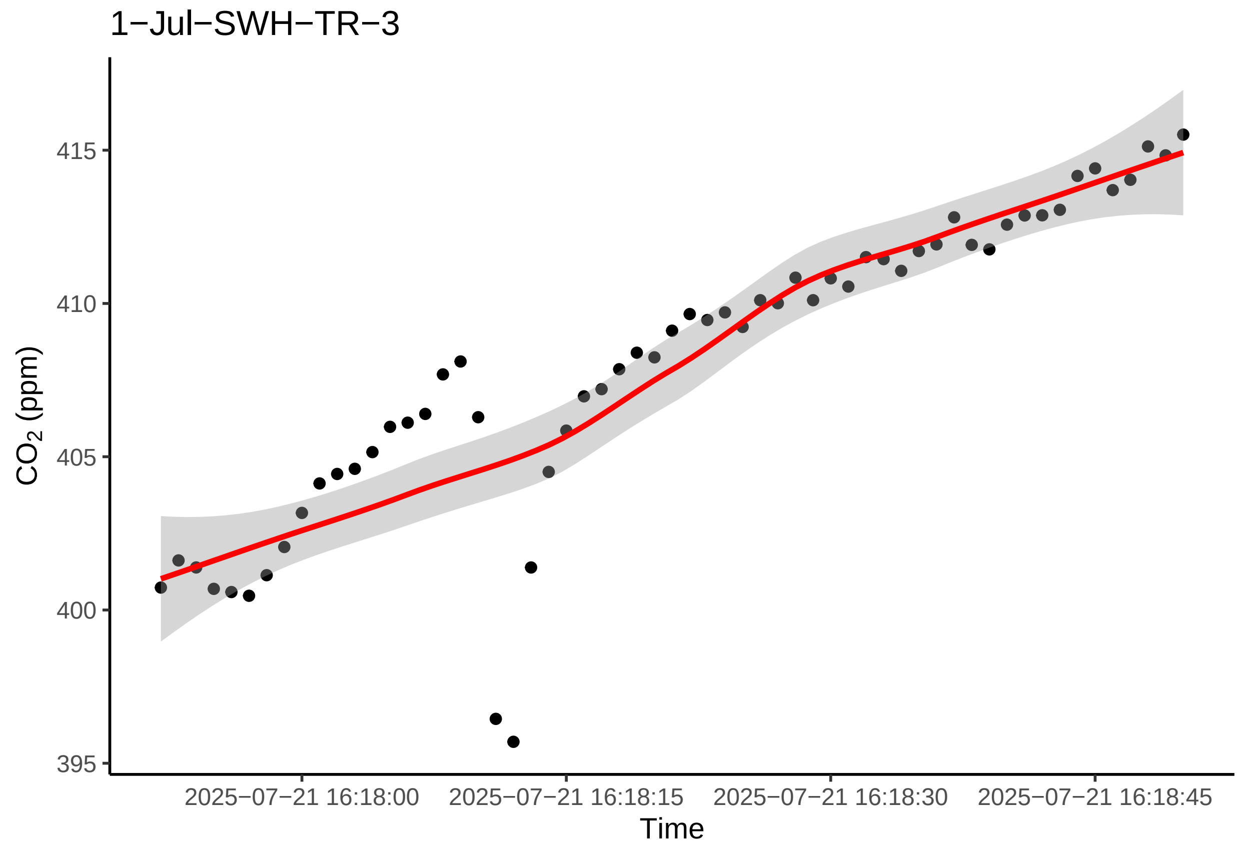


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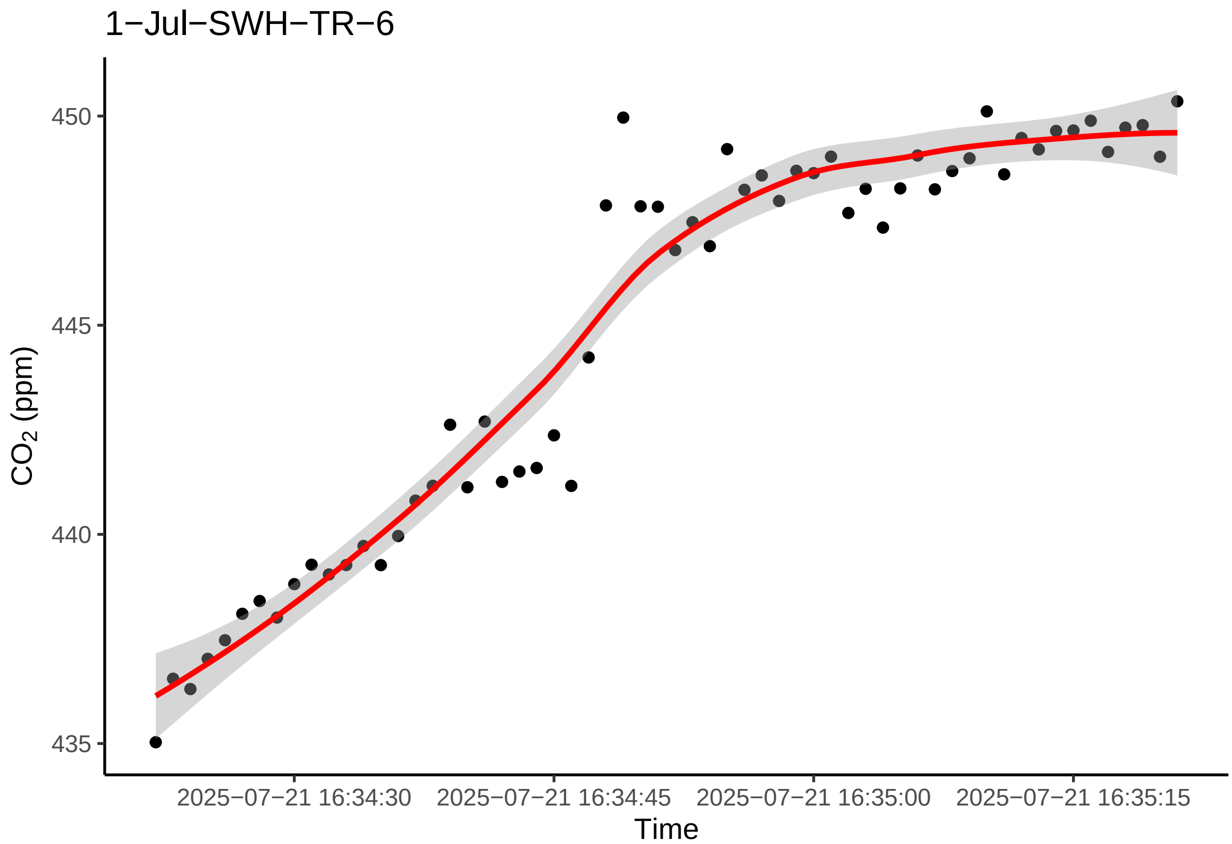


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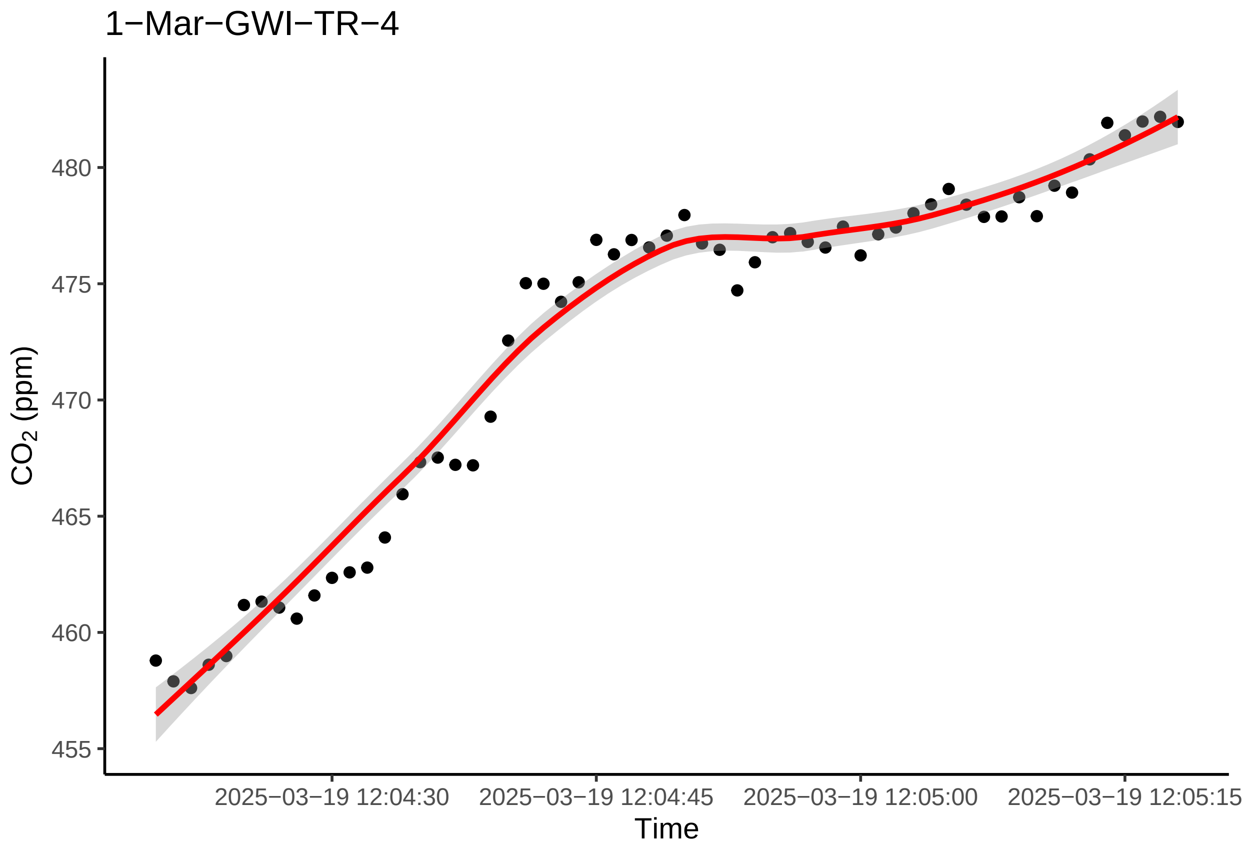


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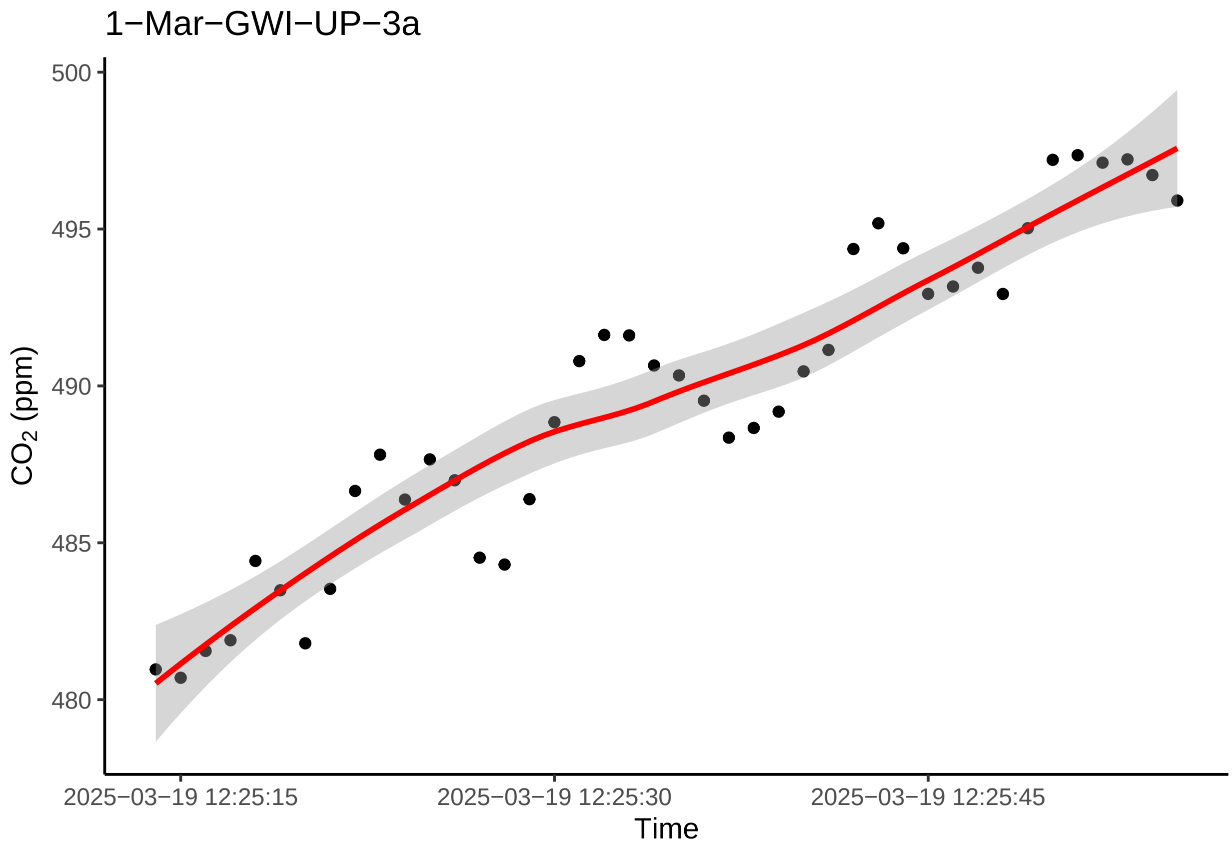


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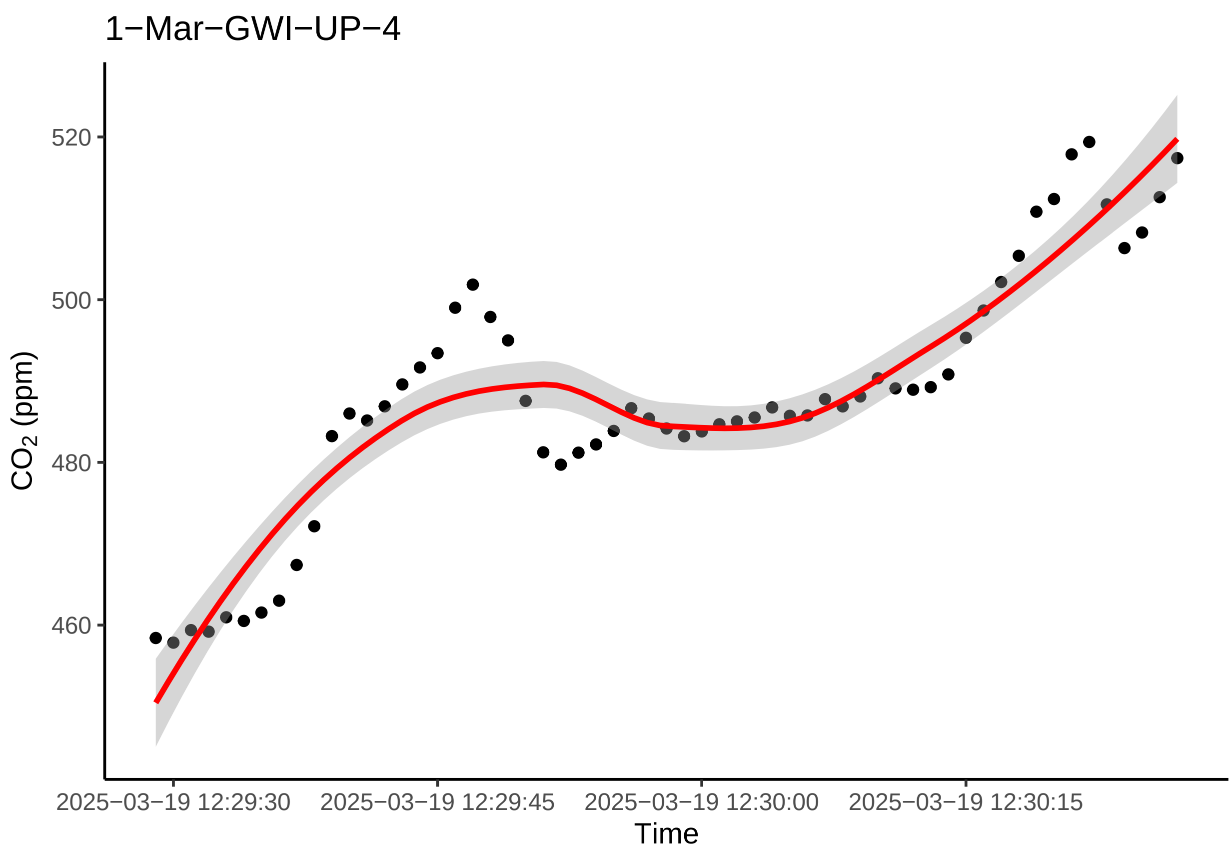


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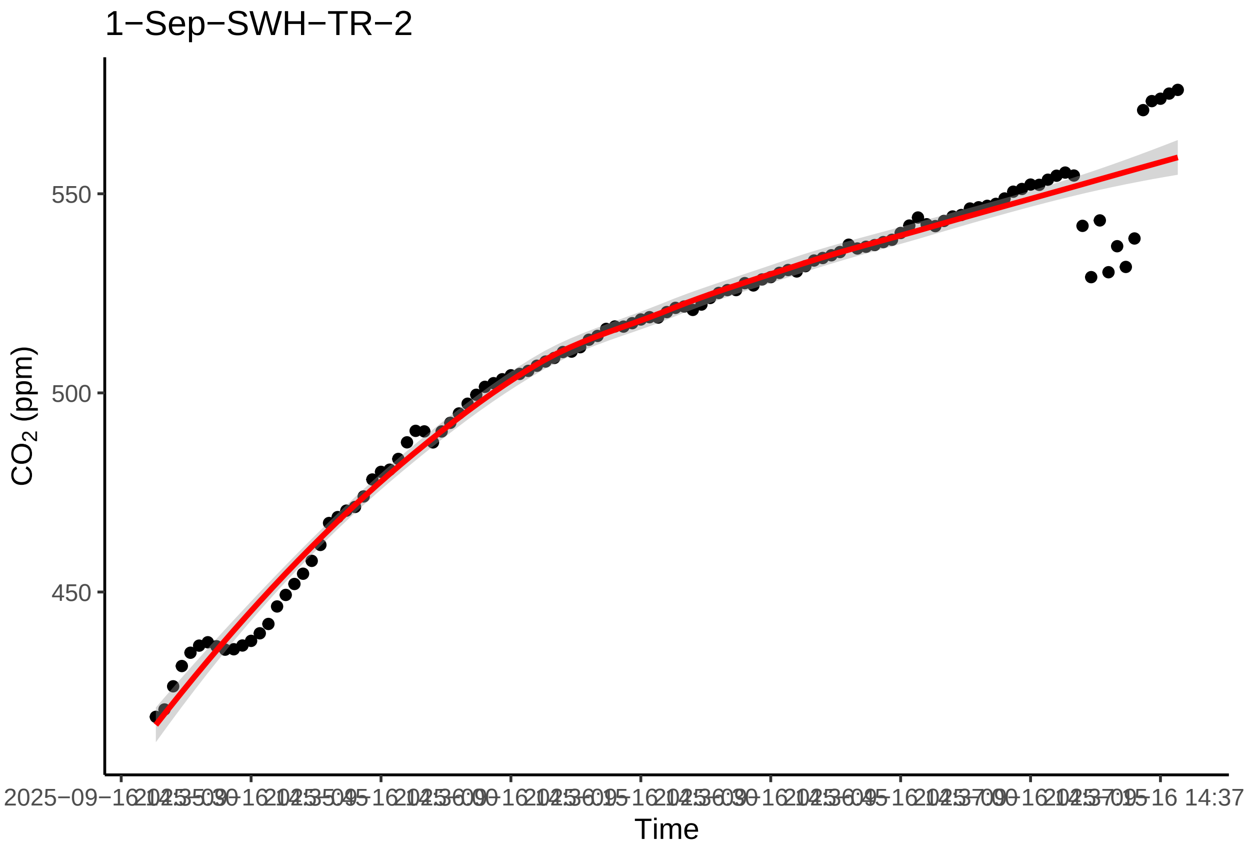
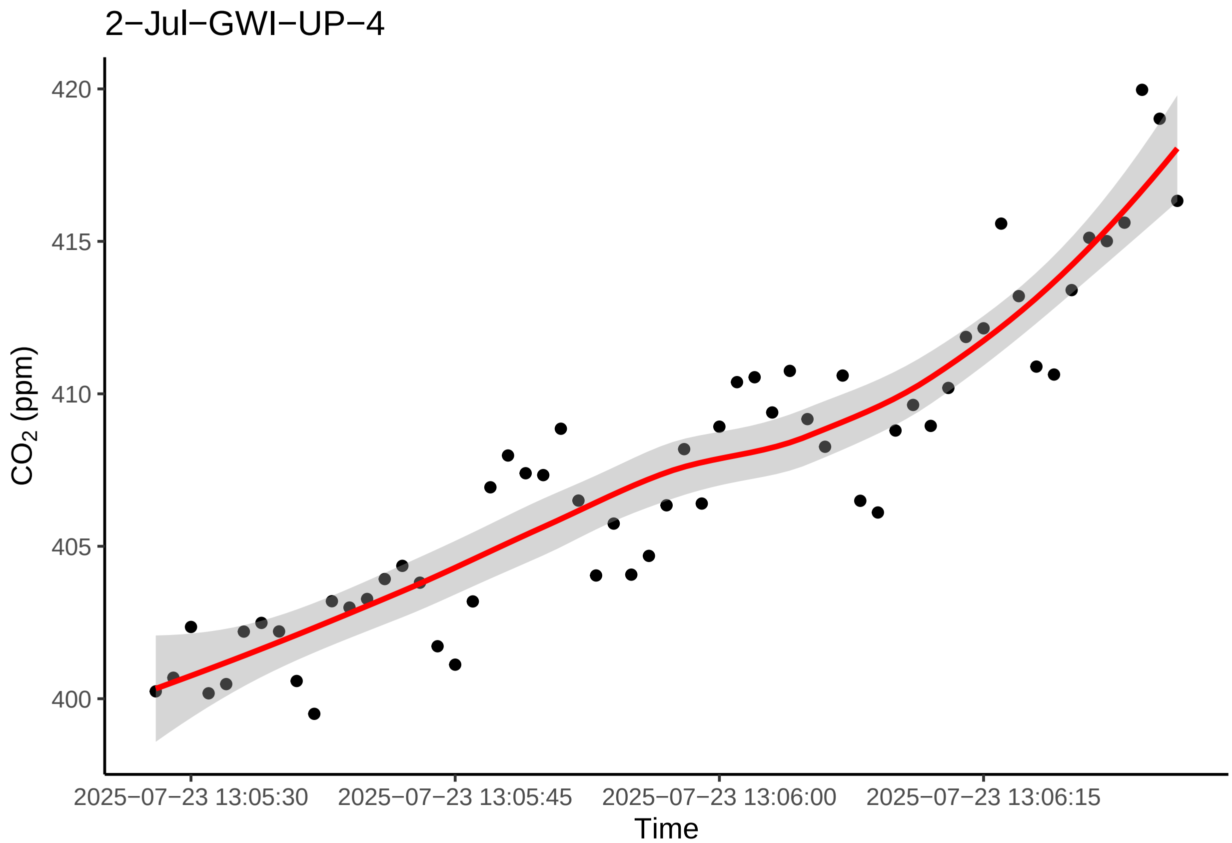
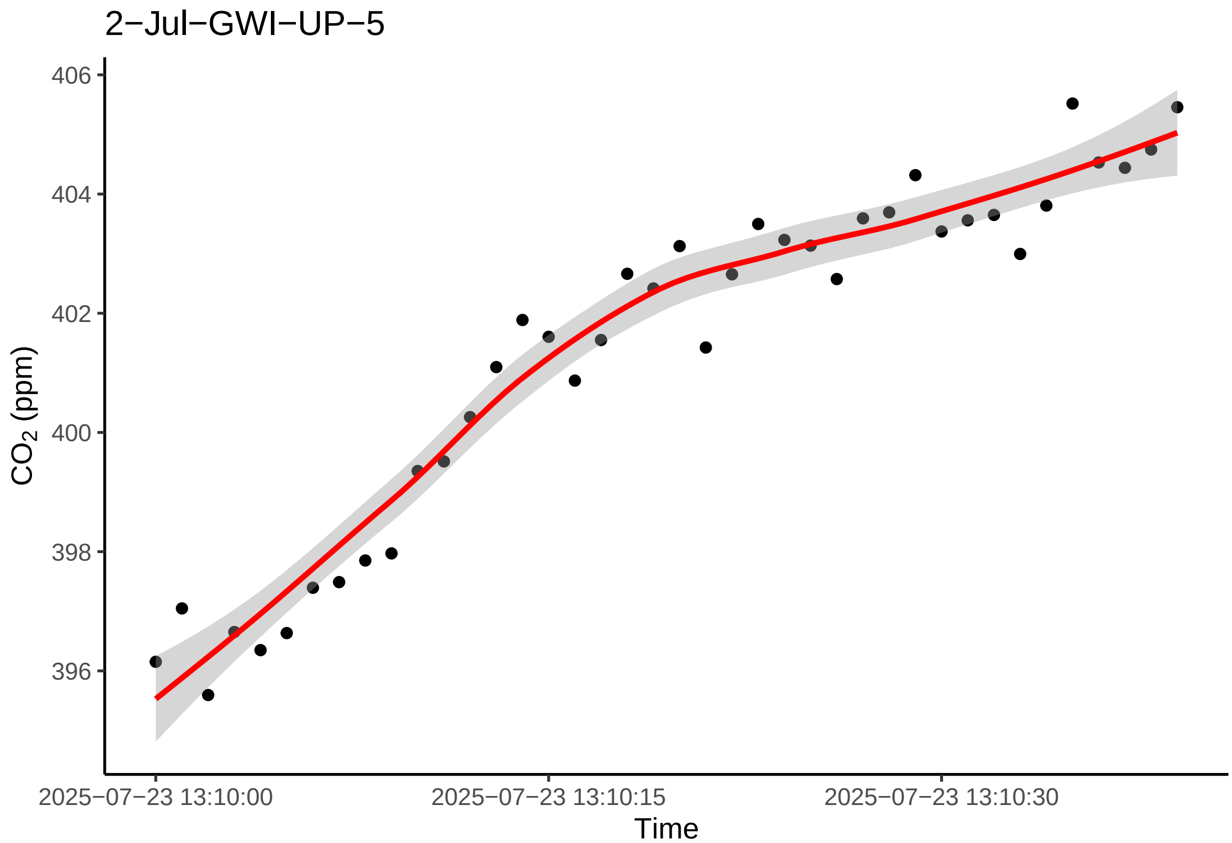


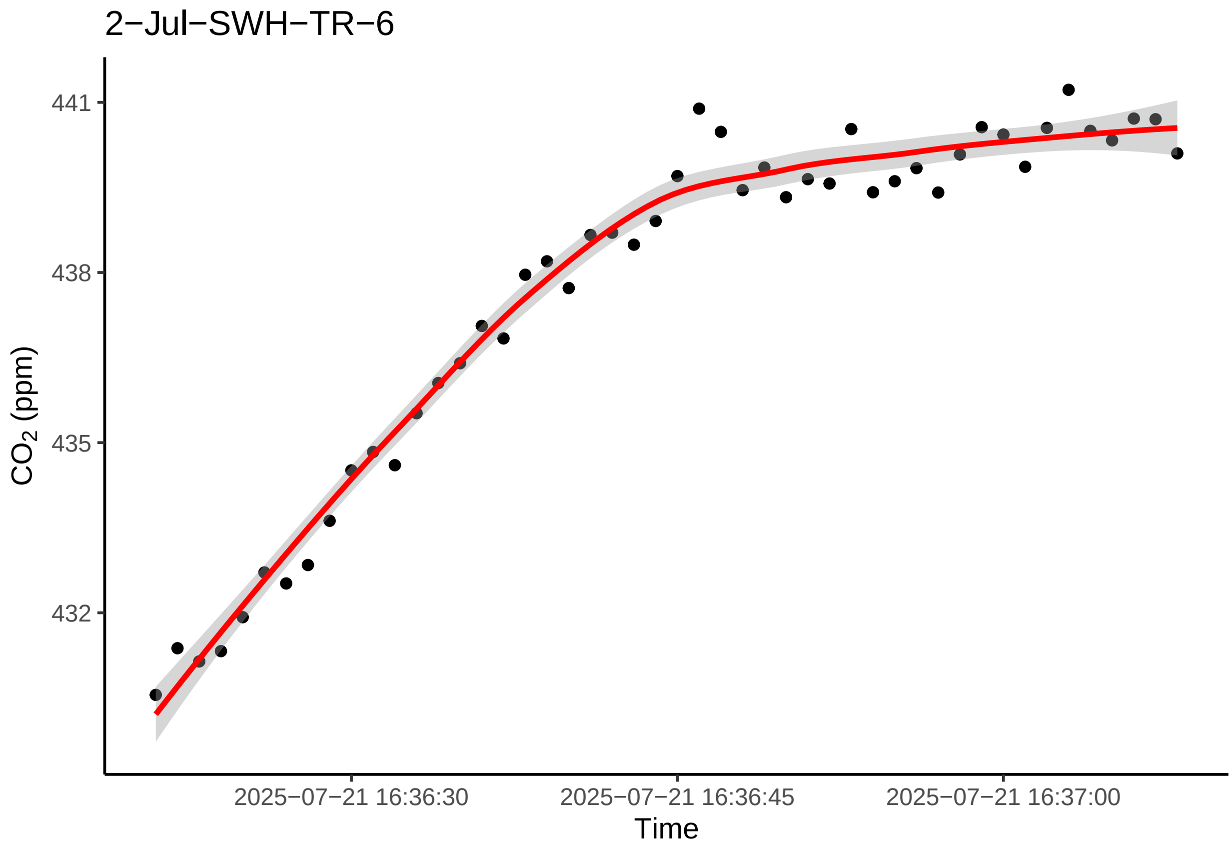

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



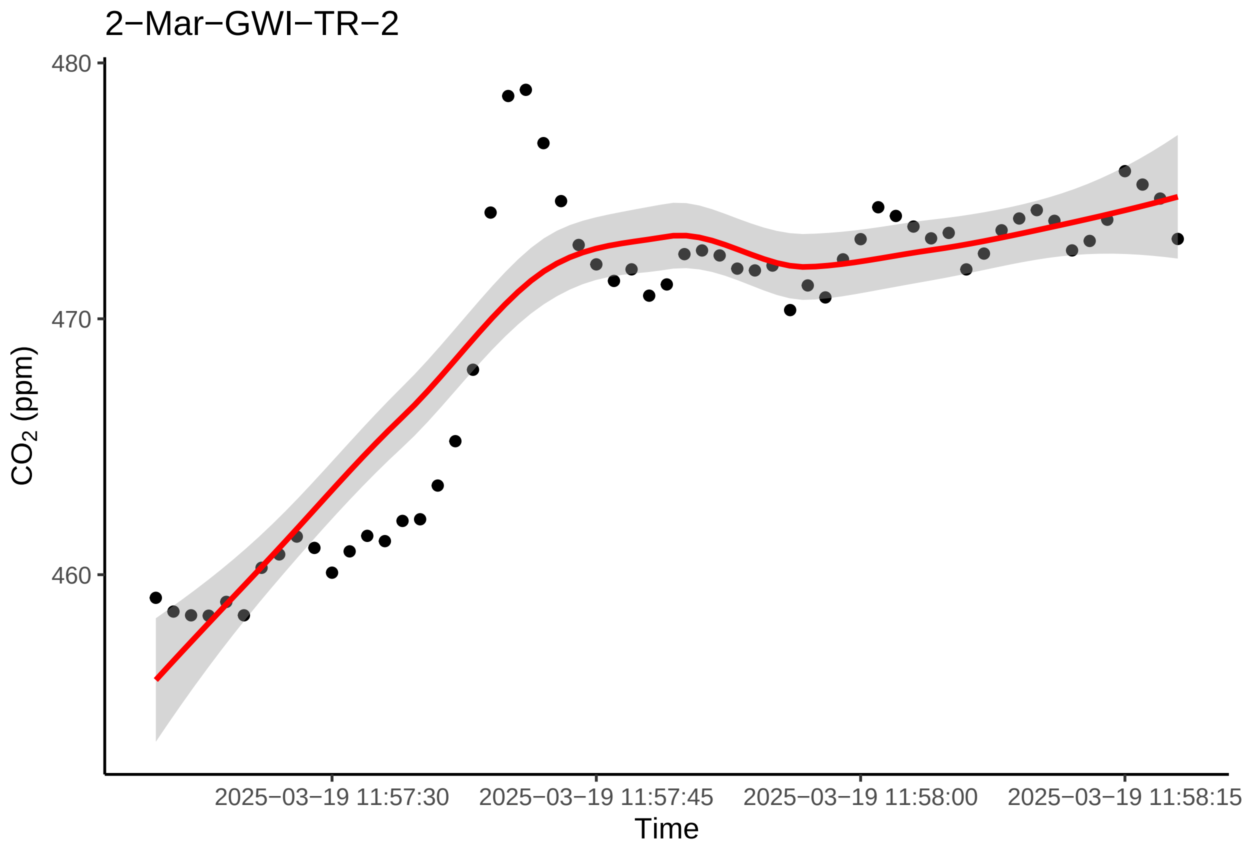

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



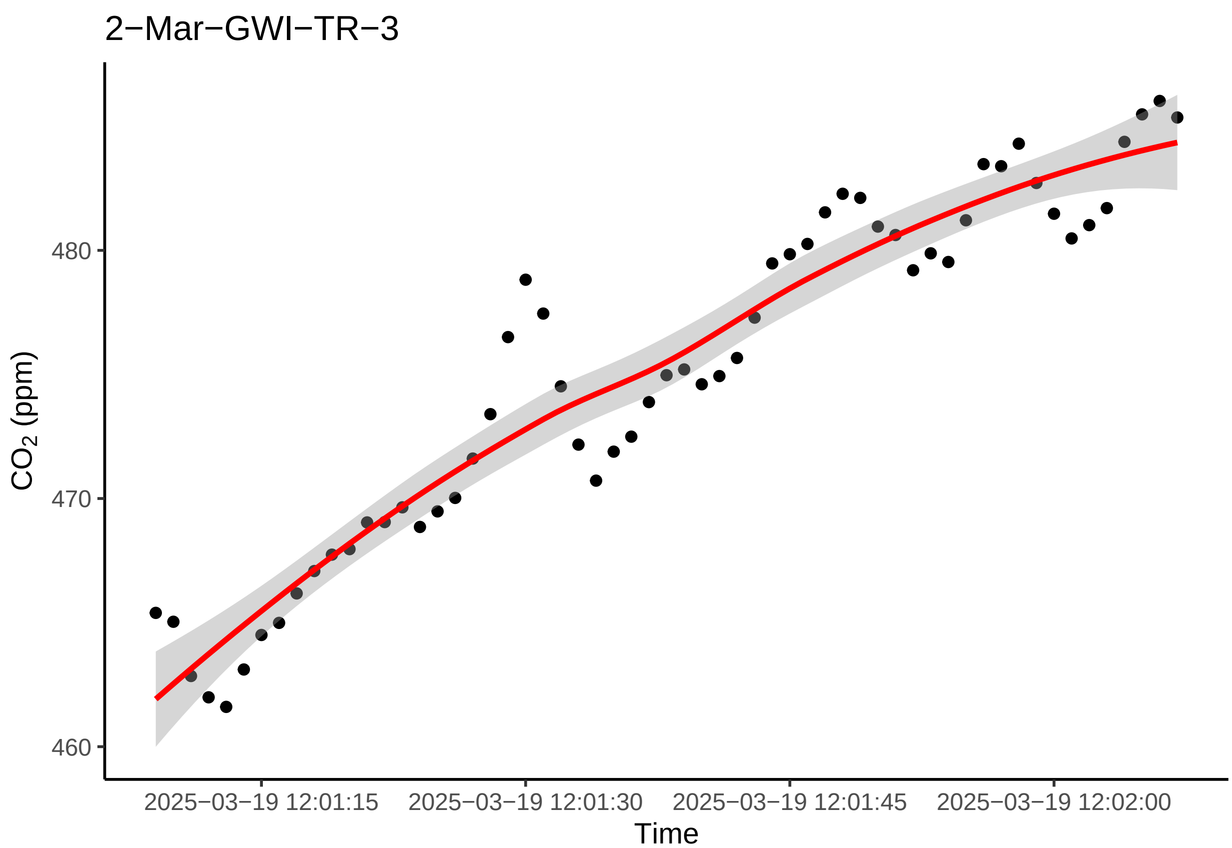

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



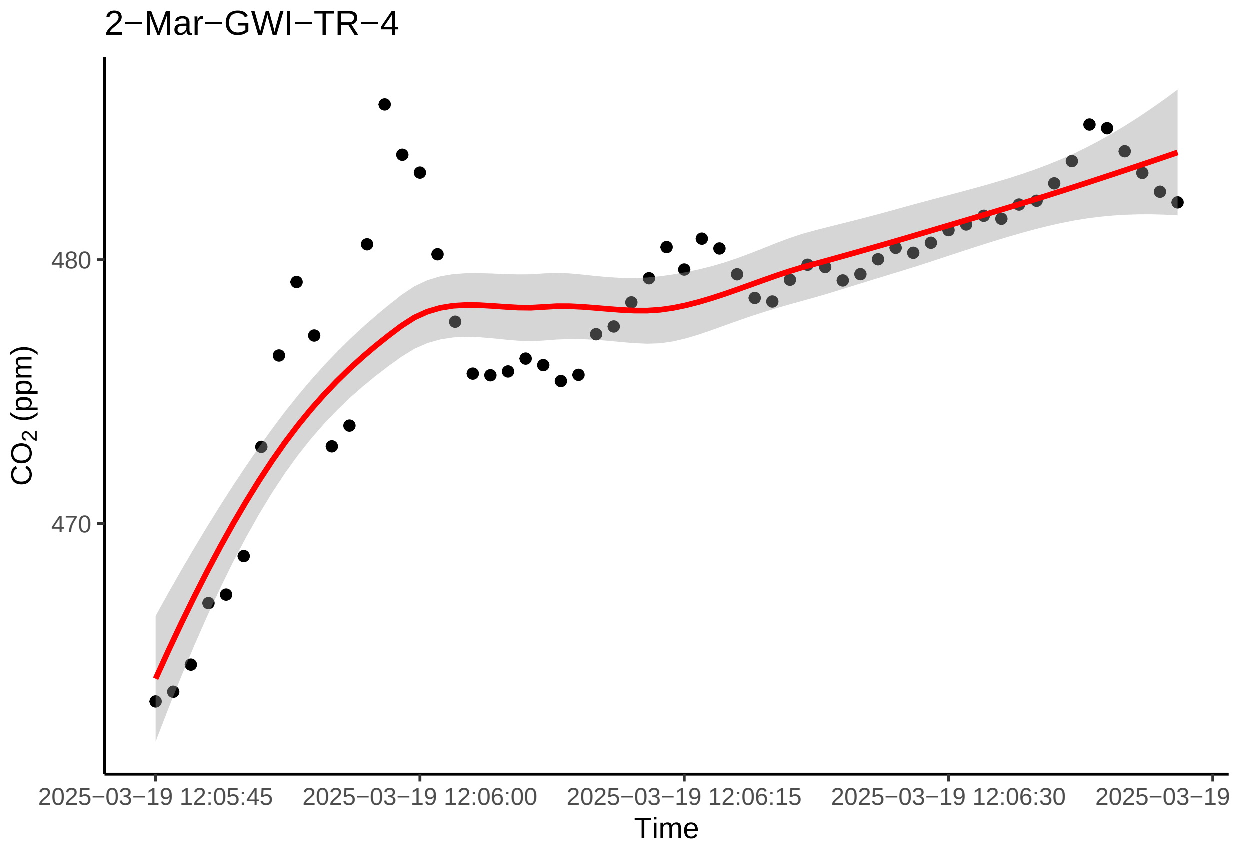

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



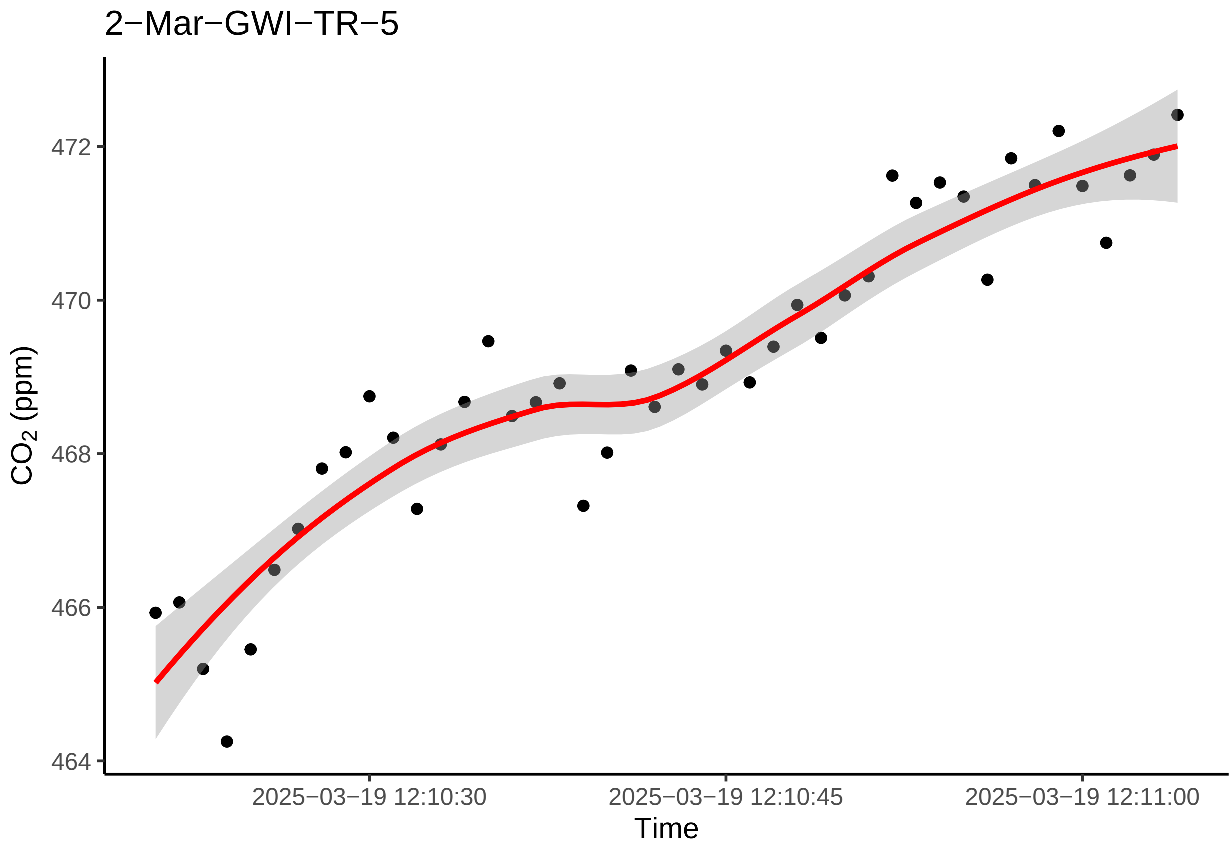

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



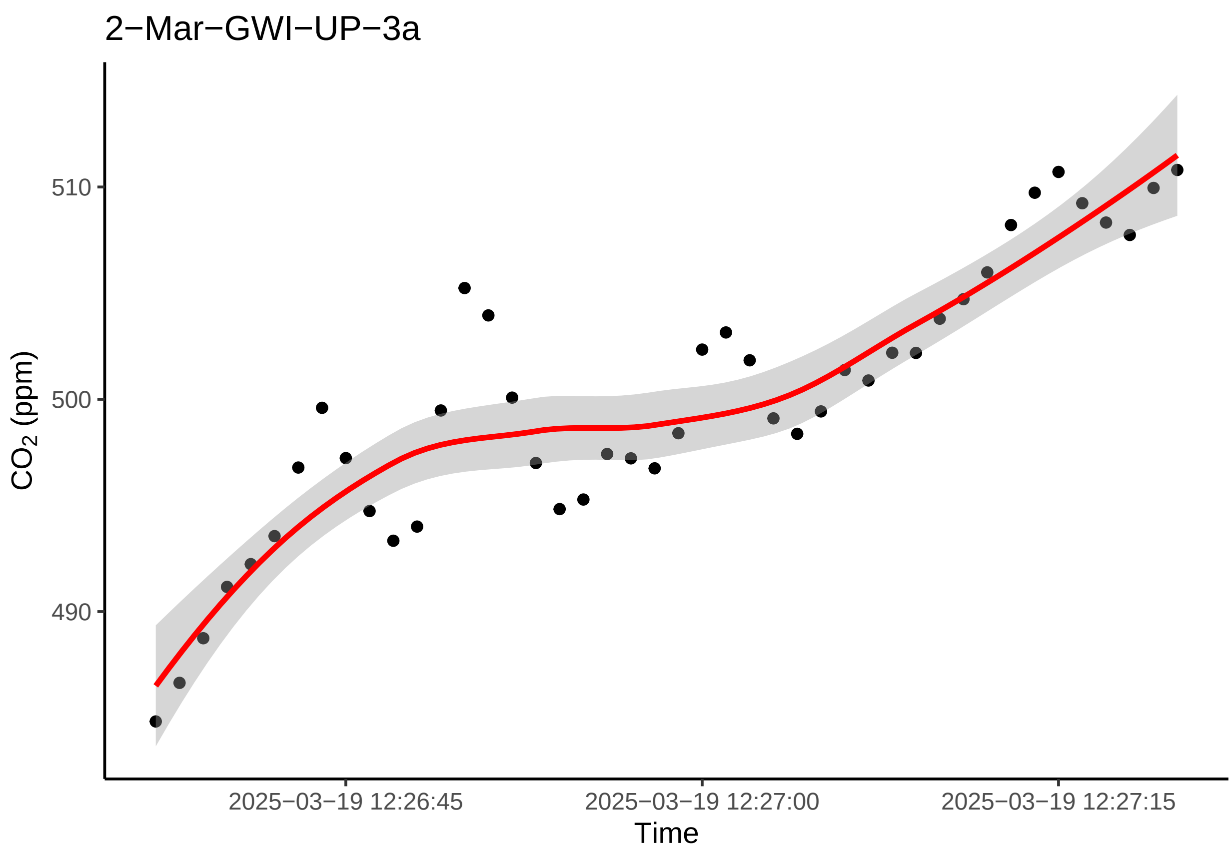

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



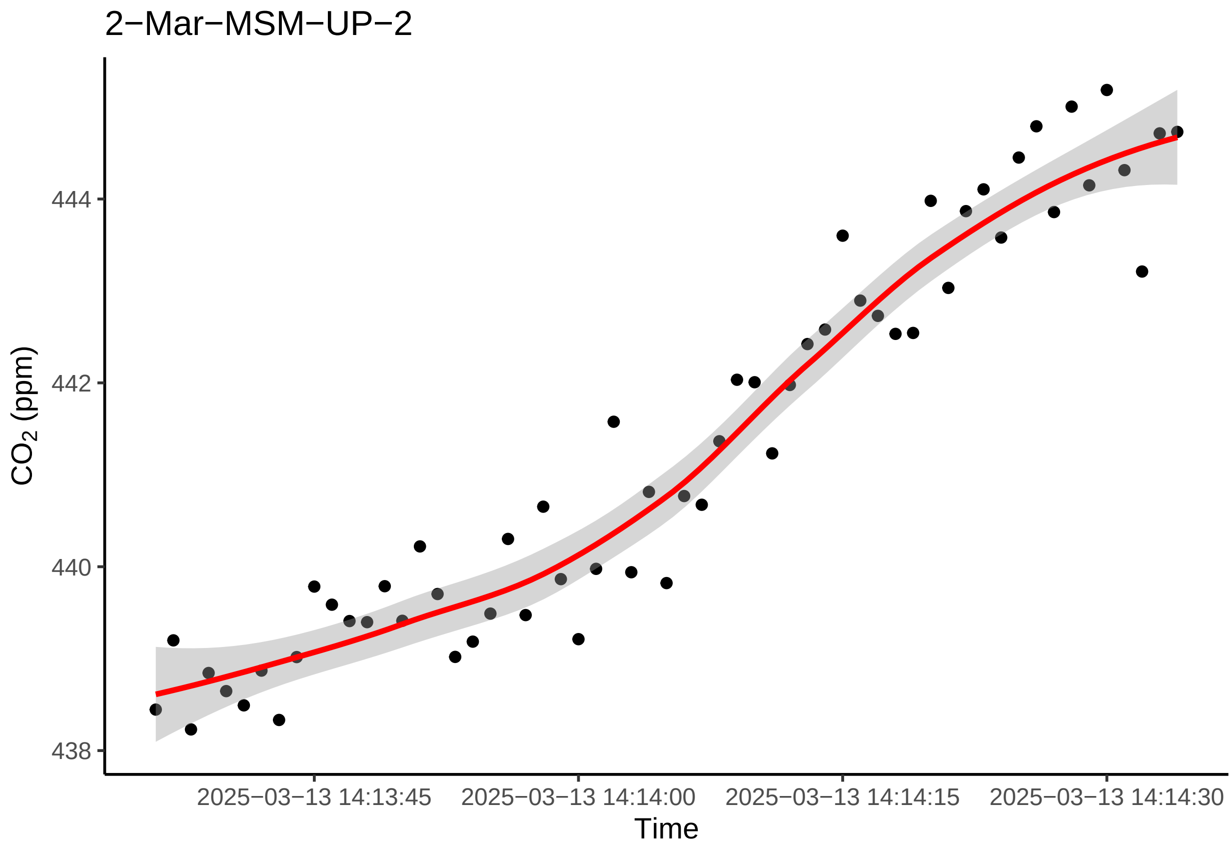

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

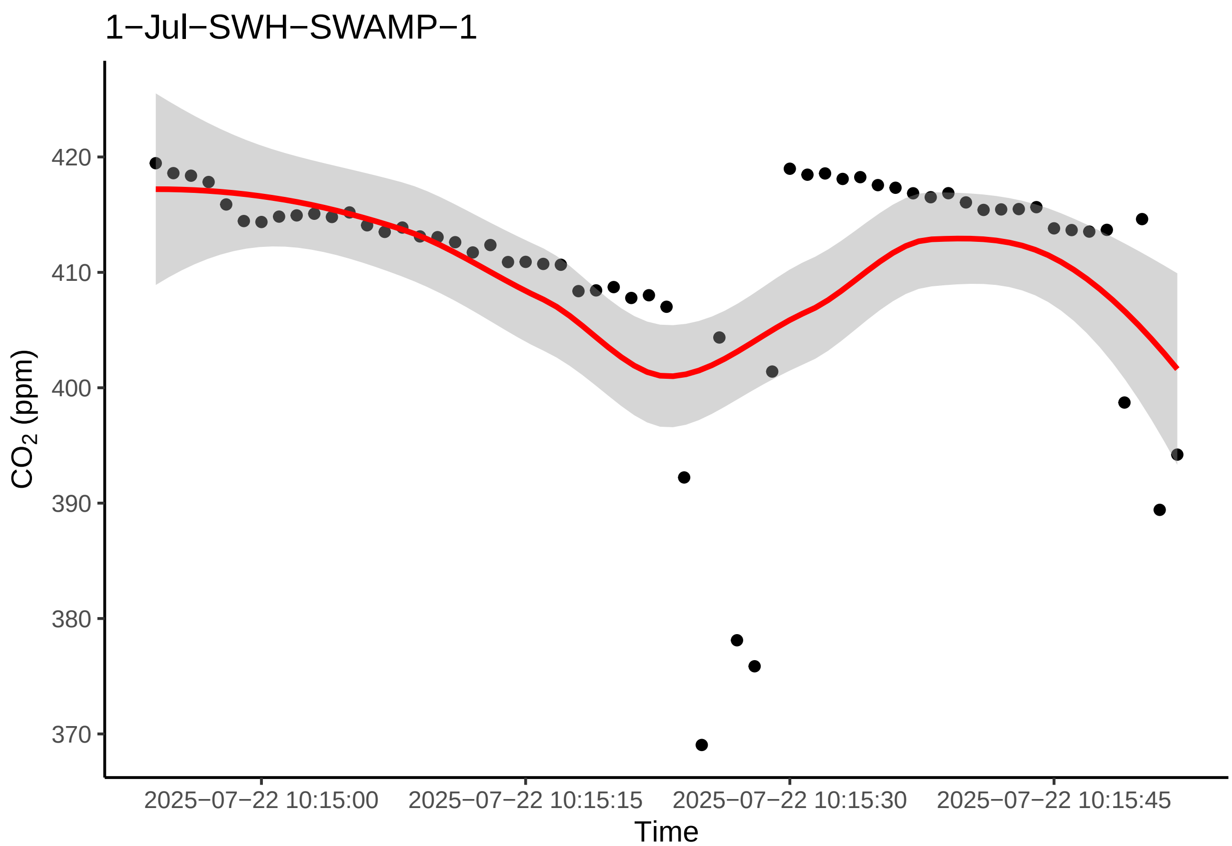



```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

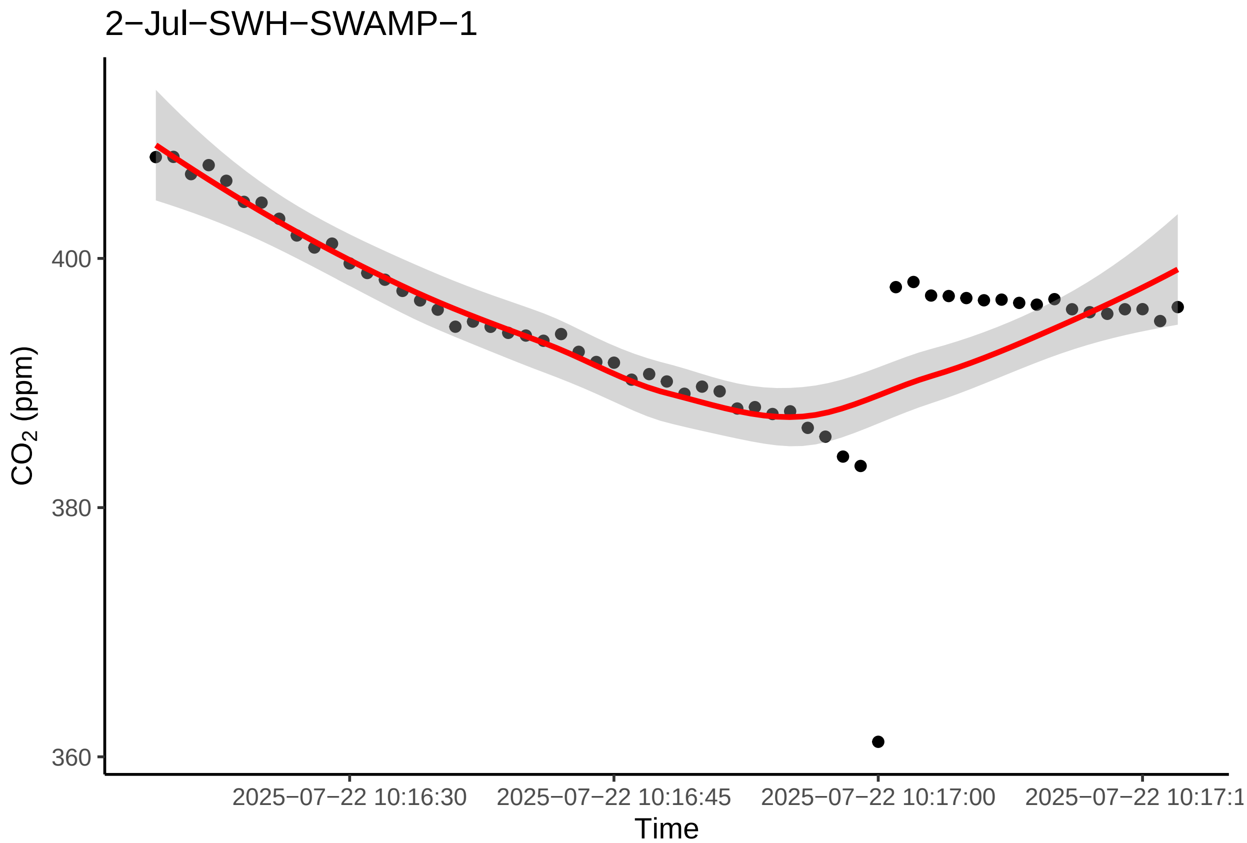

2−Mar−MSM−UP−2
444
442
440
438
CO_2 (ppm)
2025−03−13 14:13:45
2025−03−13 14:14:00
2025−03−13 14:14:15
2025−03−13 14:14:30
Time

0.14 Look at the negative CO2 fluxes

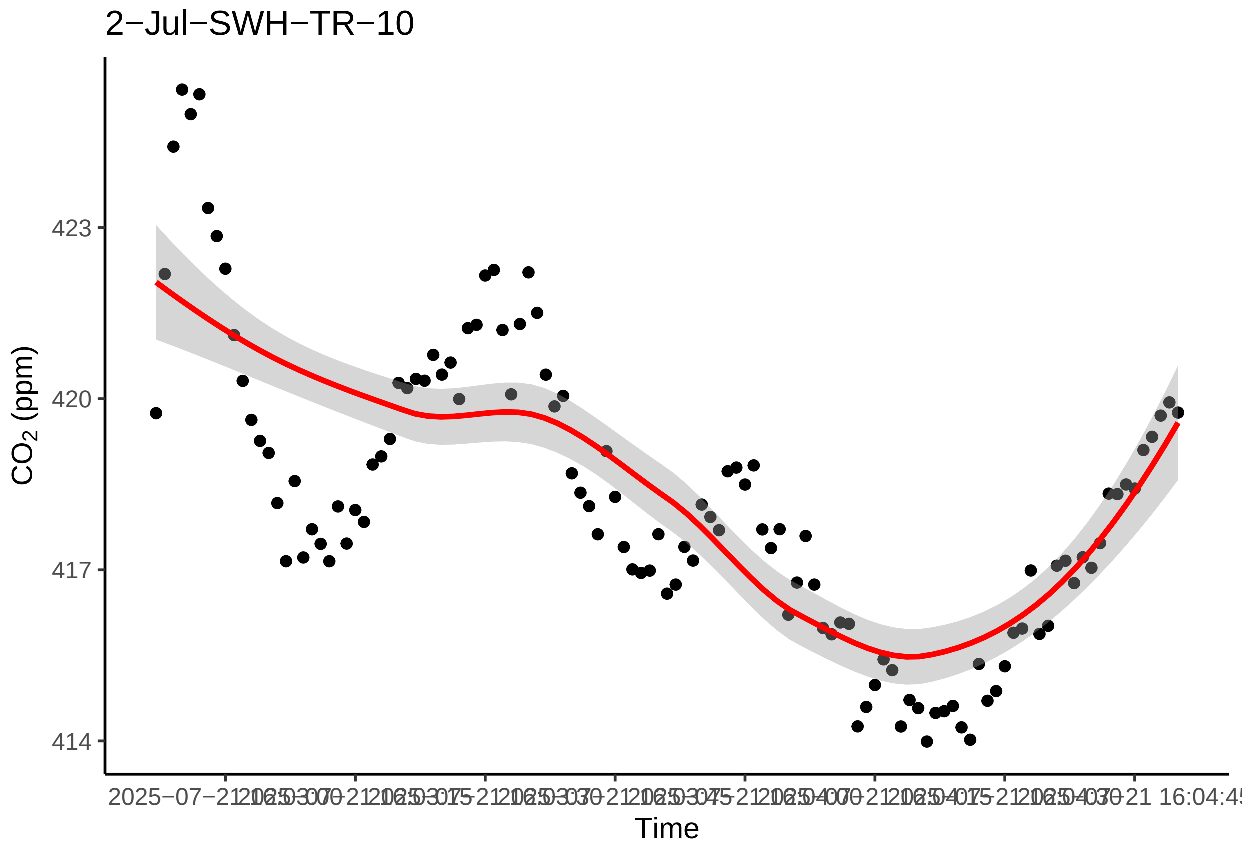
```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

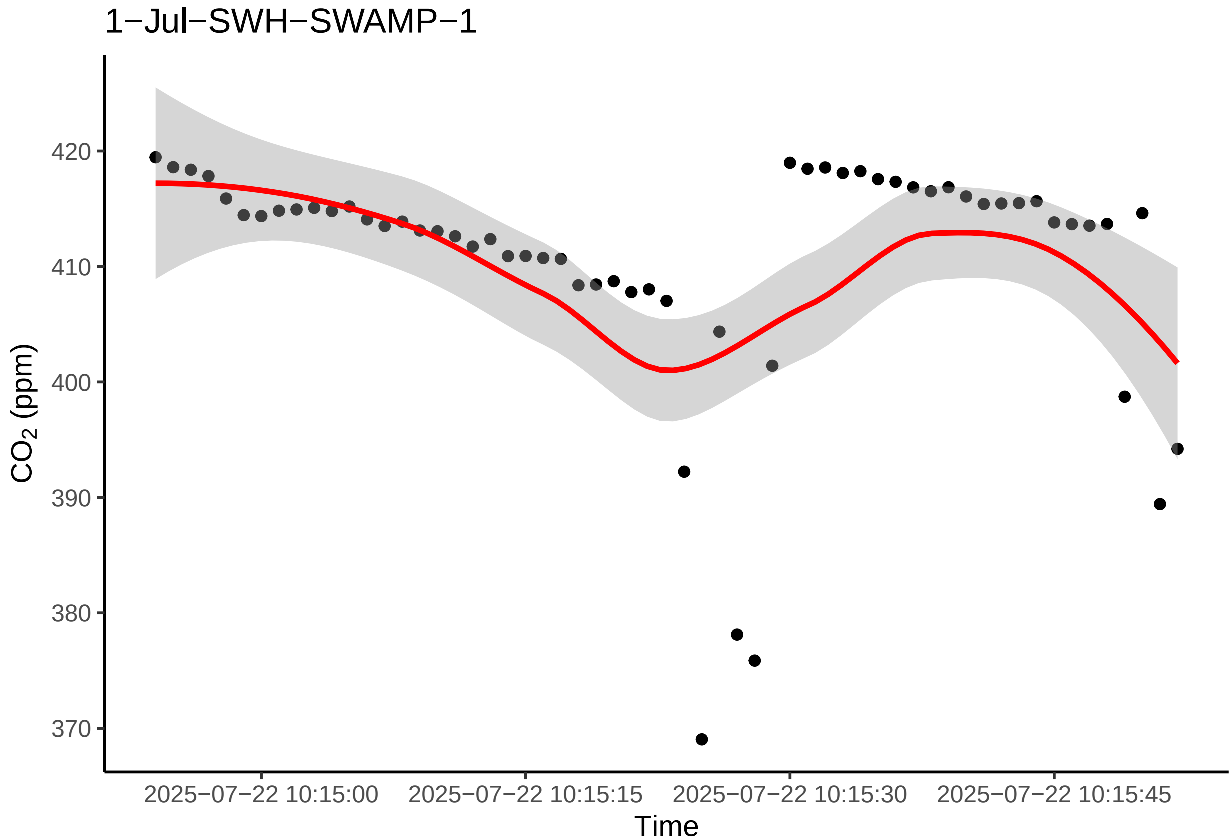



```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

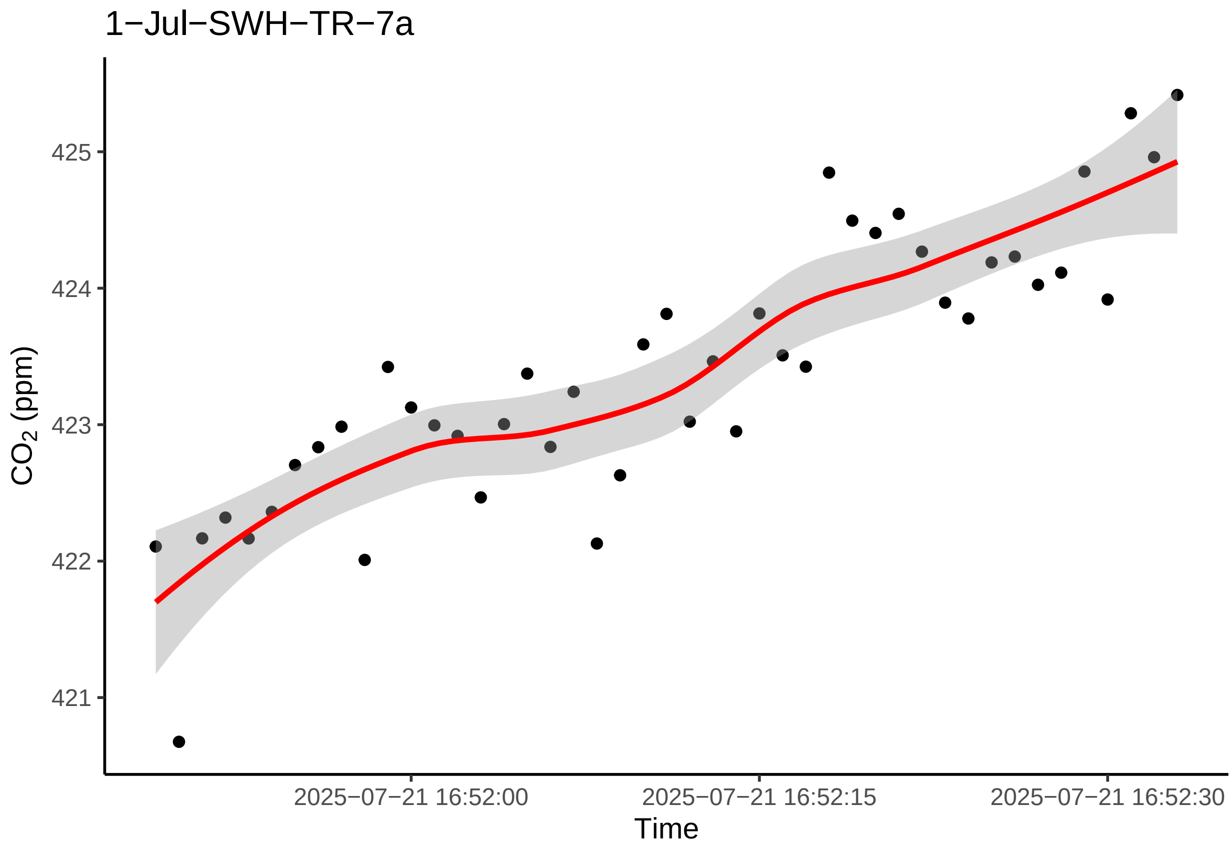

2−Jul−SWH−TR−10
CO2 (ppm)
423
420
417
414
Time

0.15 Look at the really high or really low ones to see if there is a timing issue or something

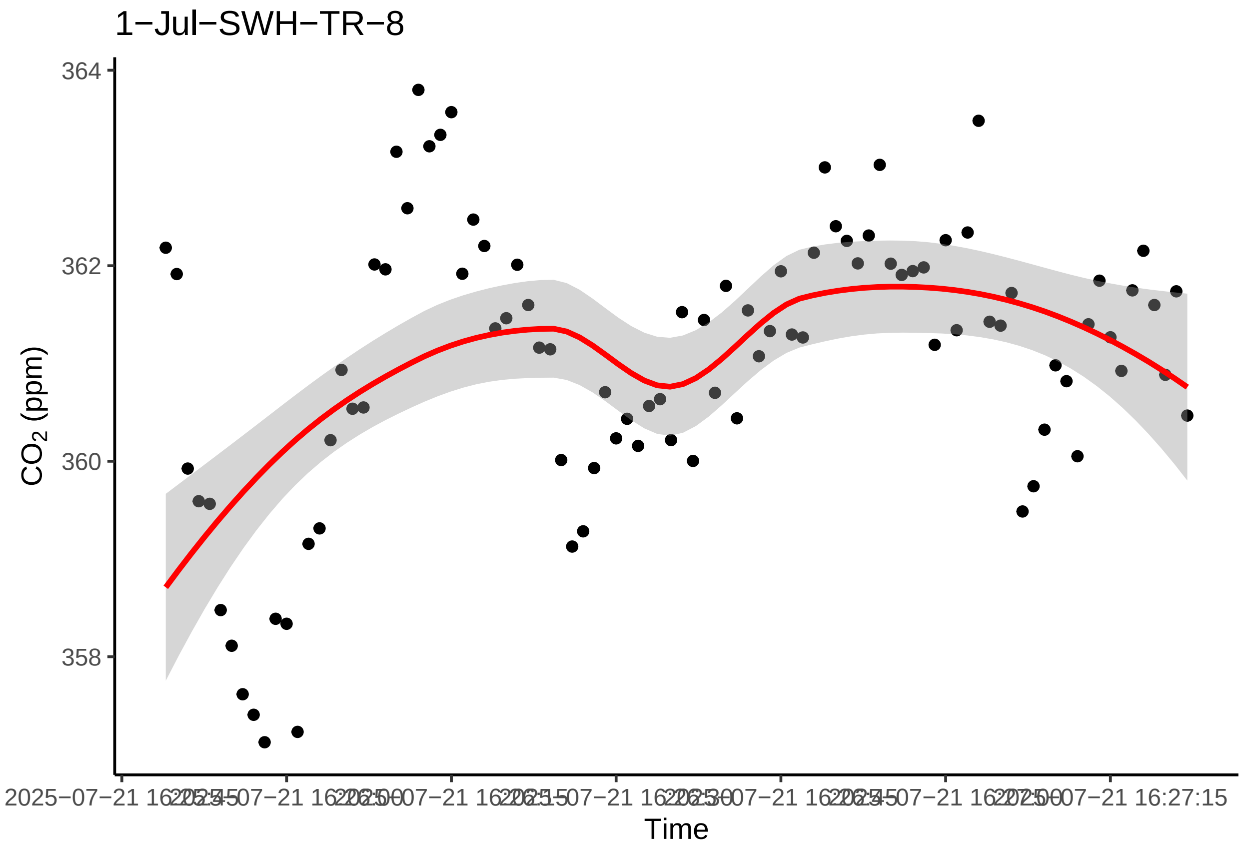
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



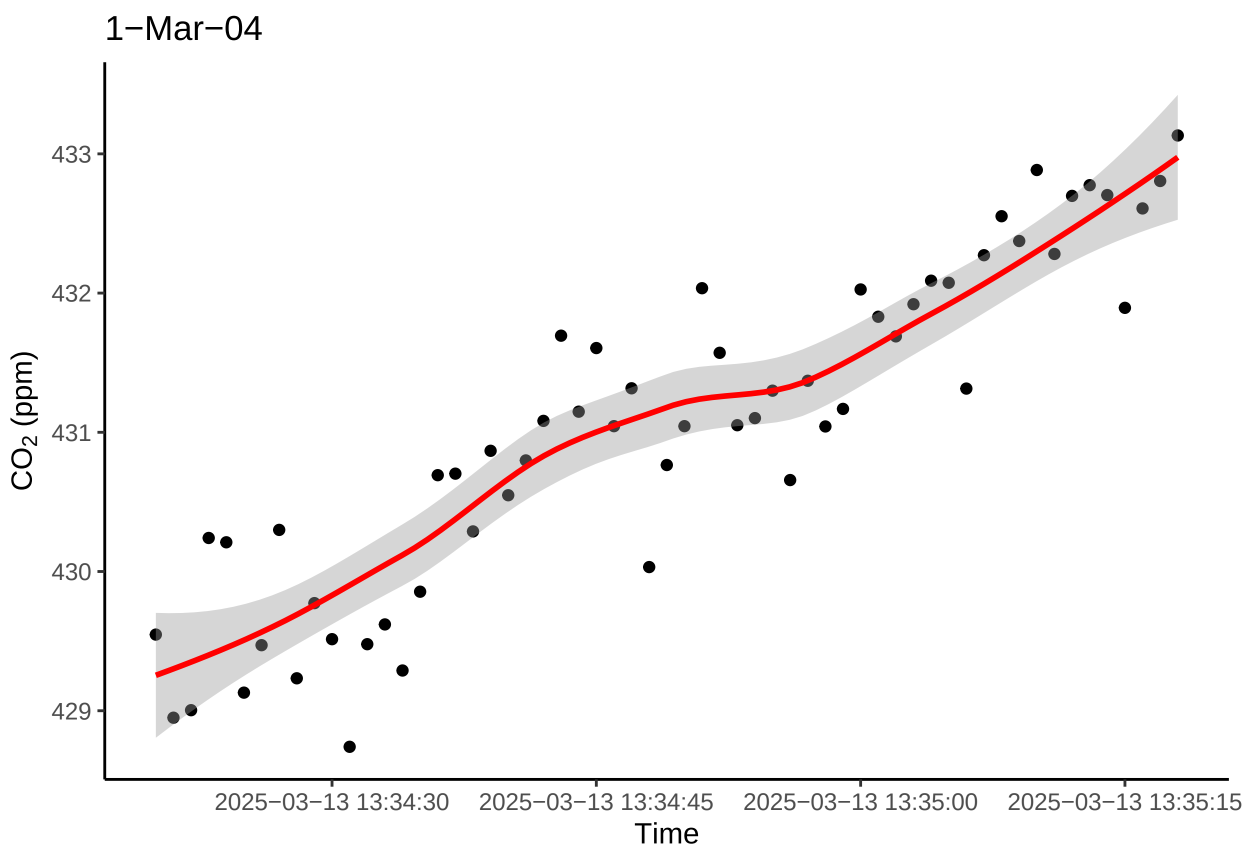

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



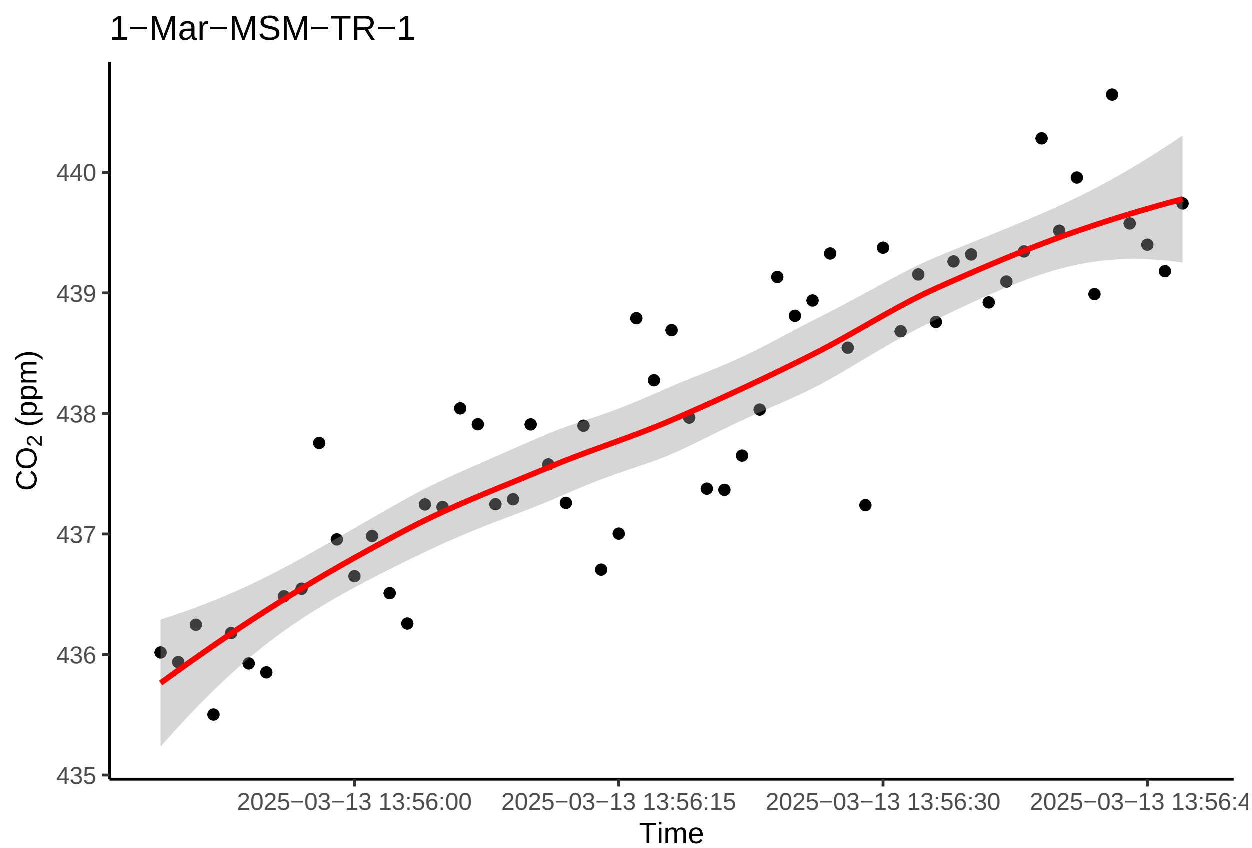

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



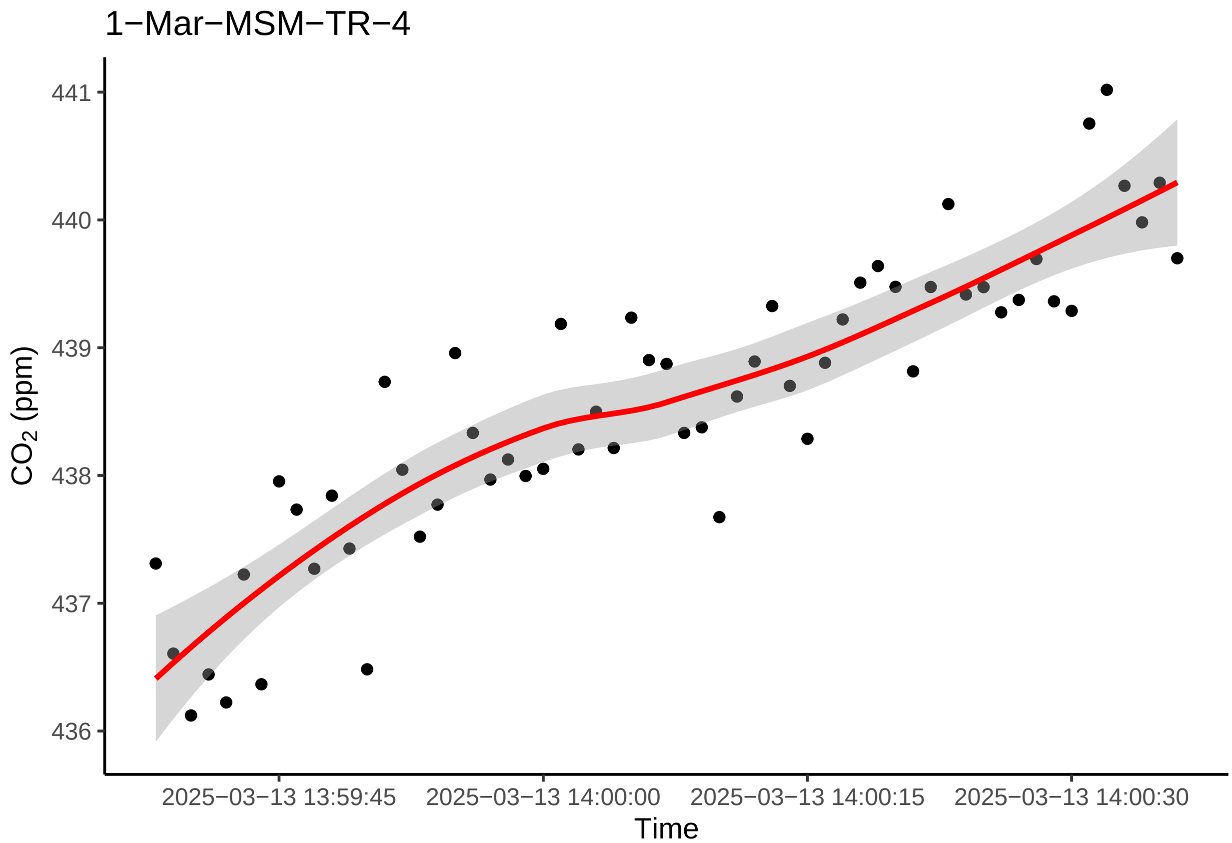

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



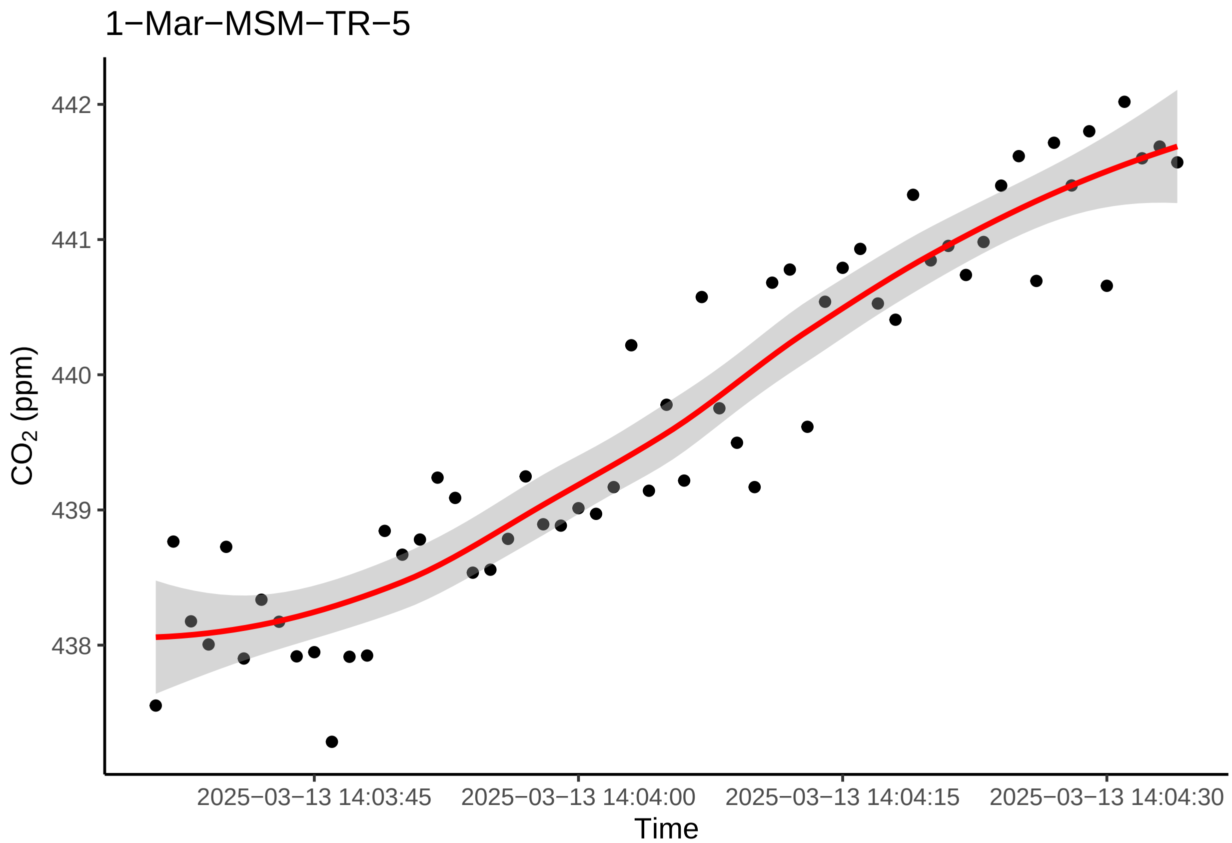

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



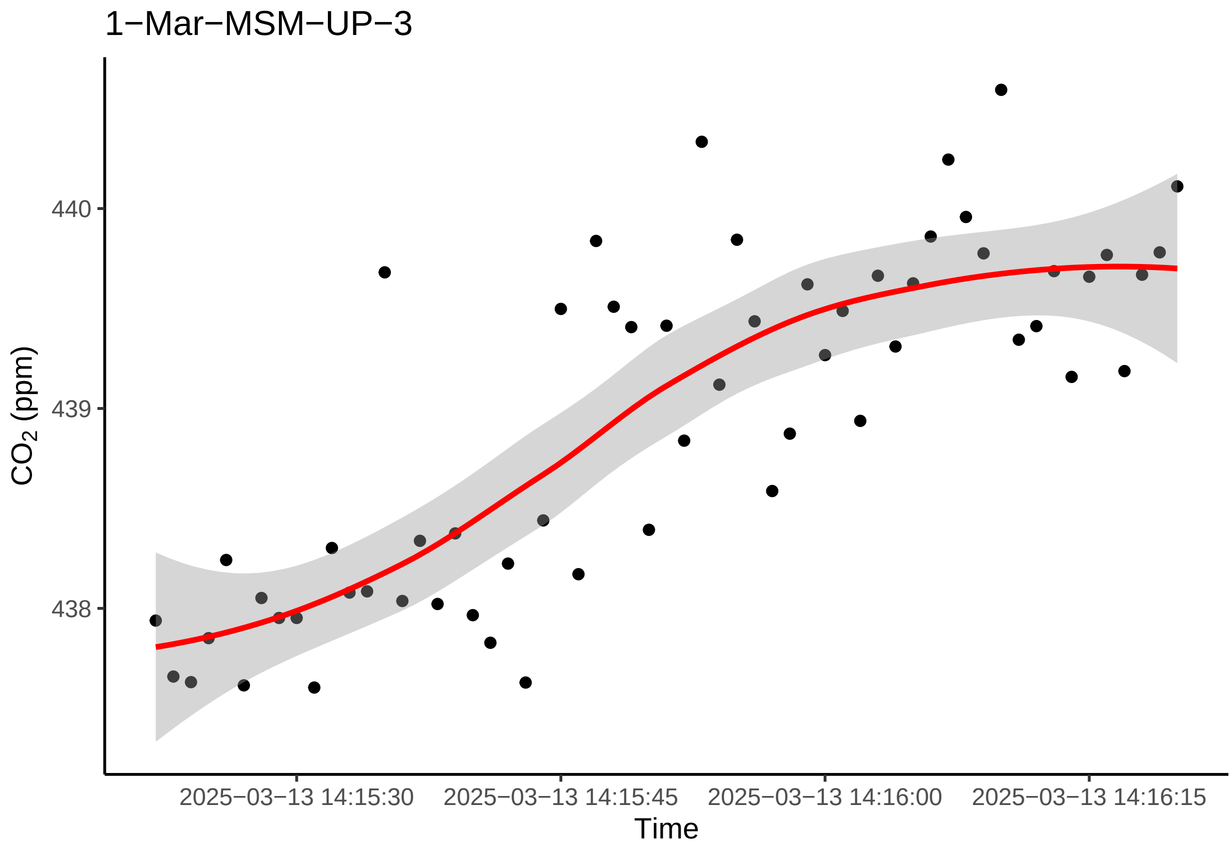

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



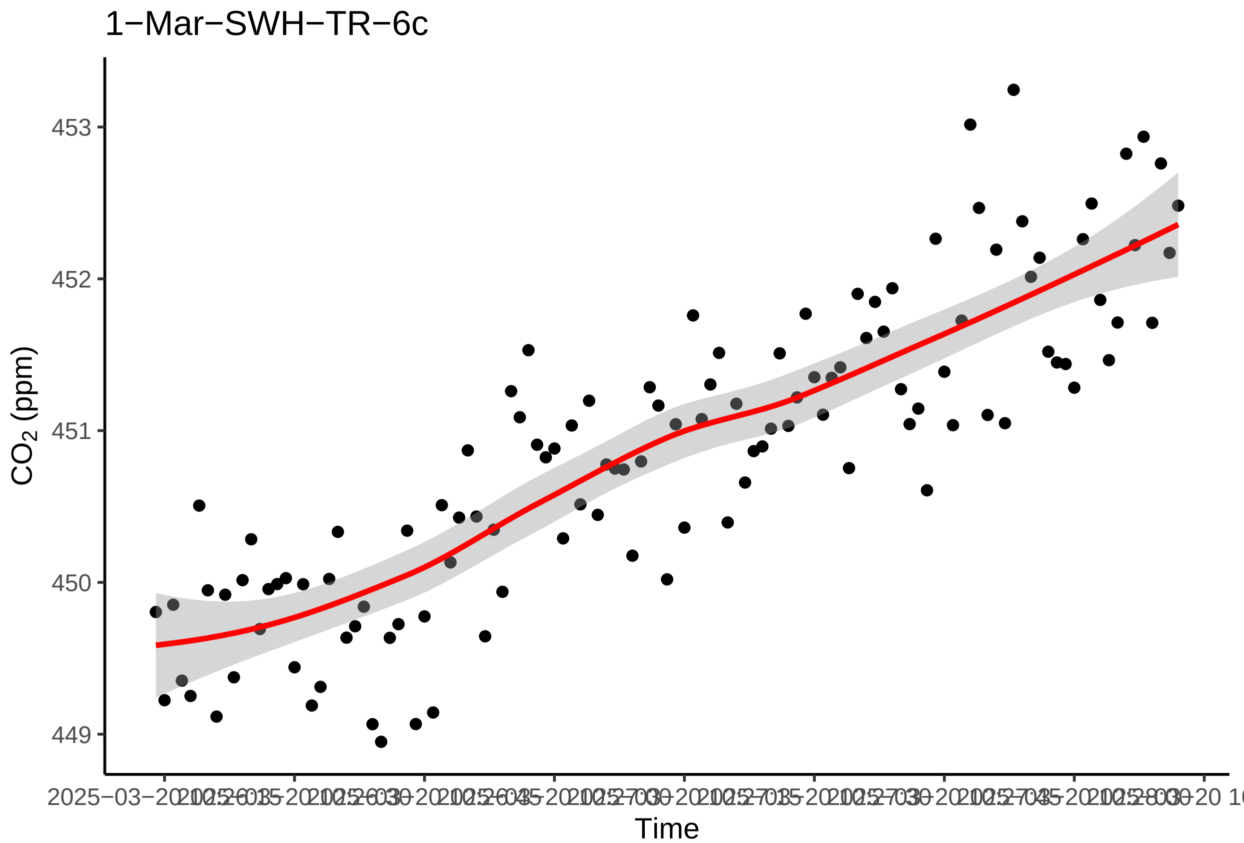

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



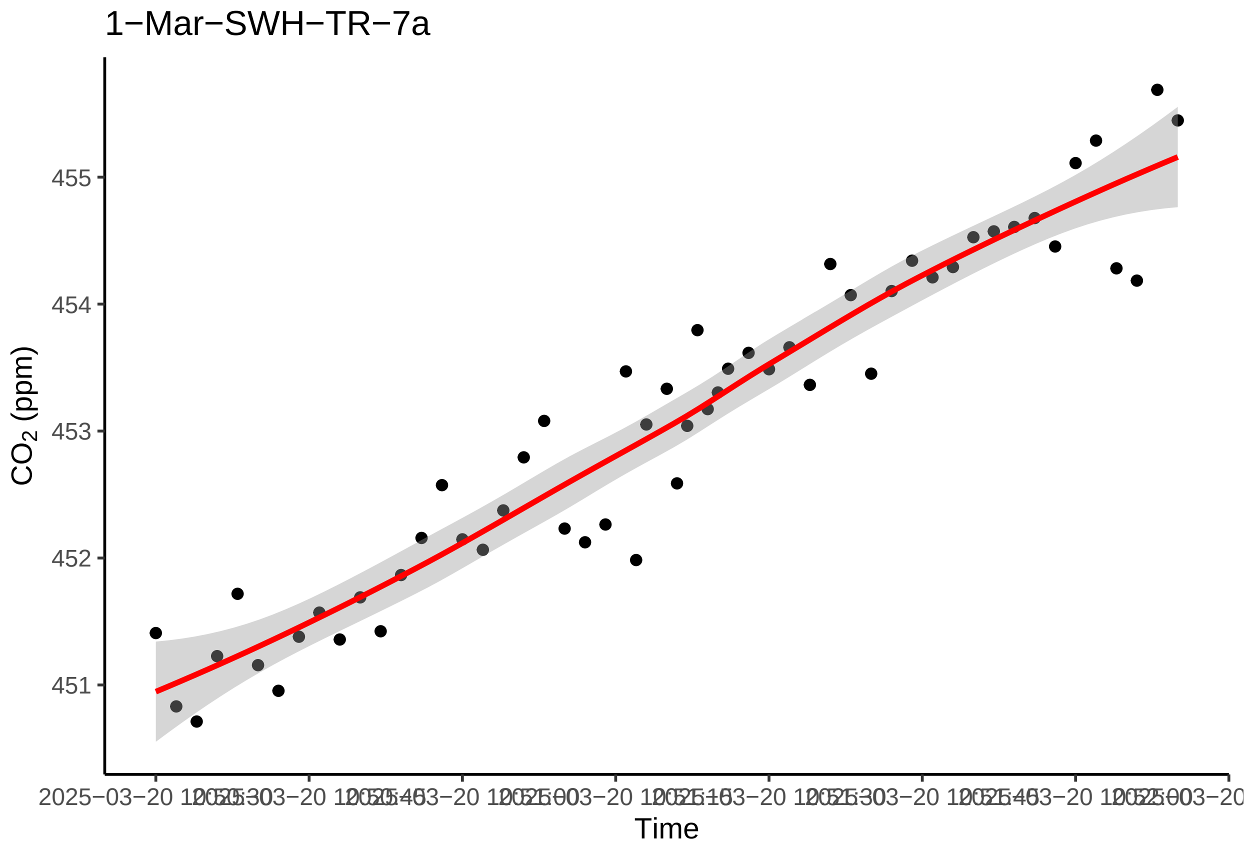

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



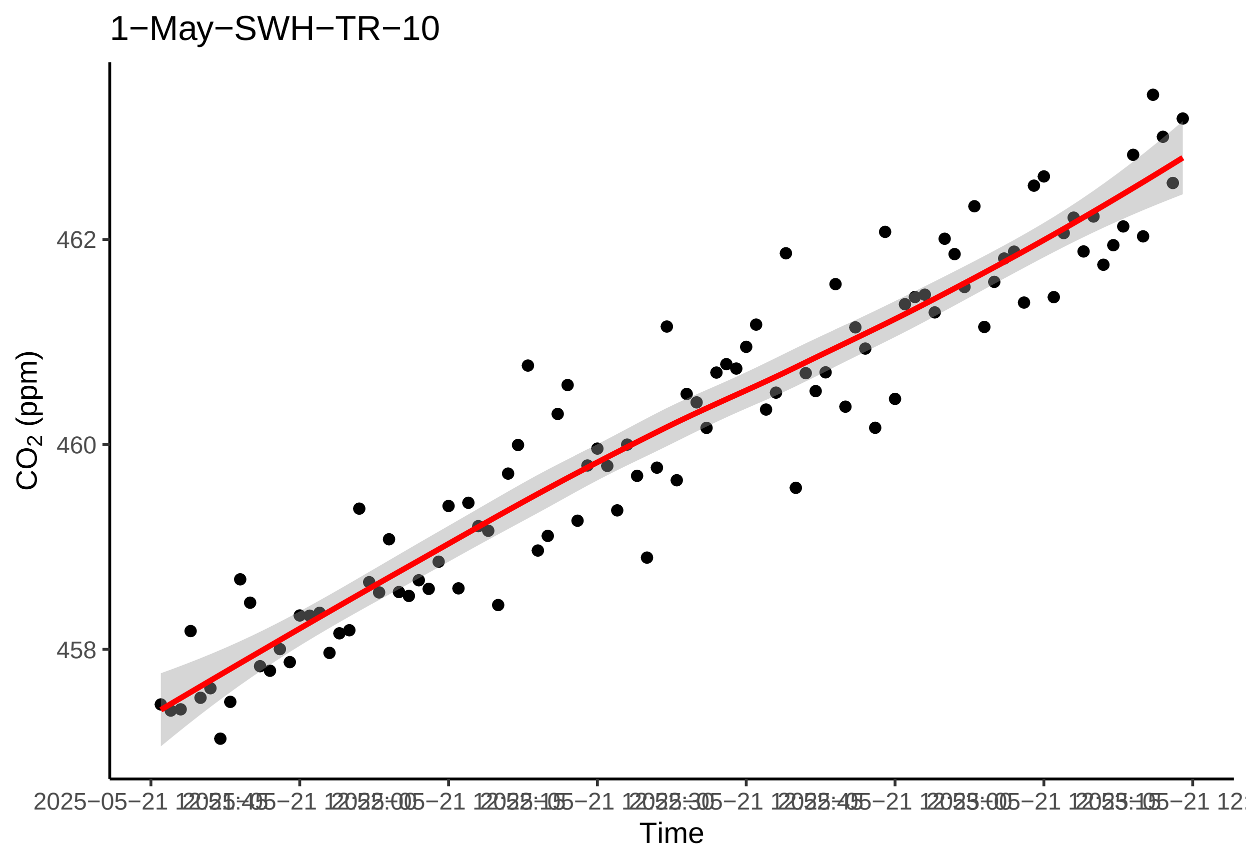

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



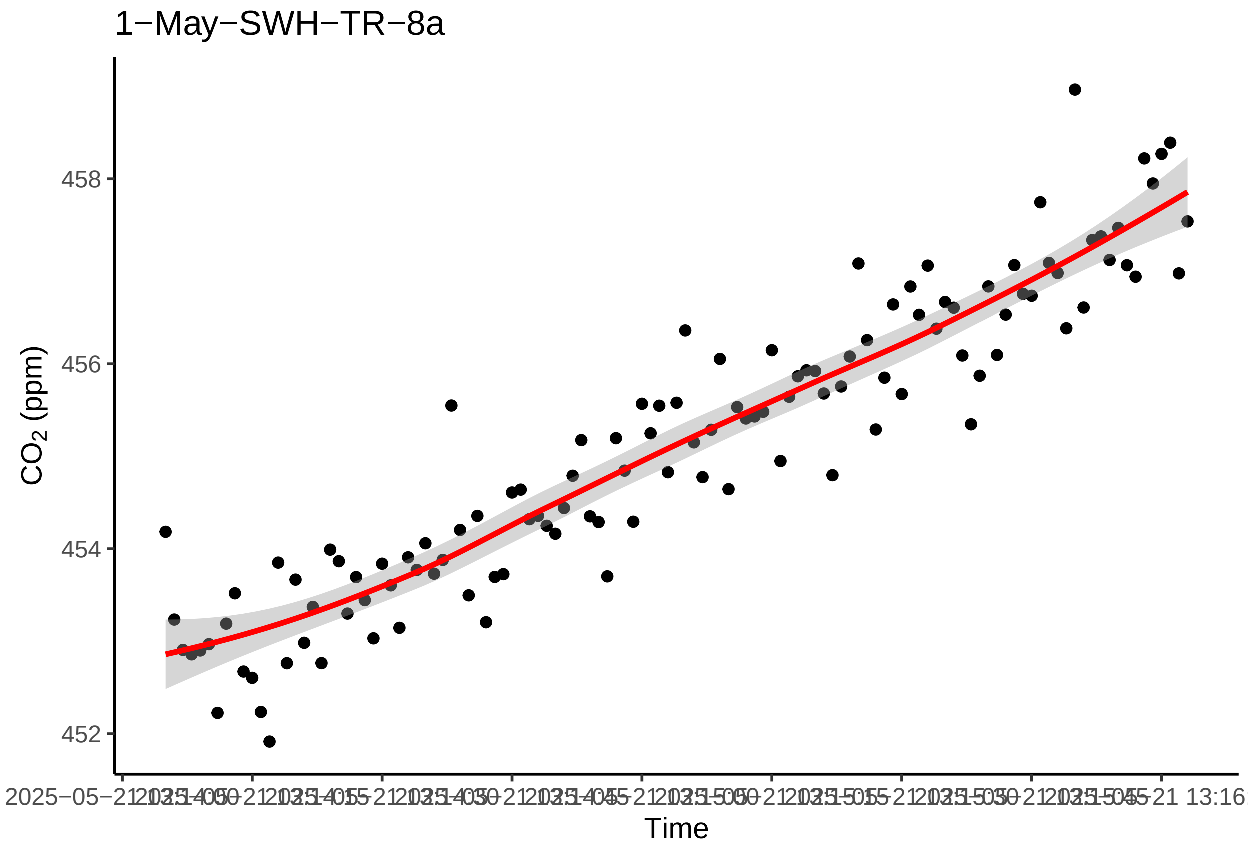

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



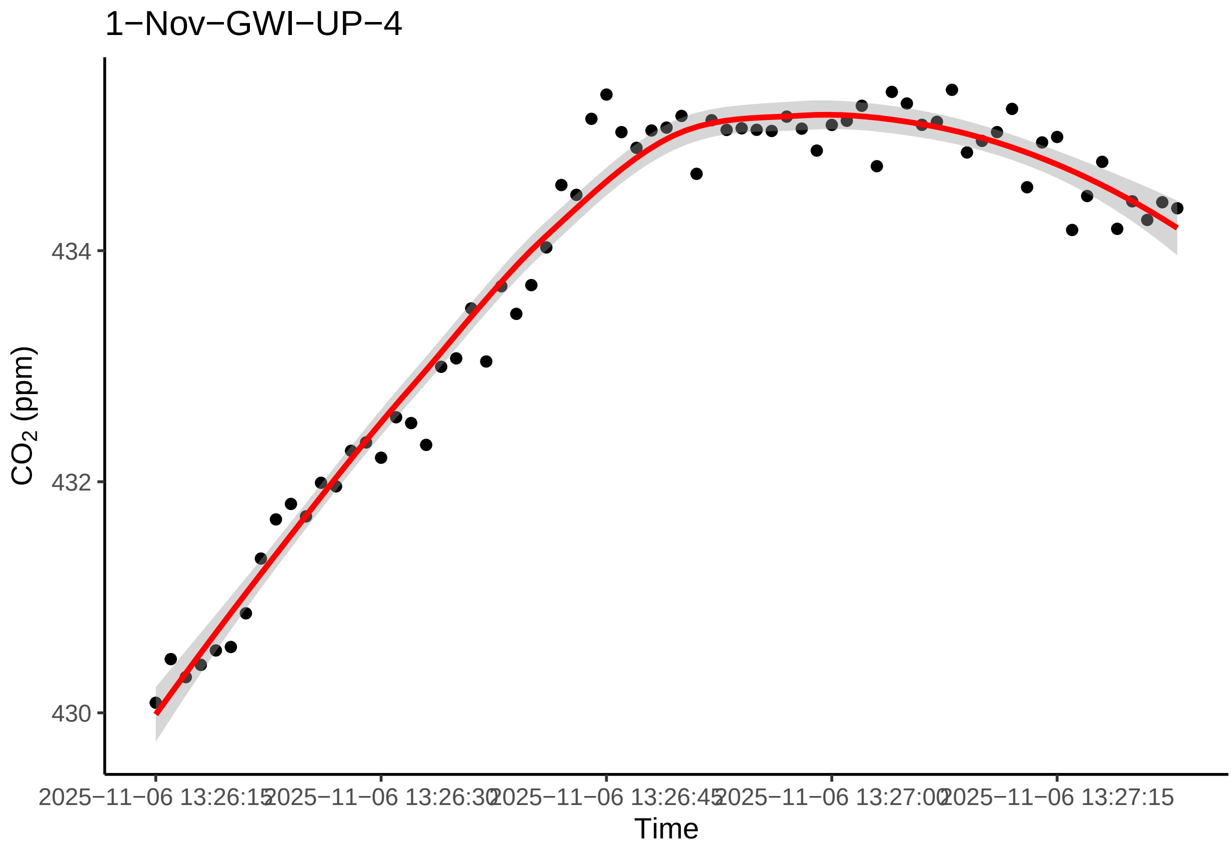

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



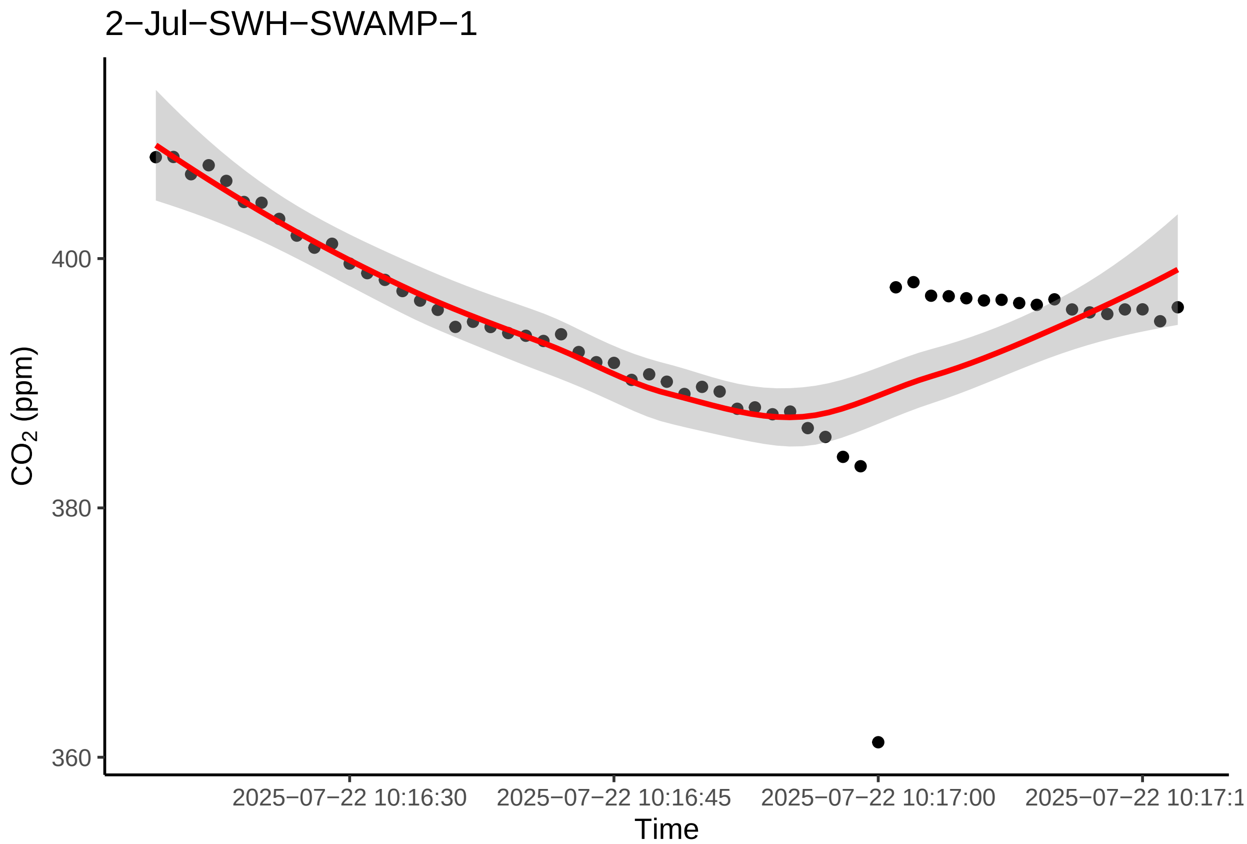

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



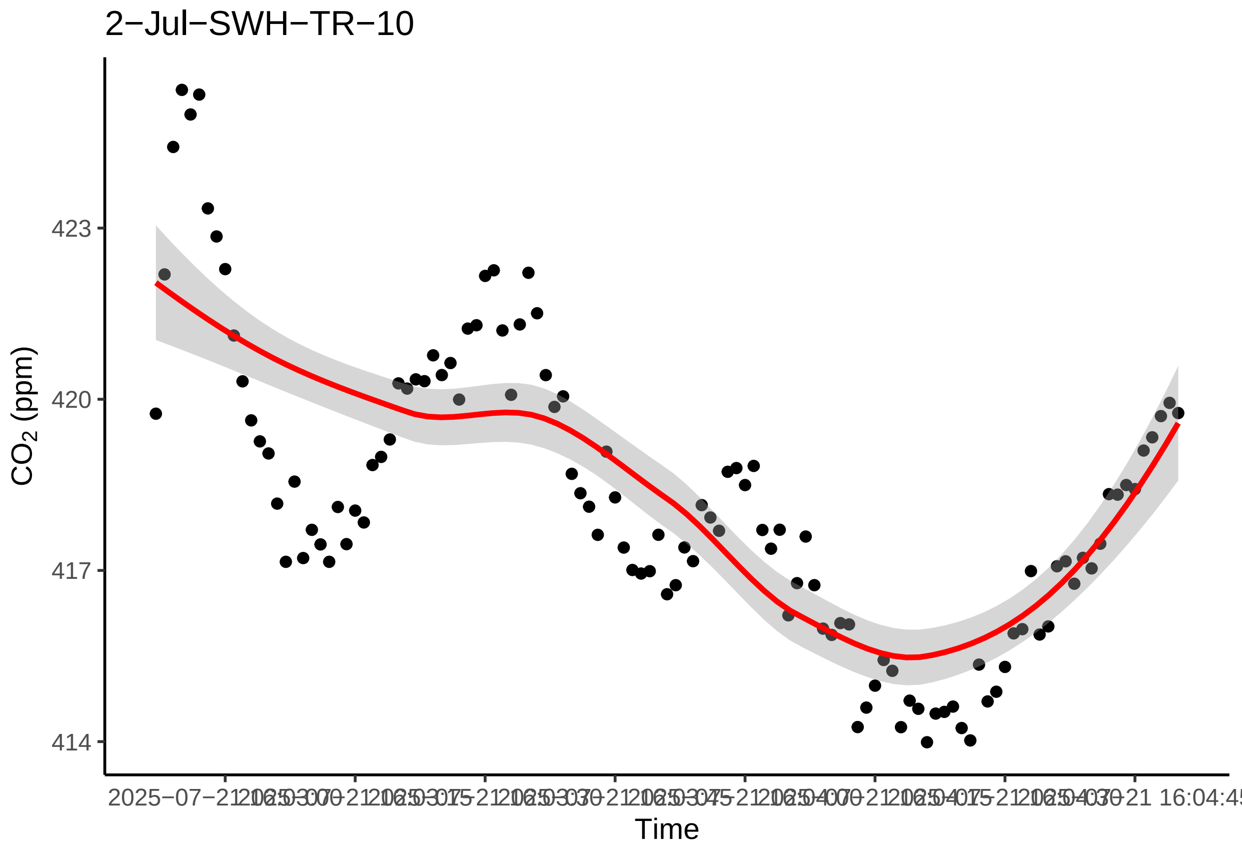

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



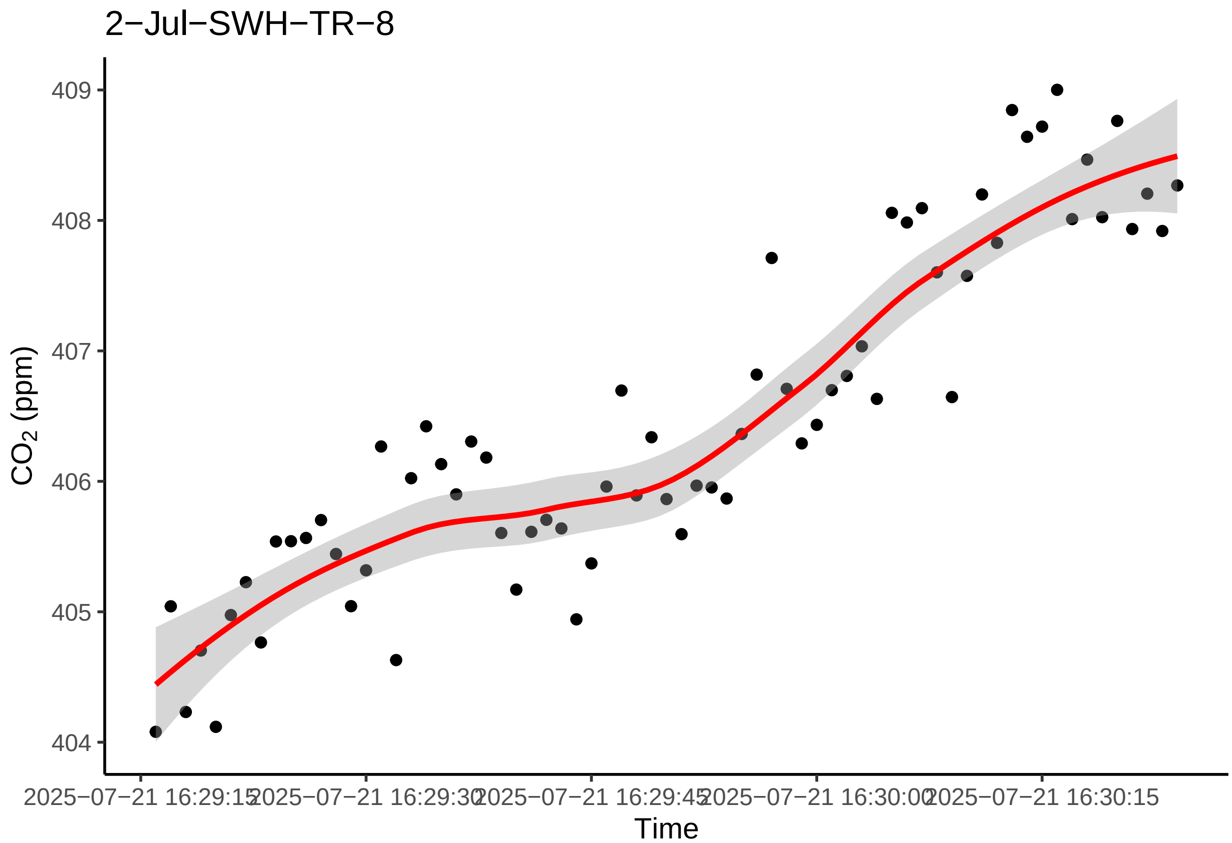

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



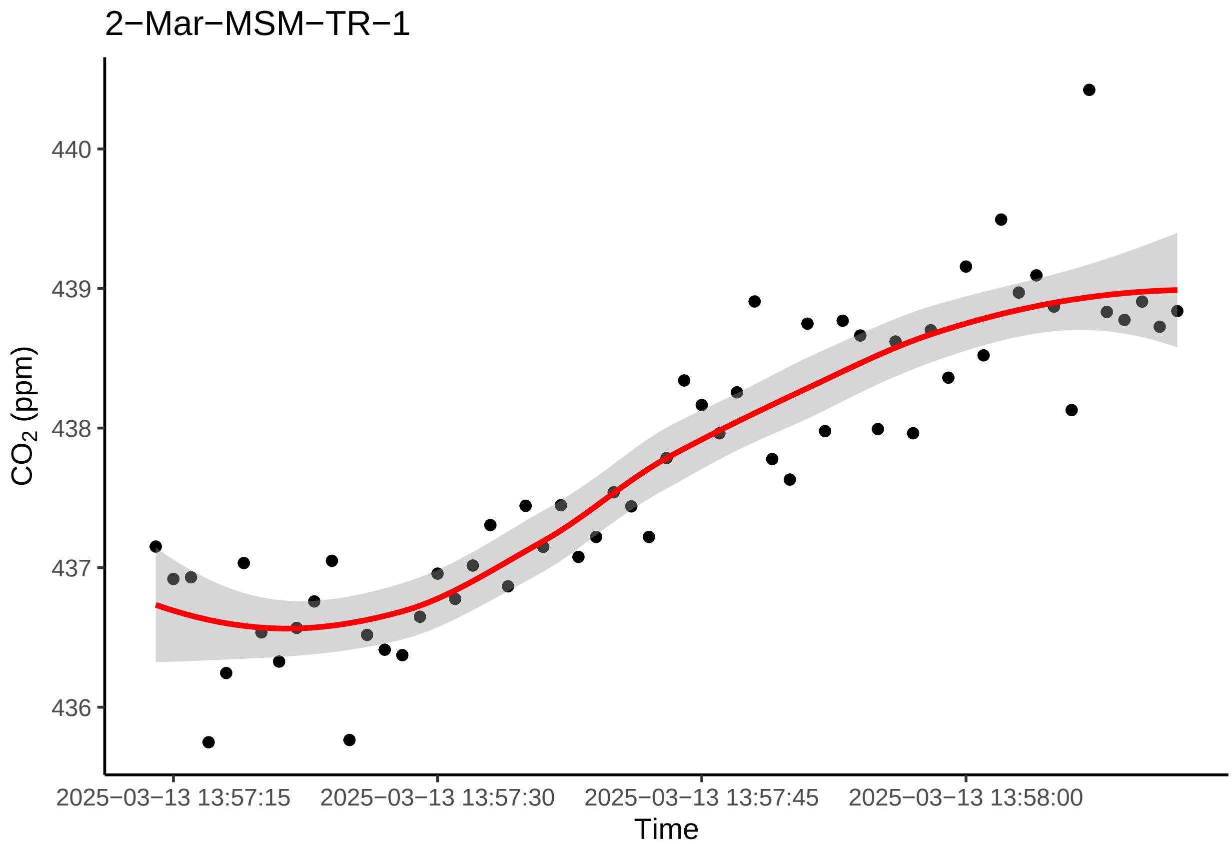

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



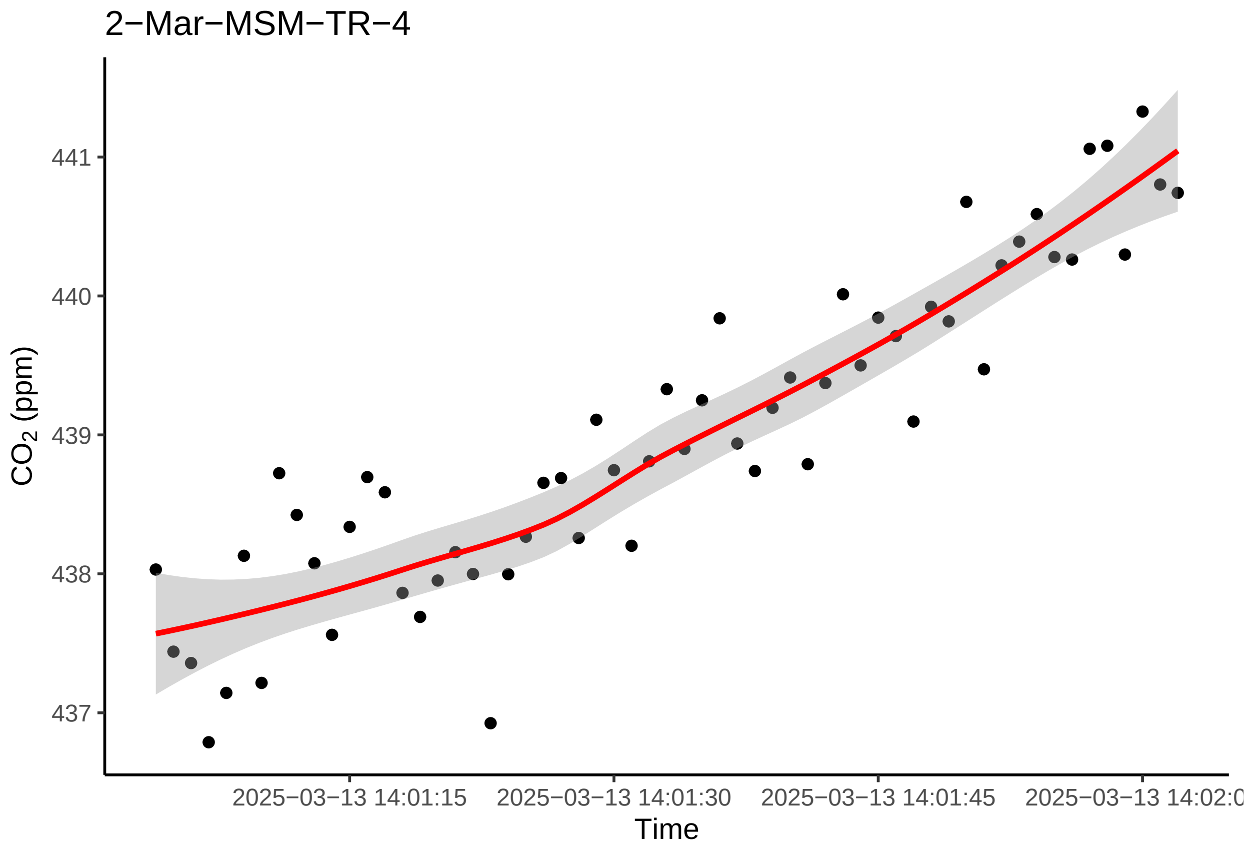
‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’



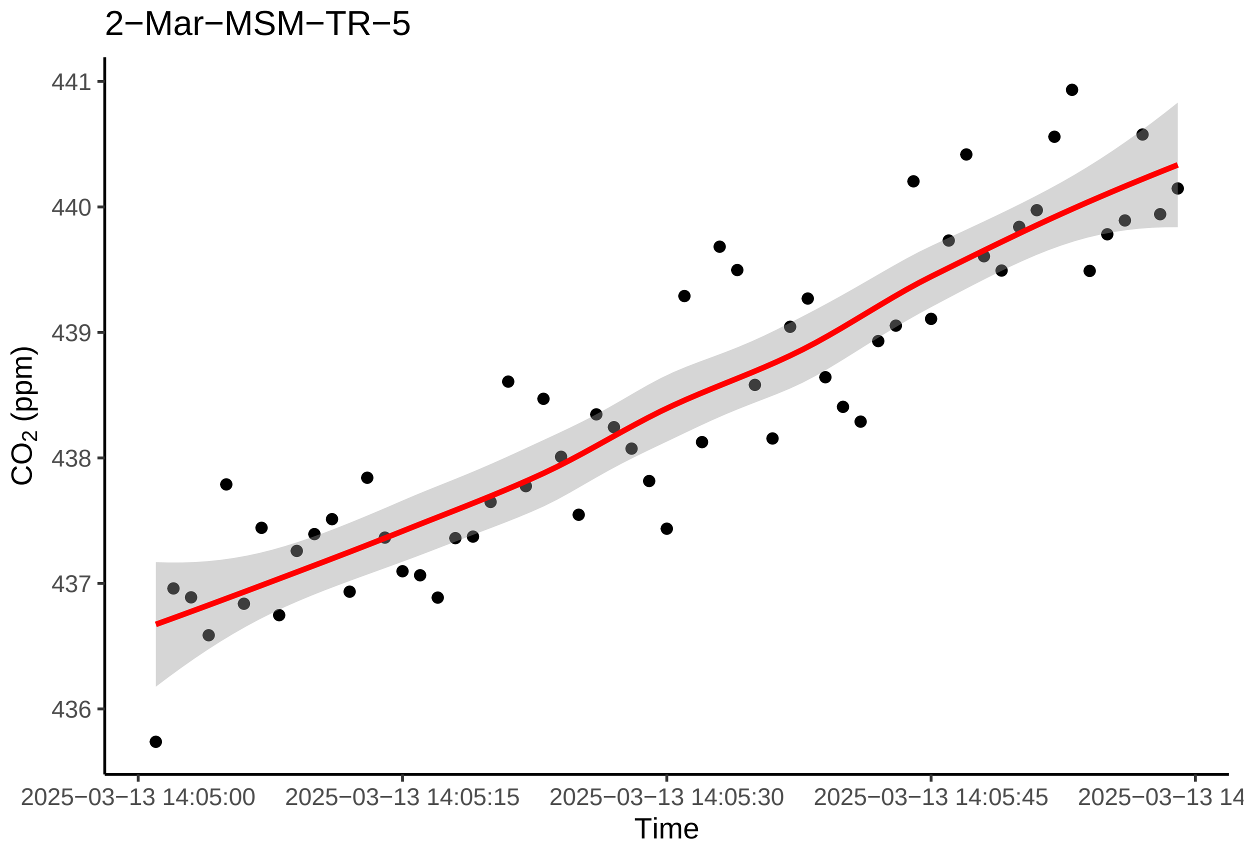

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



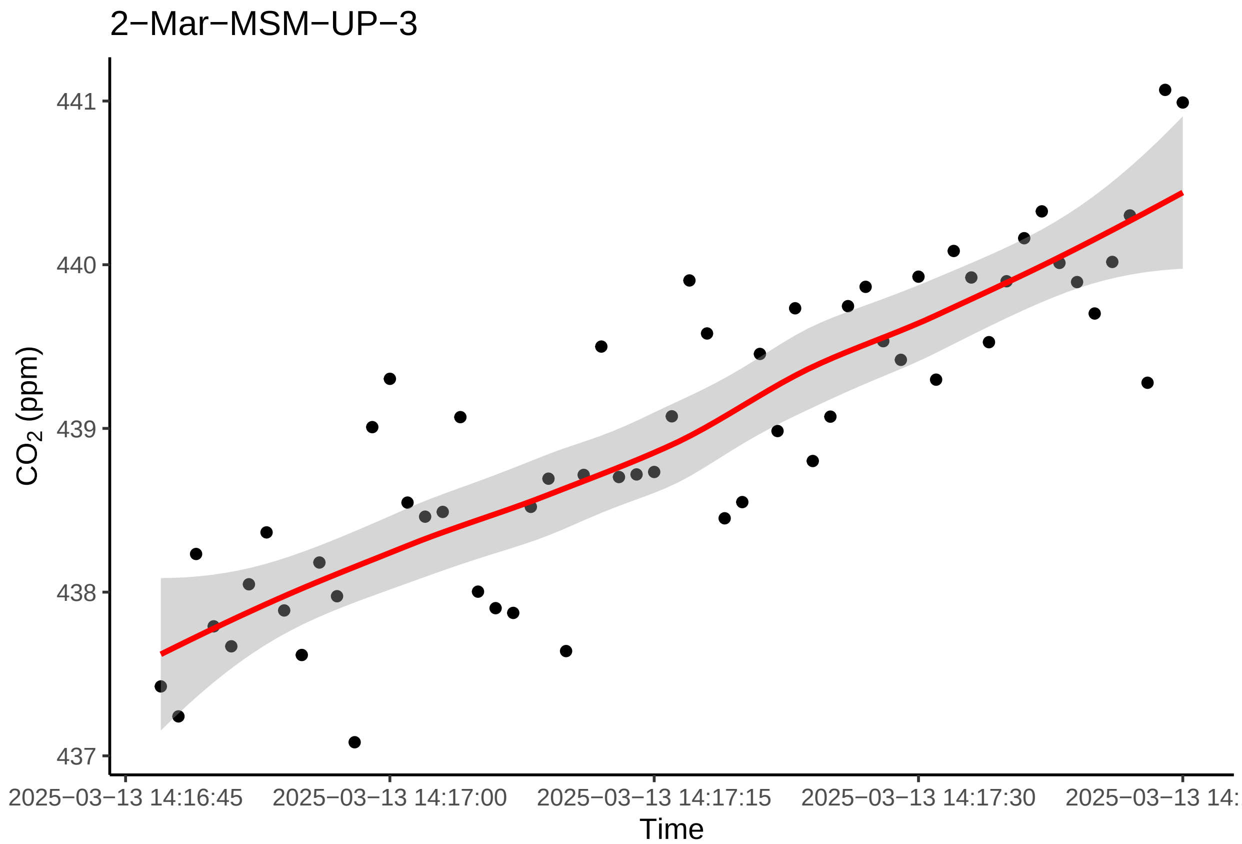

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



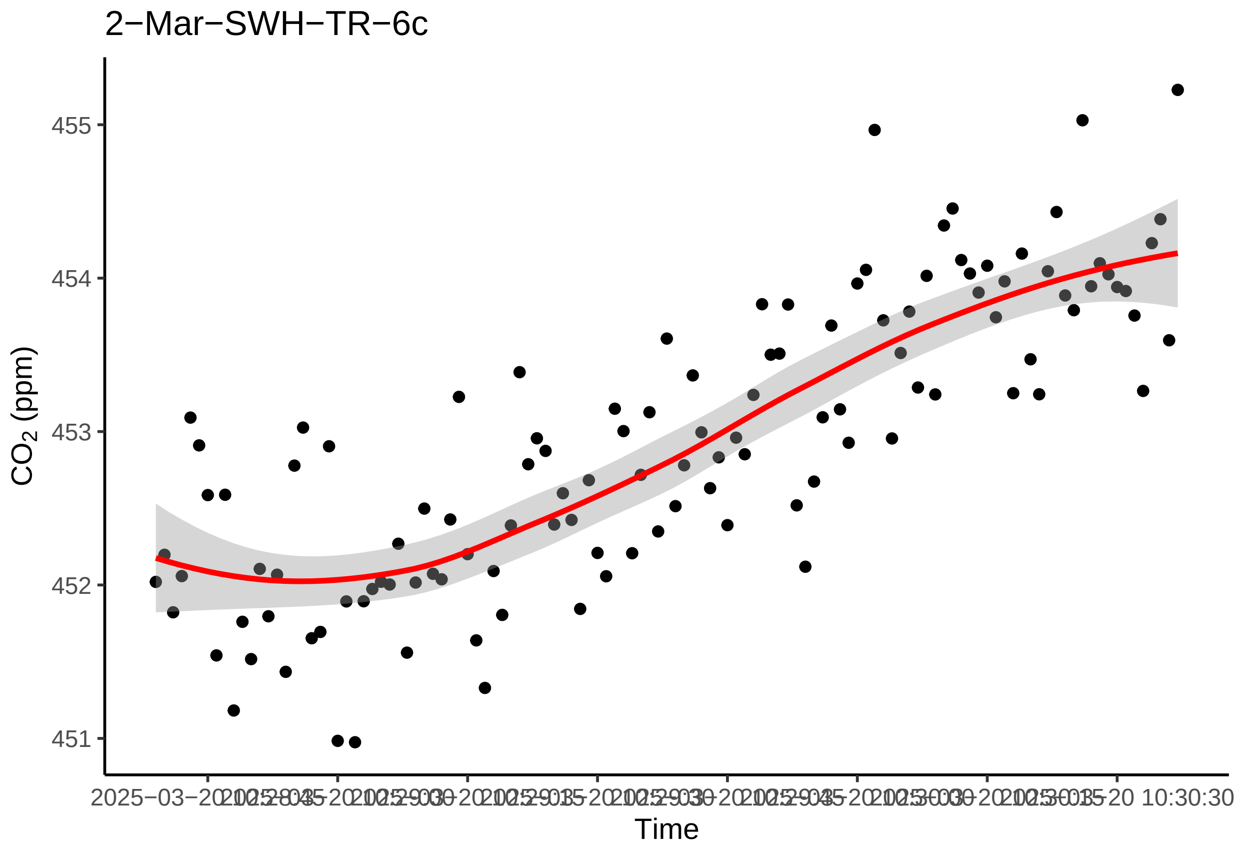

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



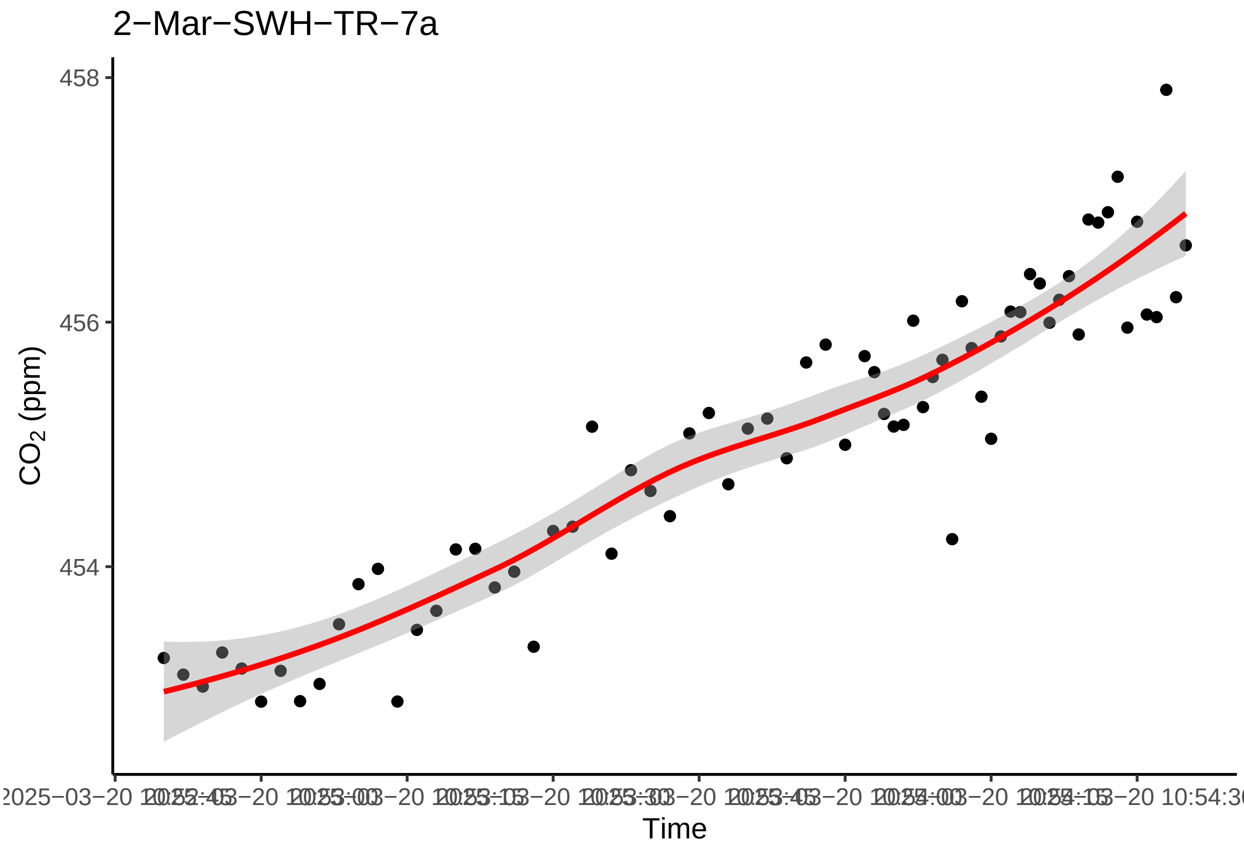

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



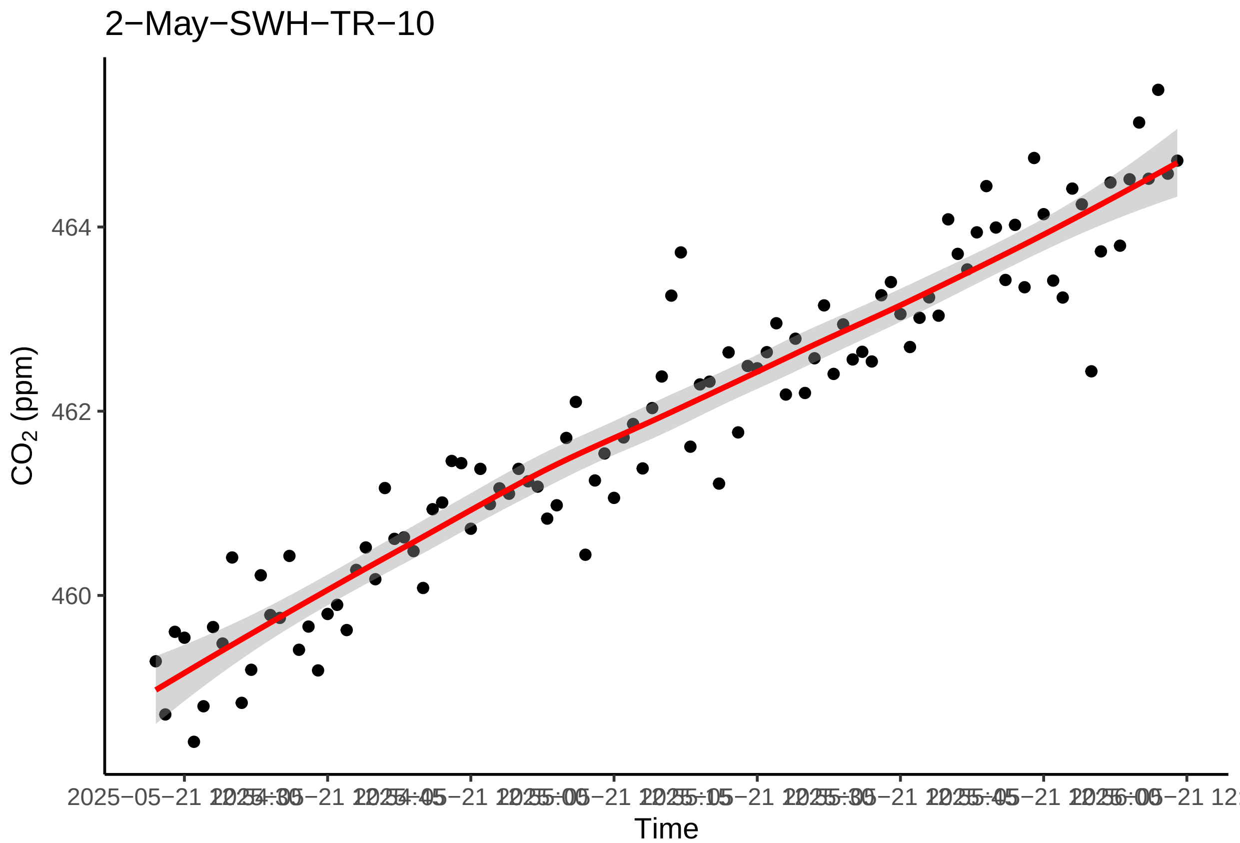

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



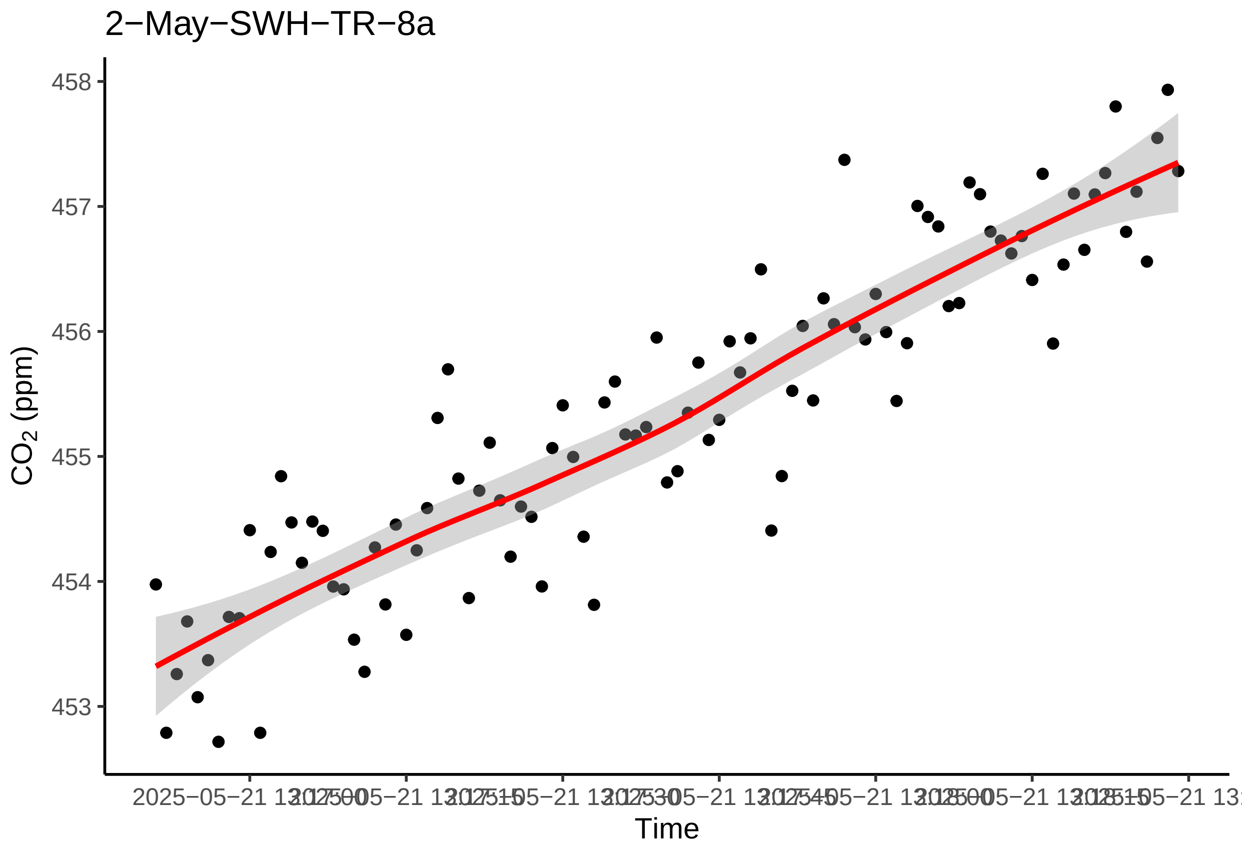

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



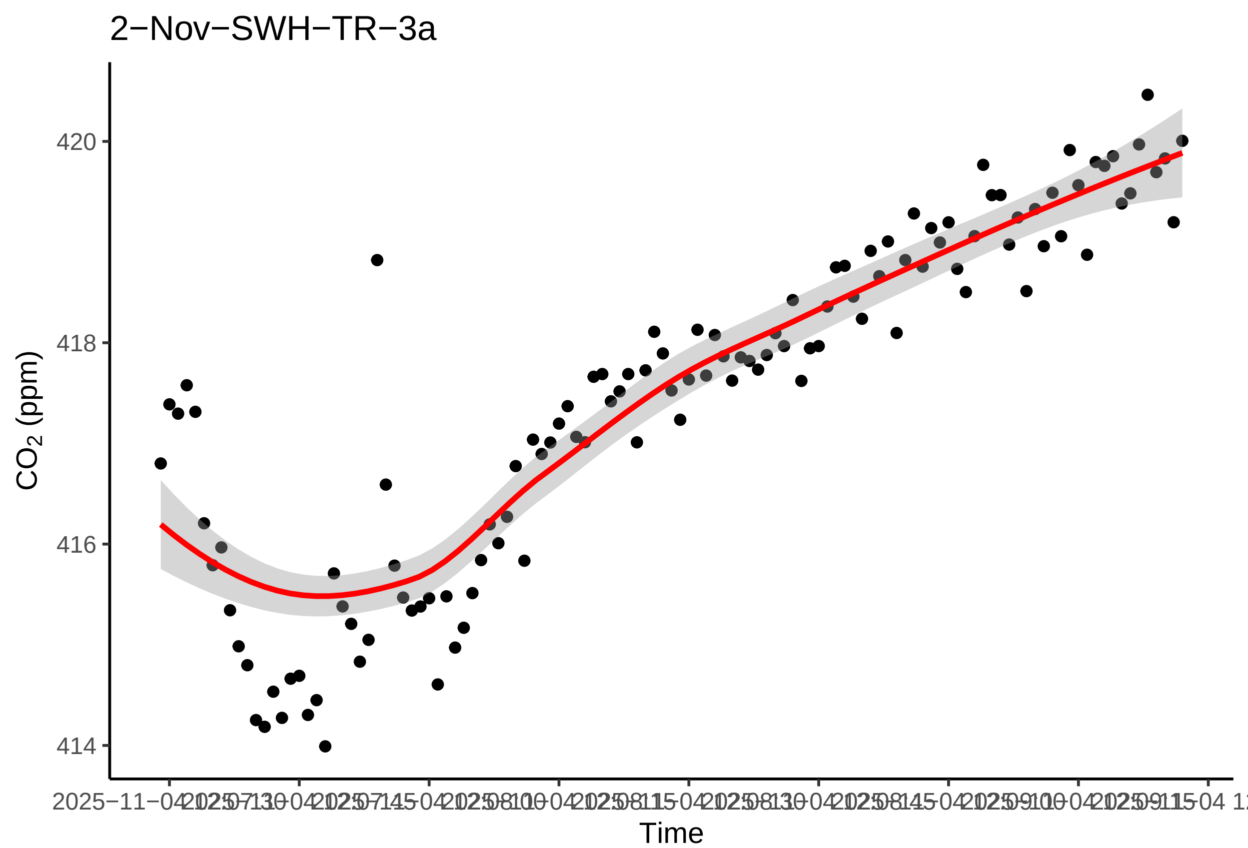

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



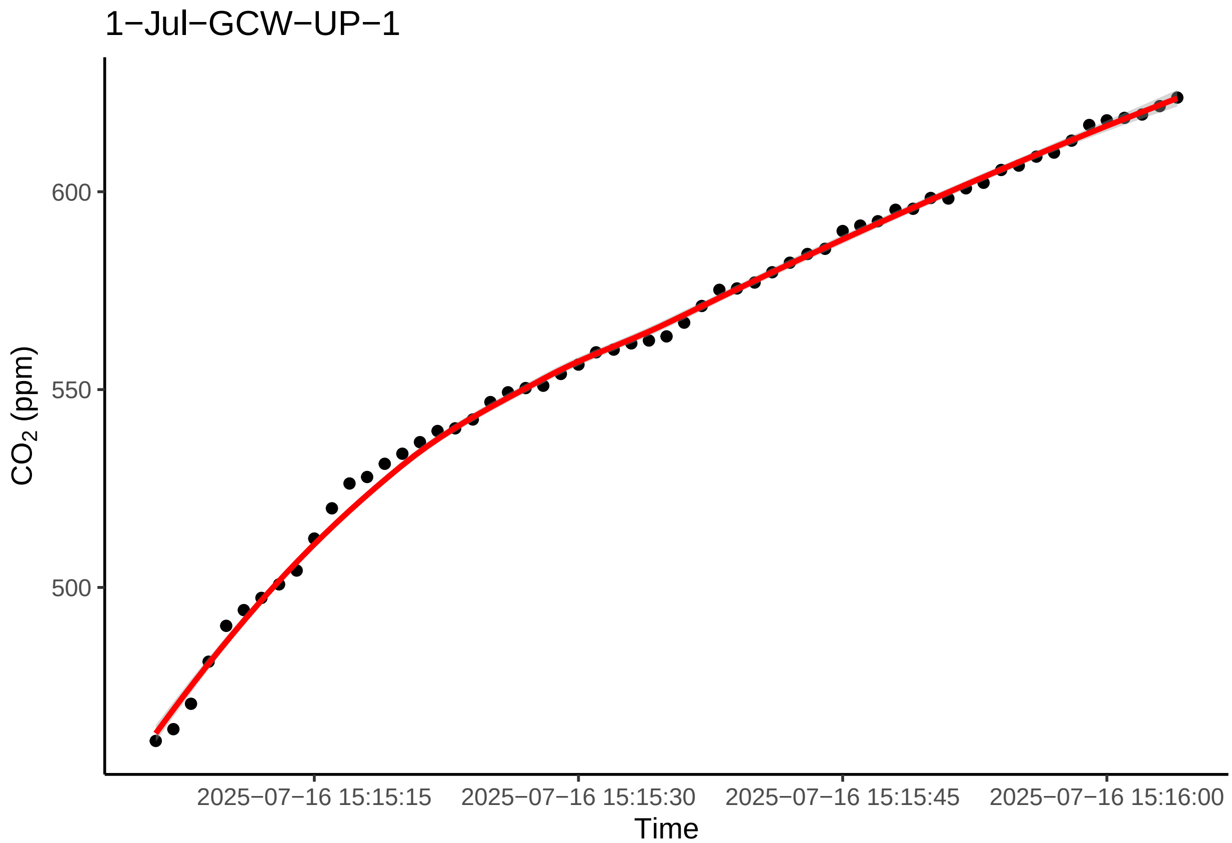

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



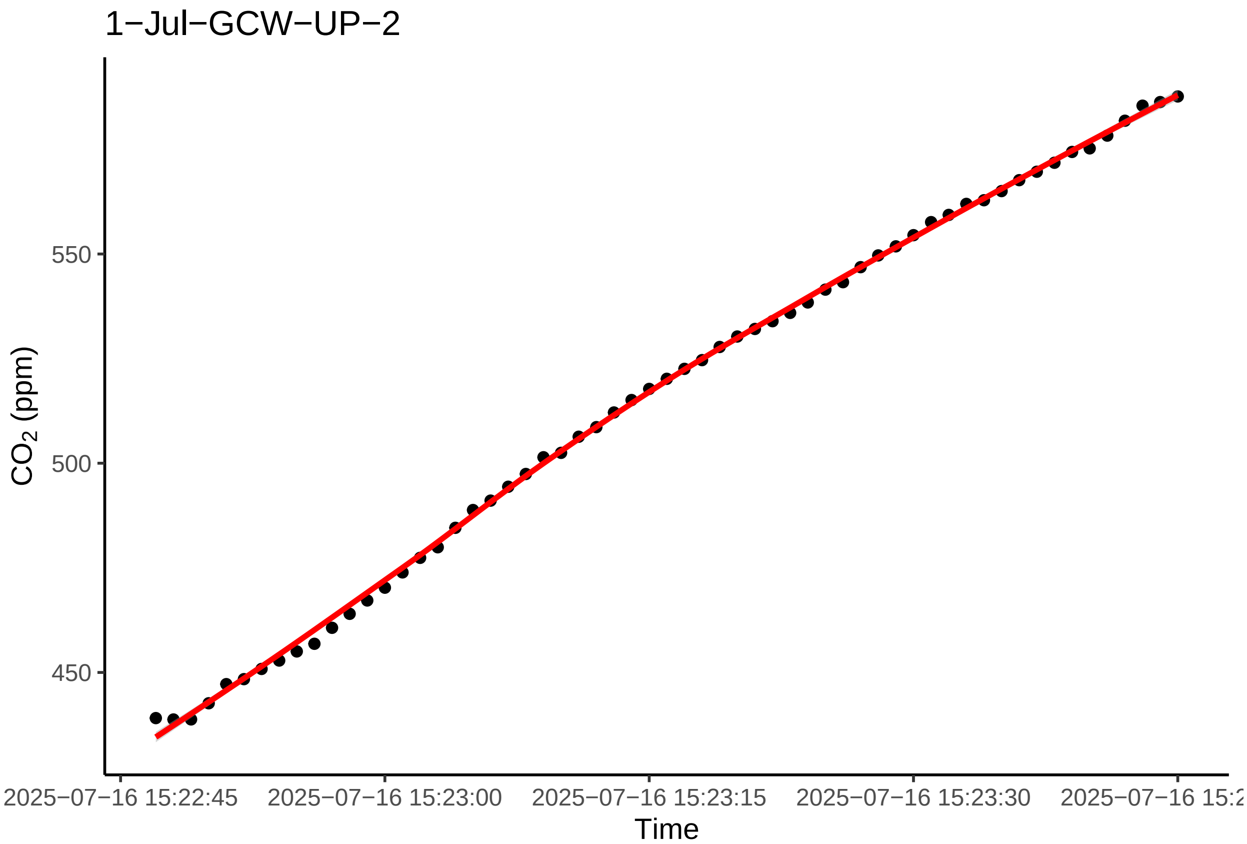

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



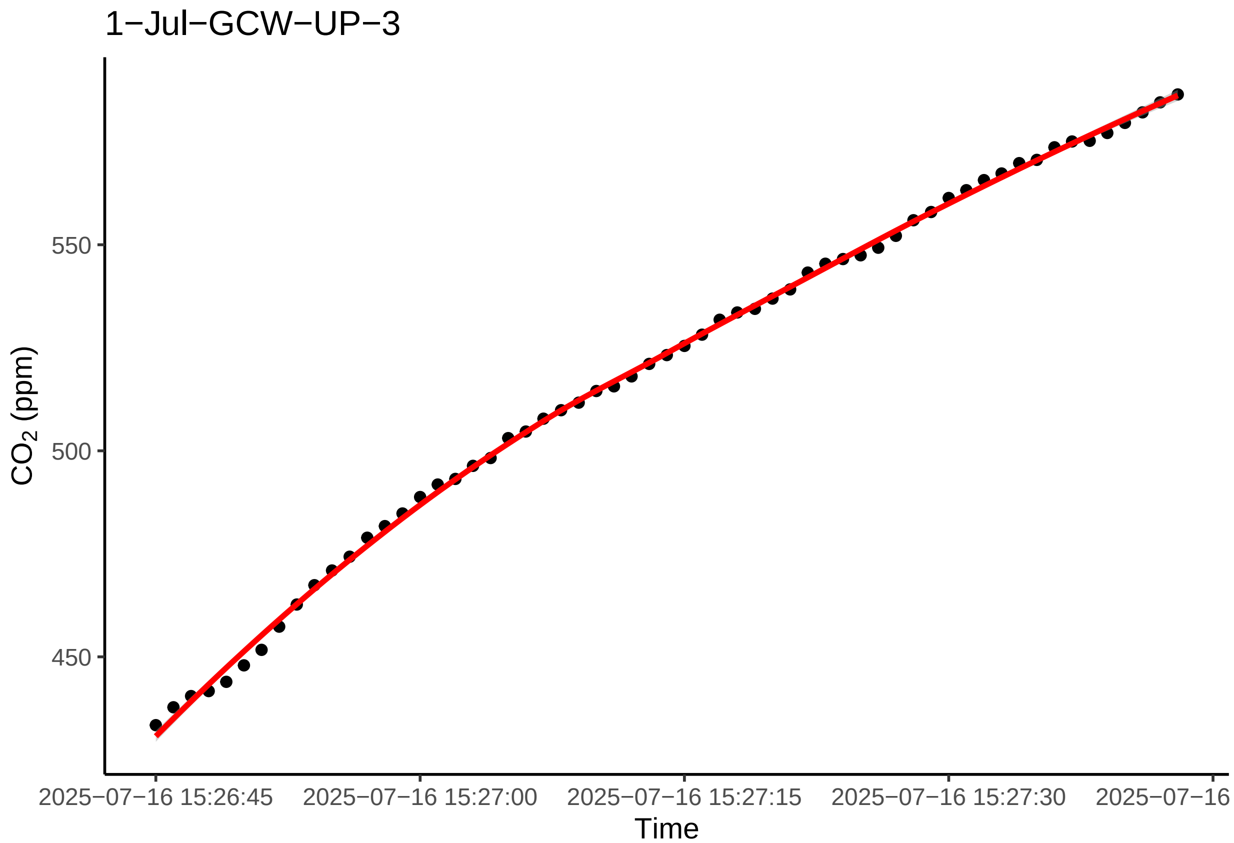

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



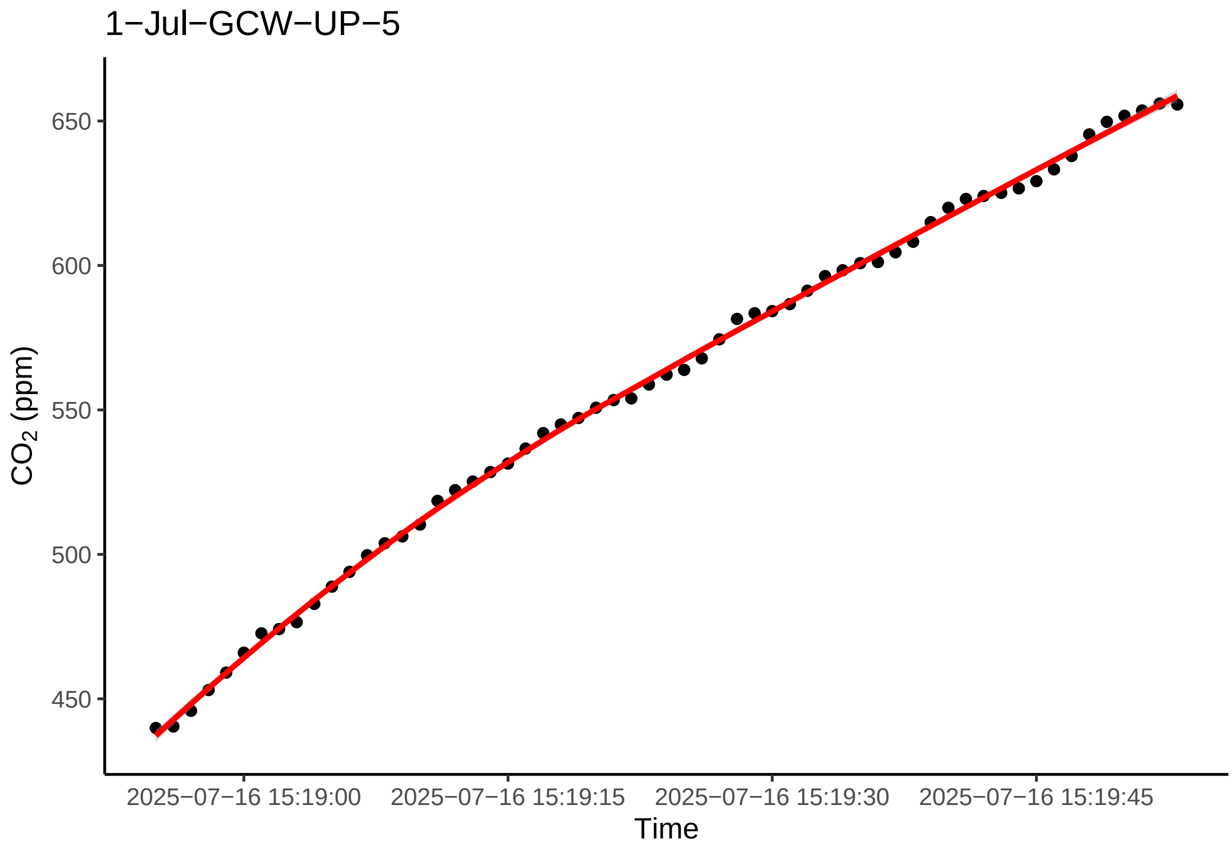

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



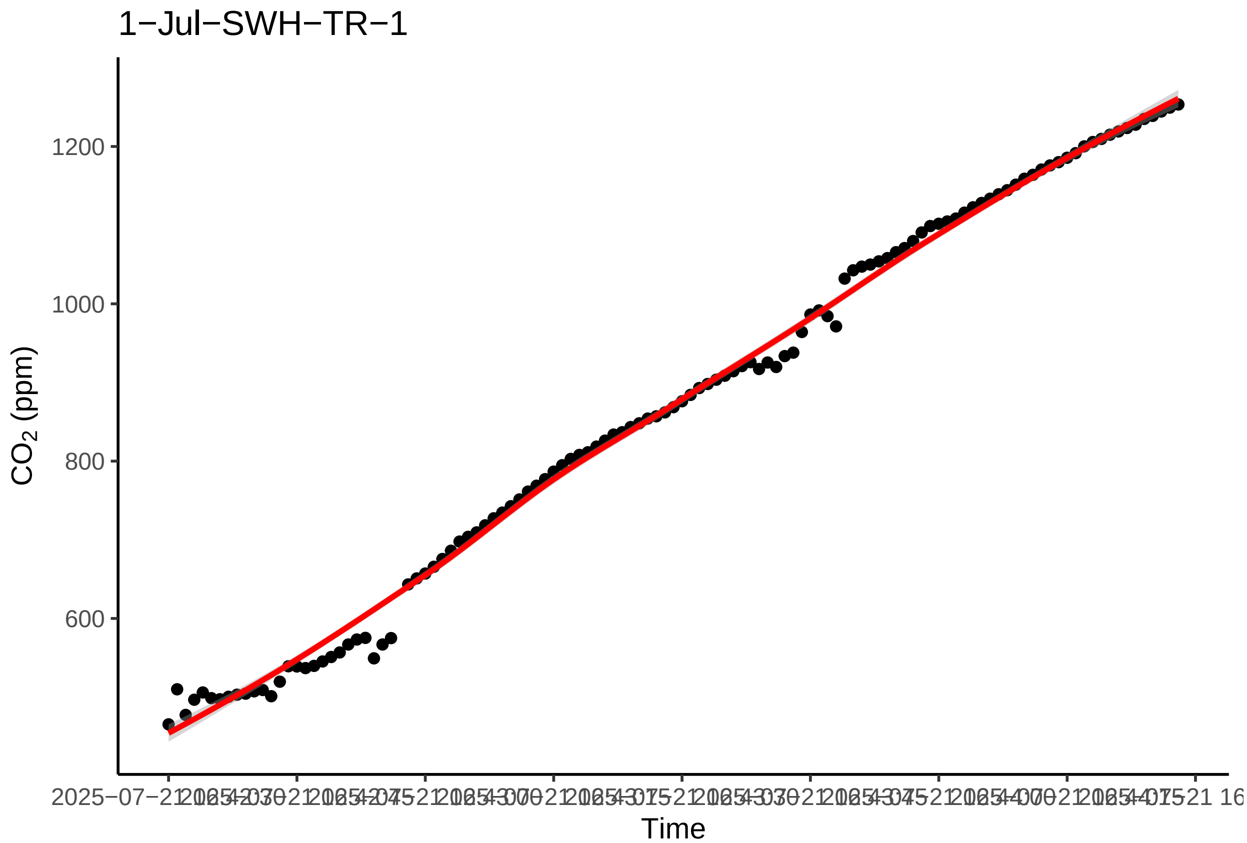

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



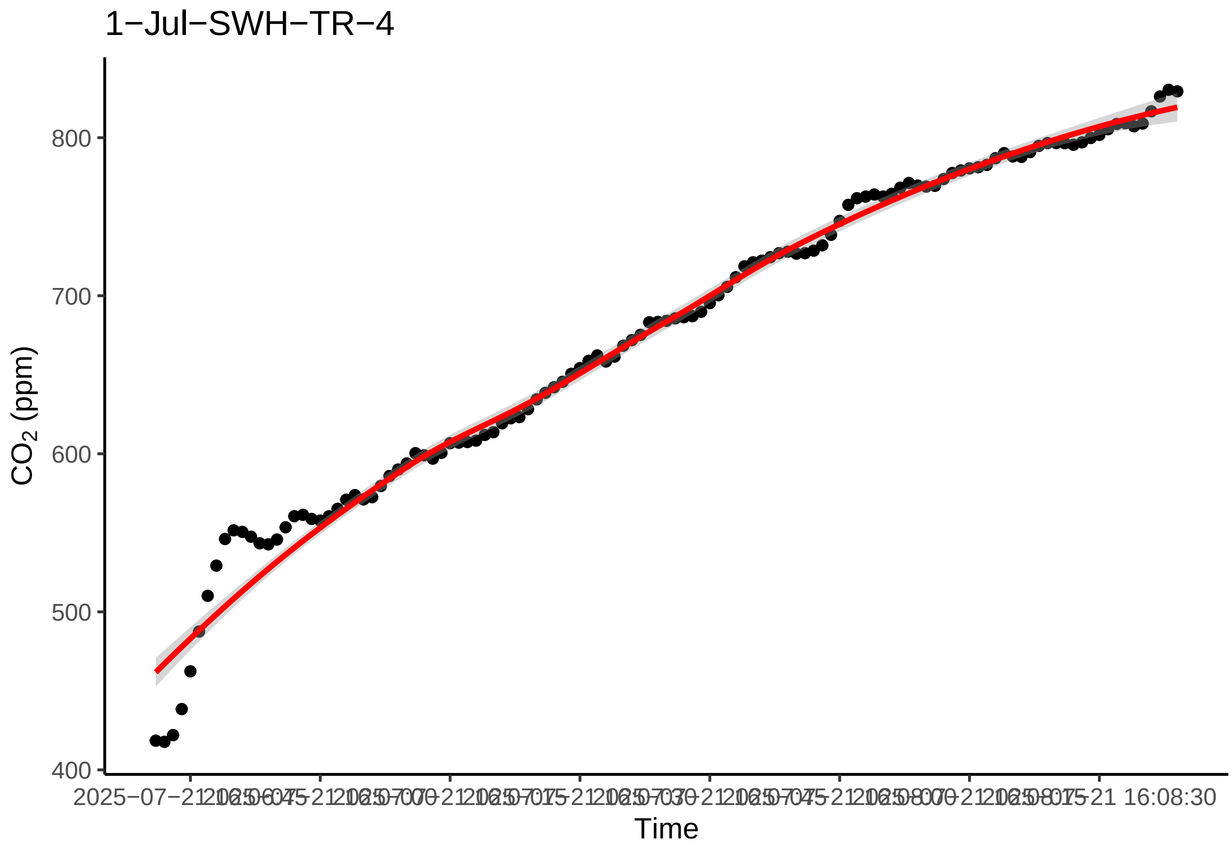

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



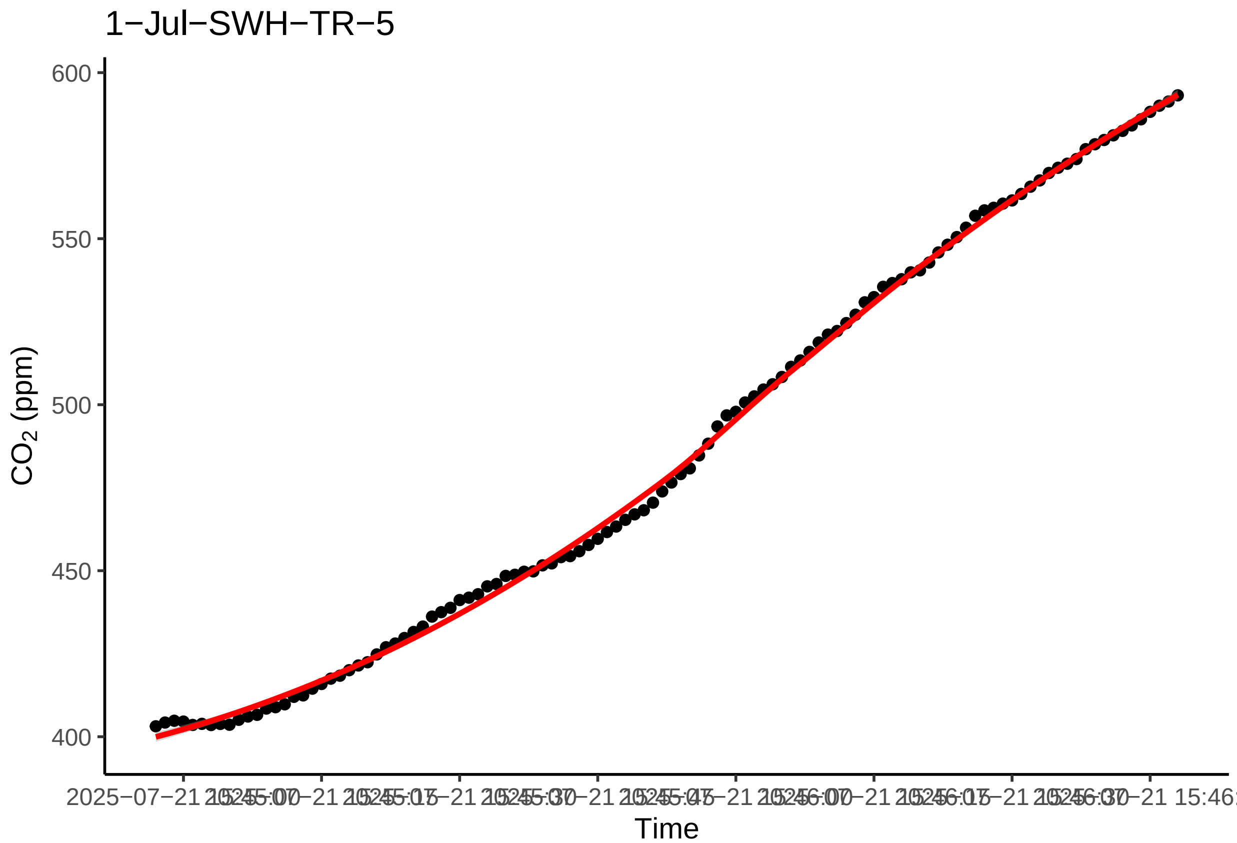

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



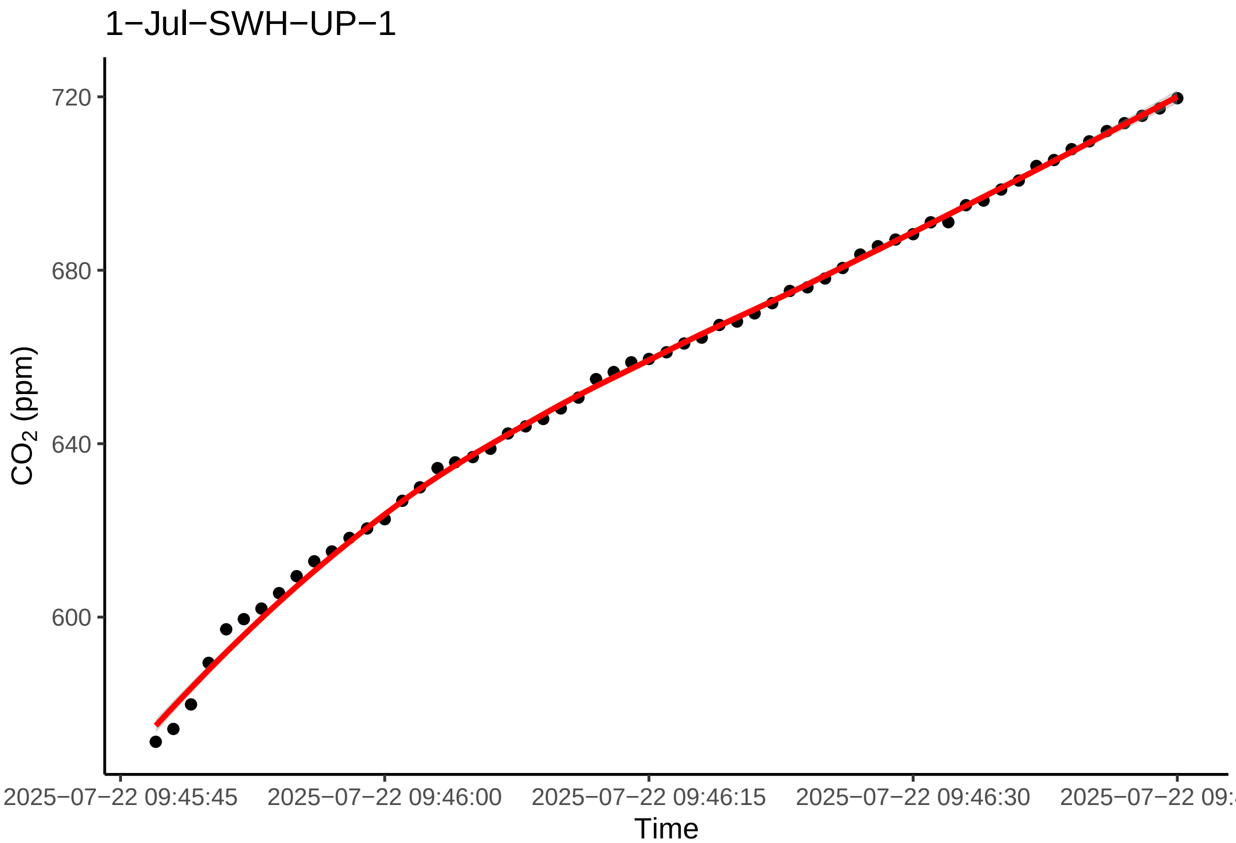

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



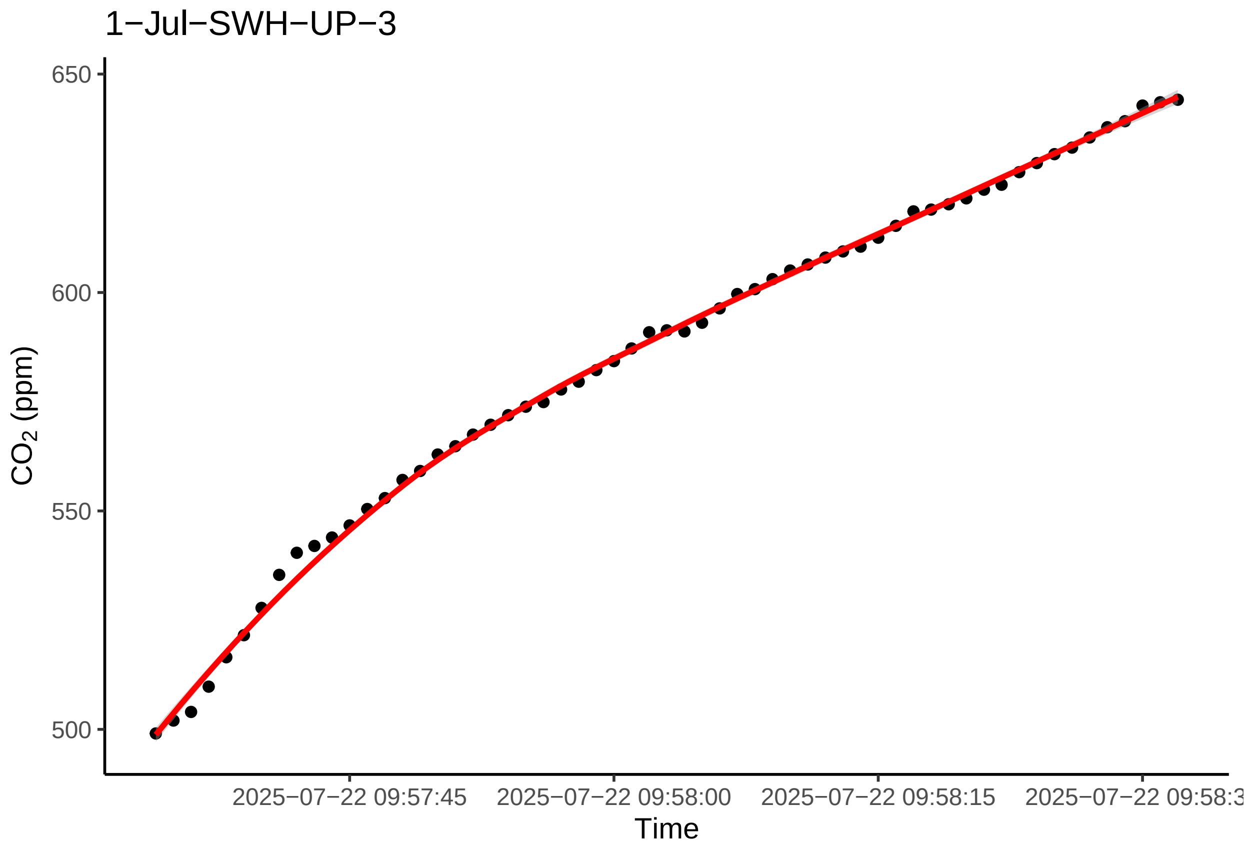
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'



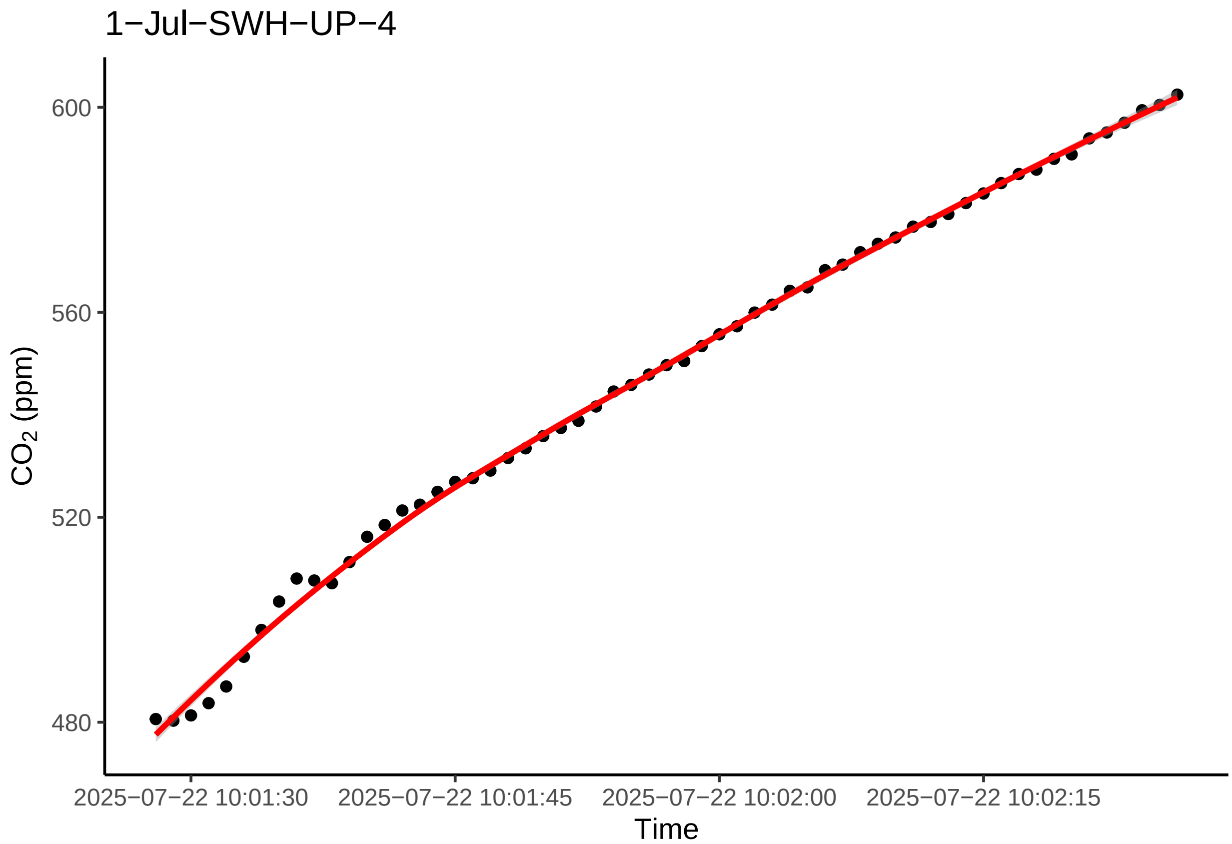

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



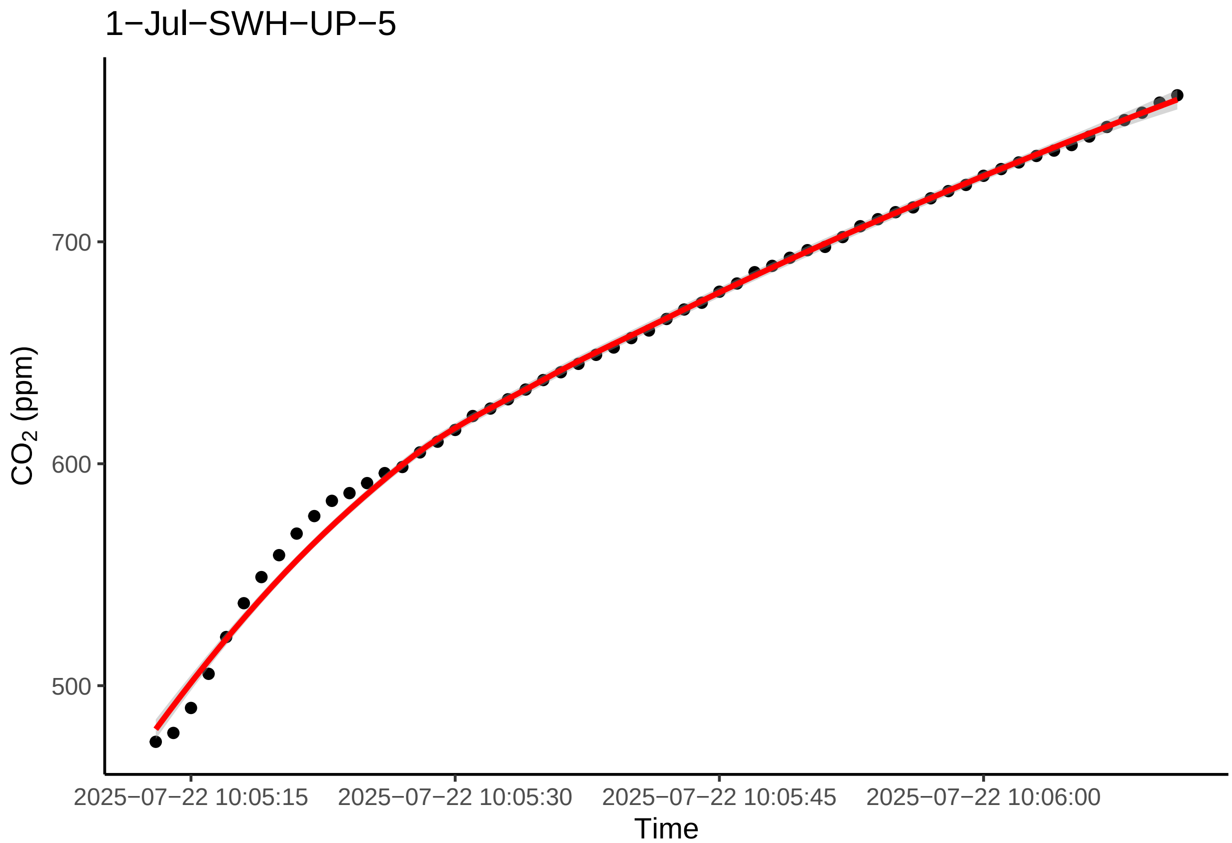

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



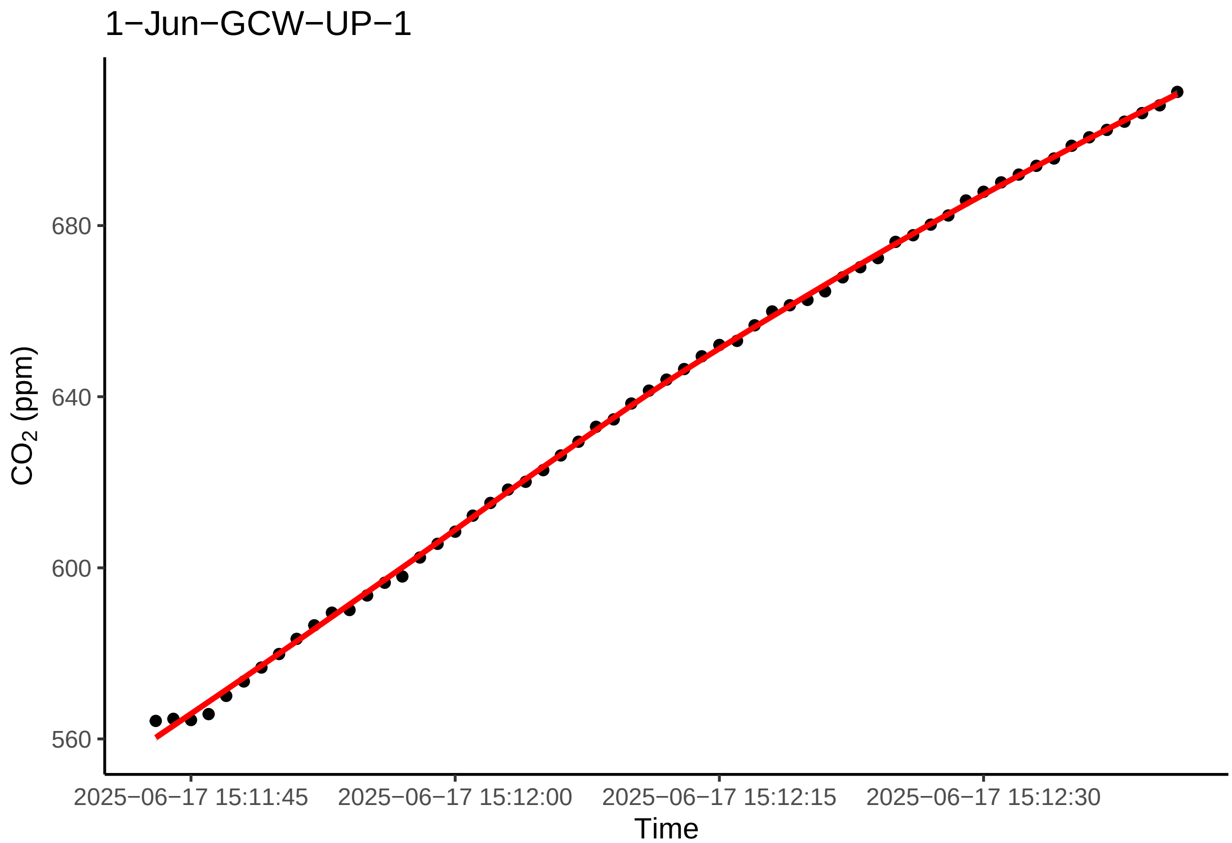

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



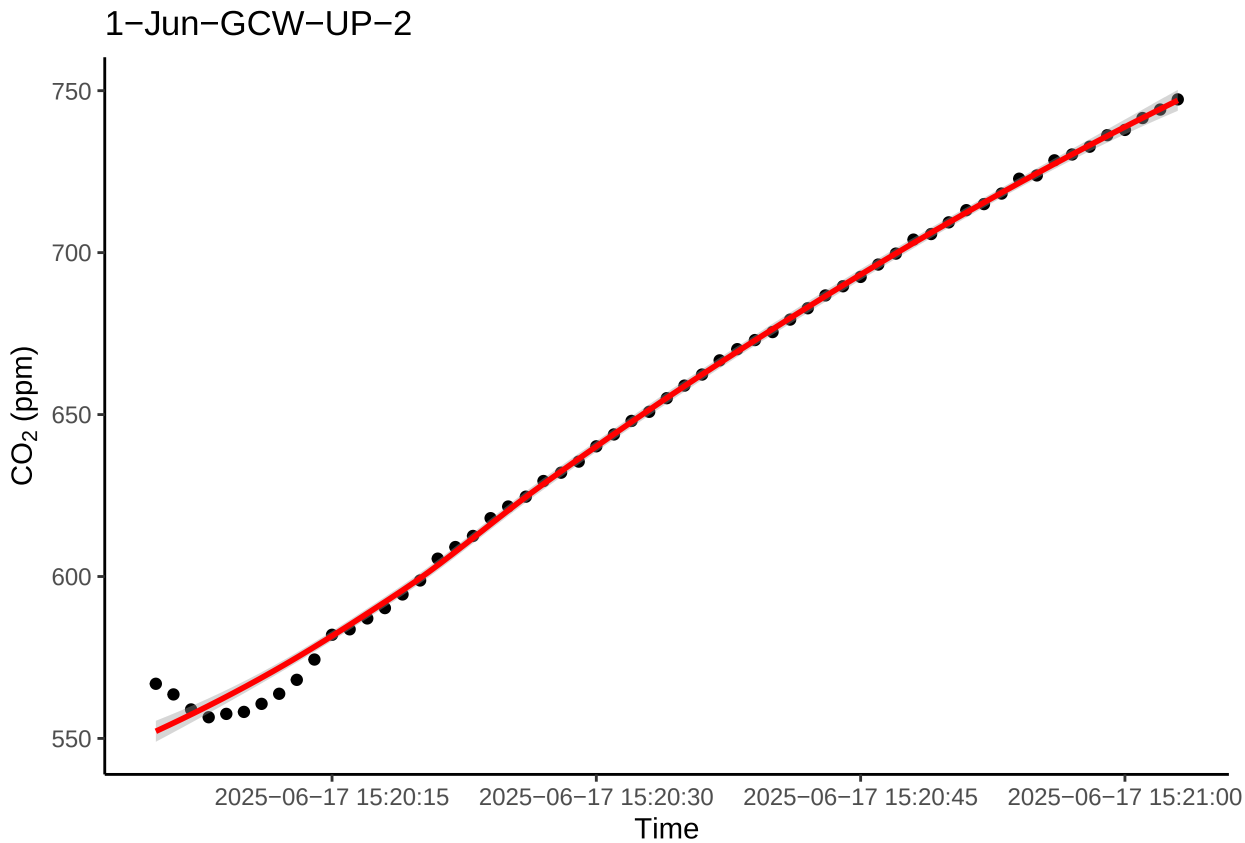

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



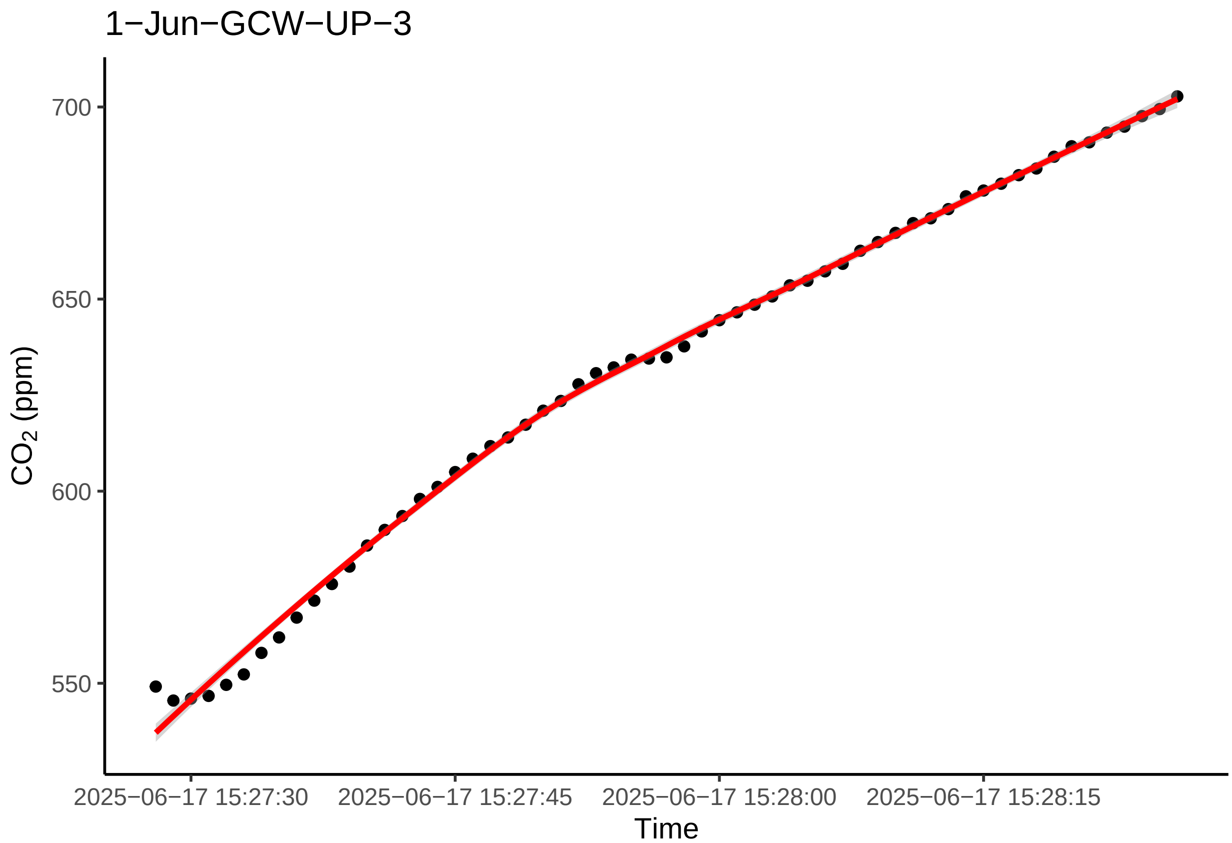

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



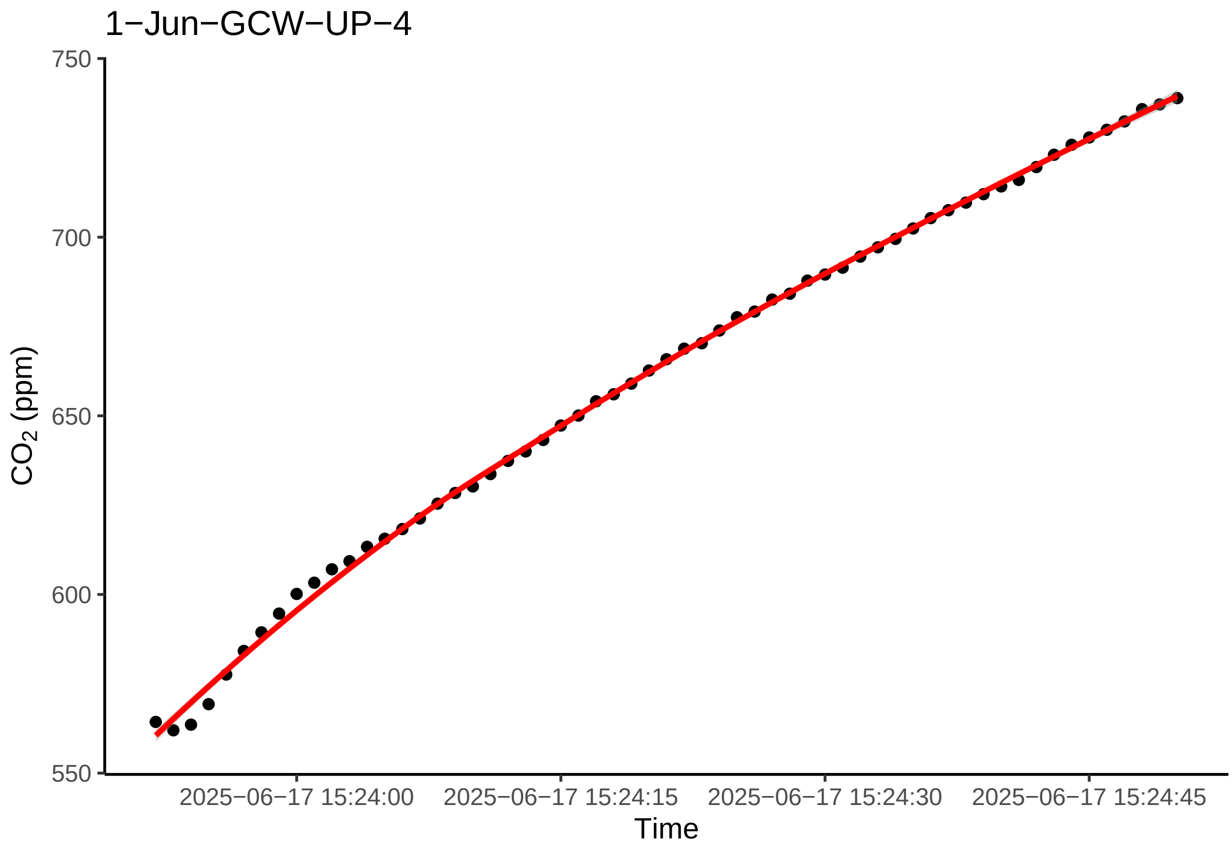

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



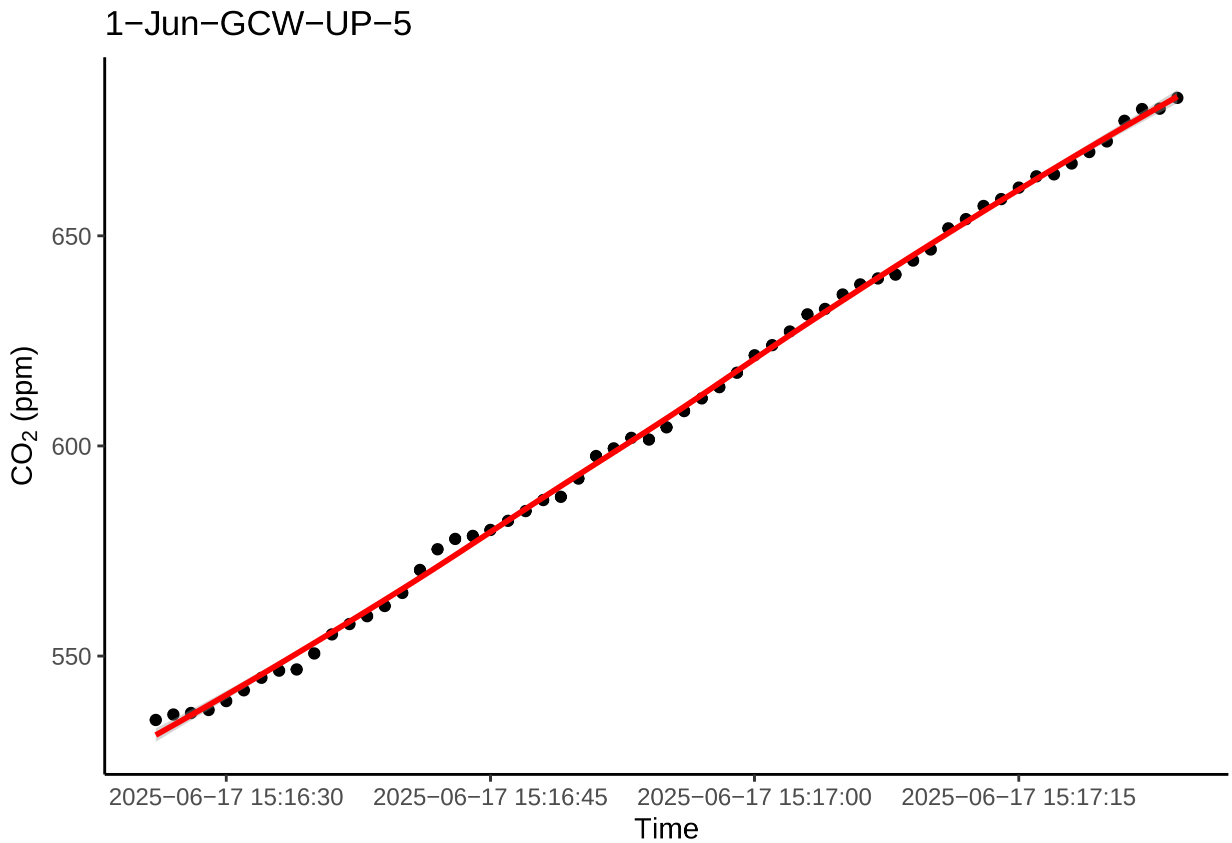

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



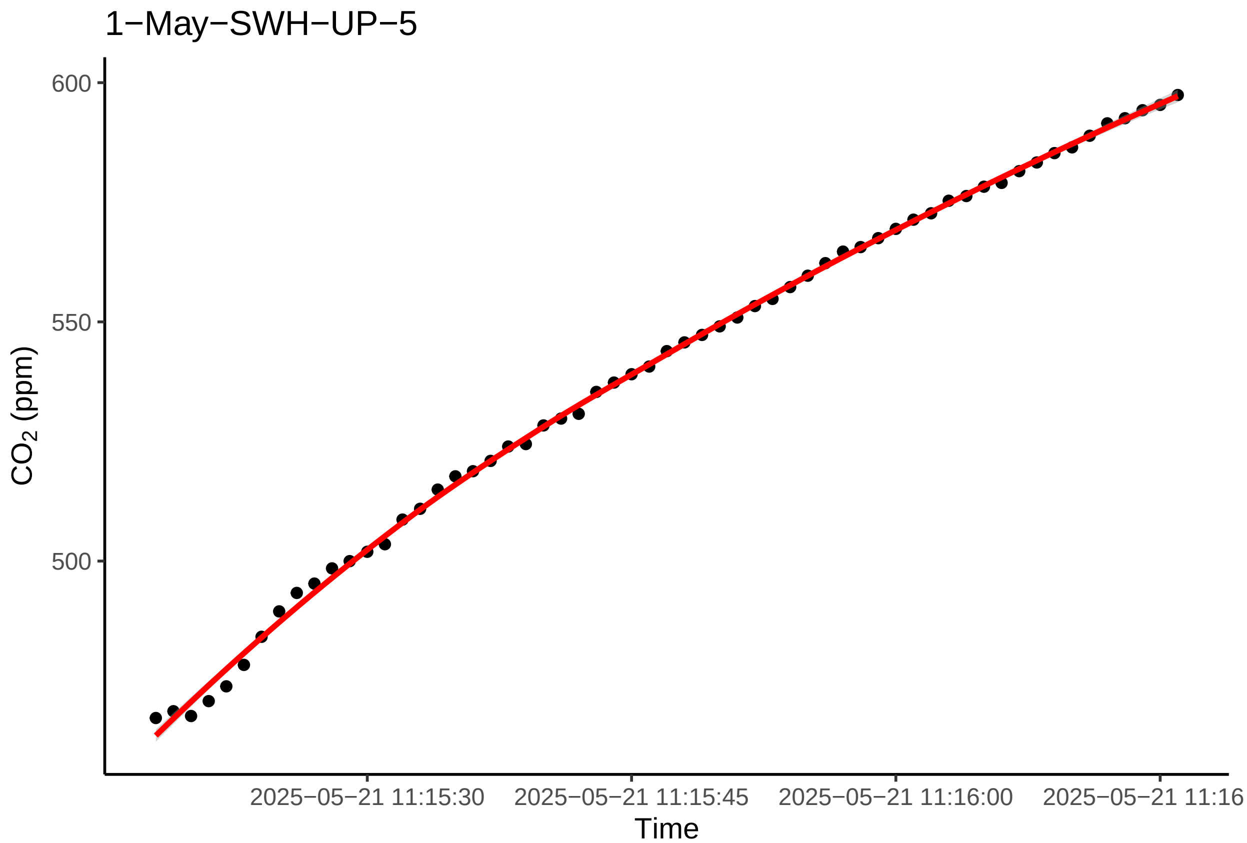

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



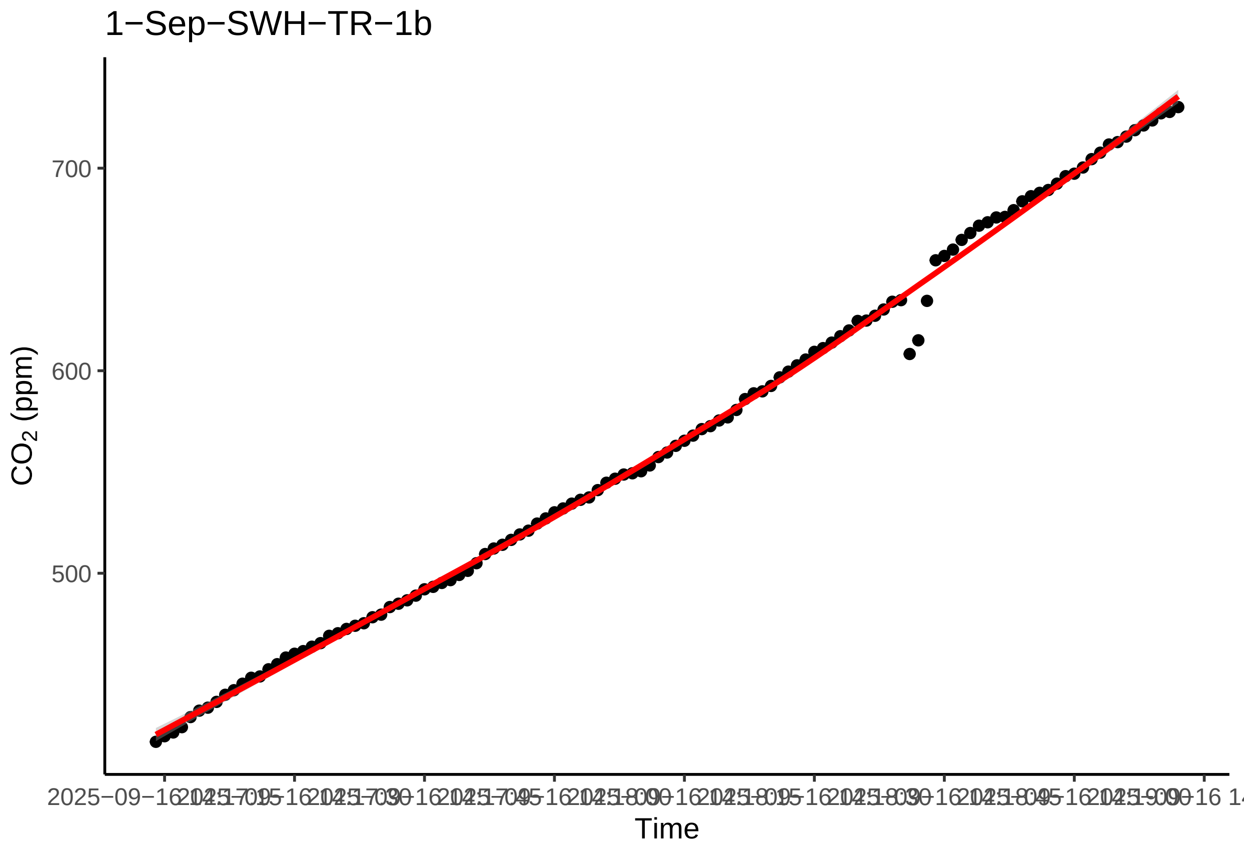

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



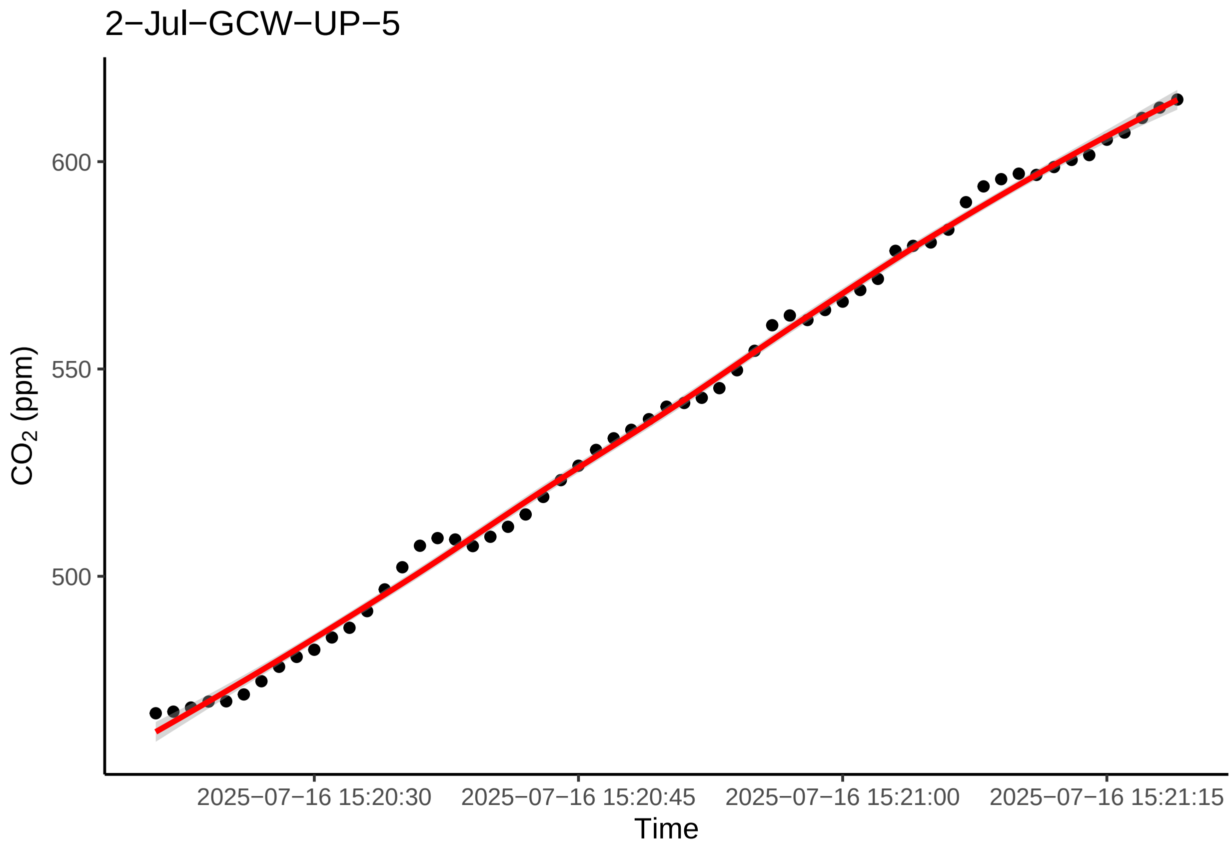

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



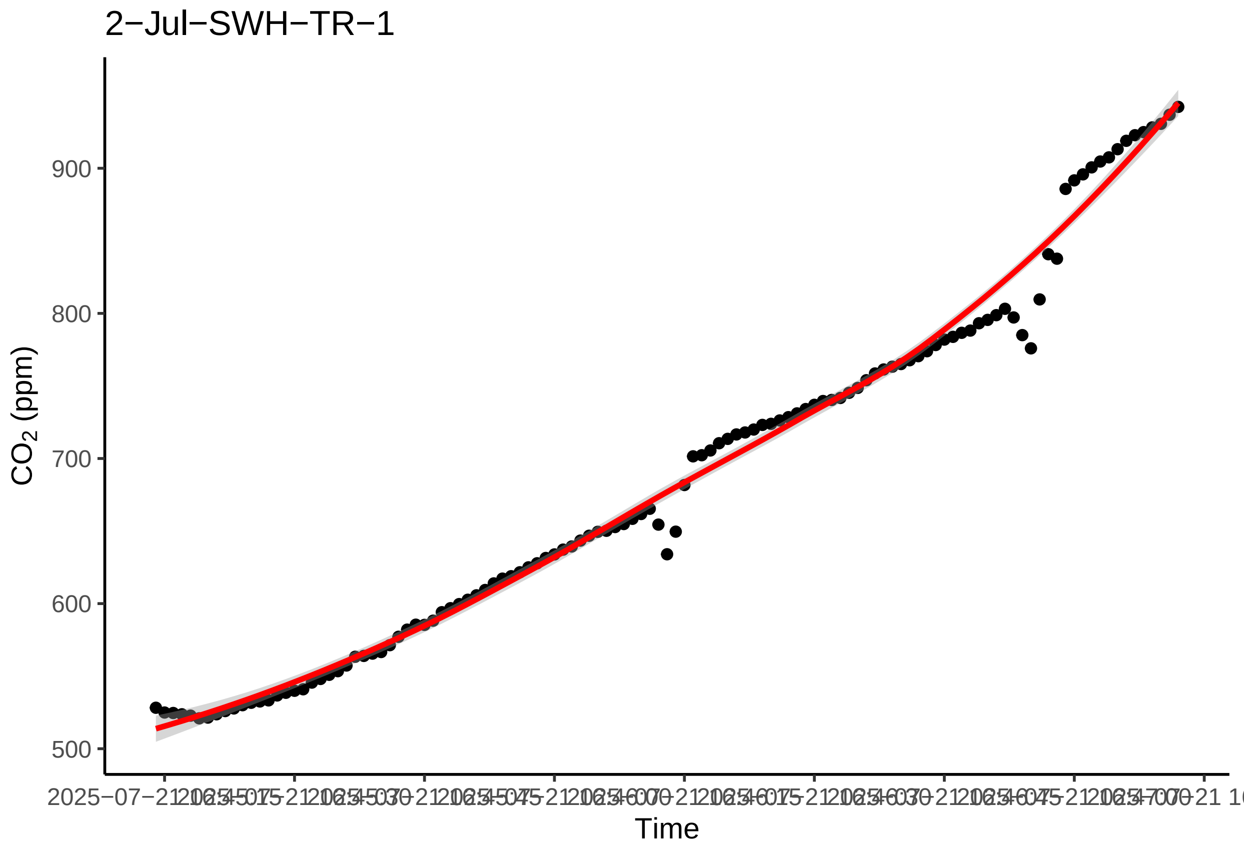

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



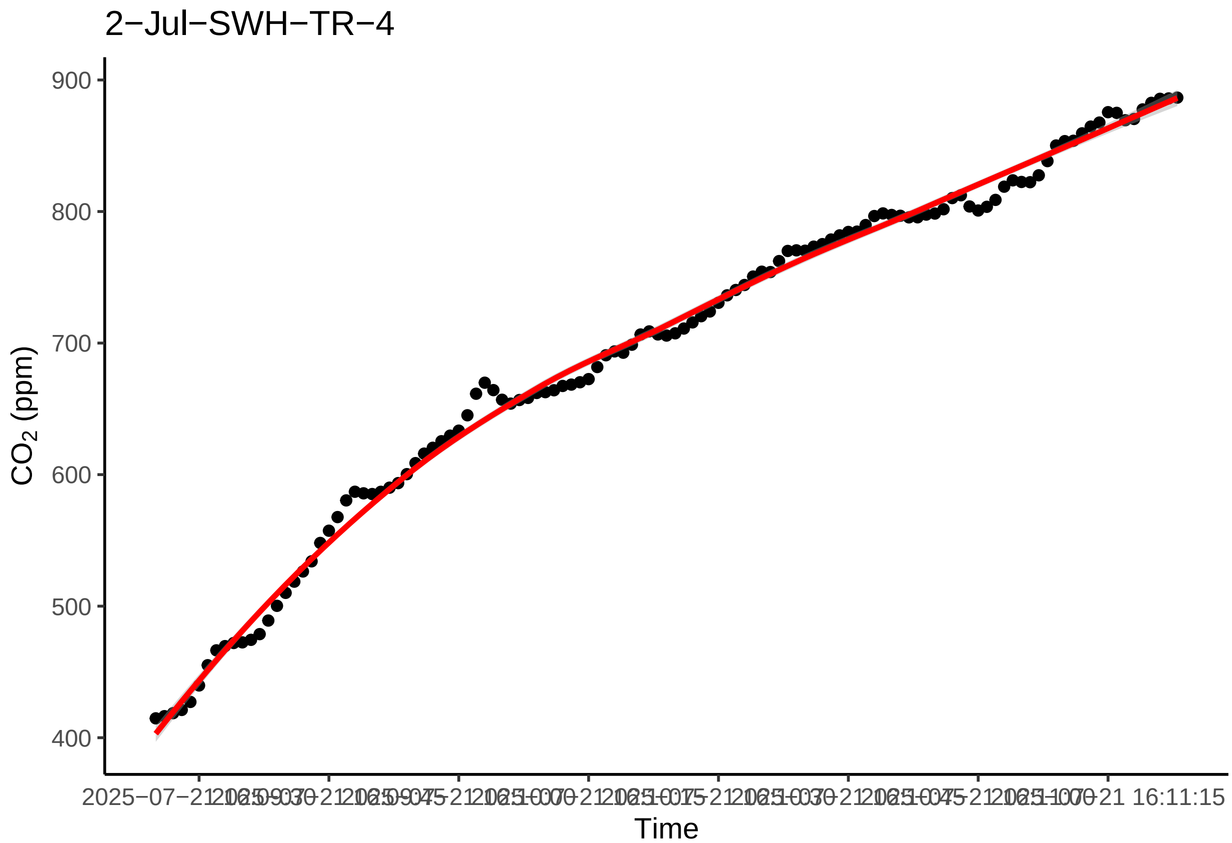

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



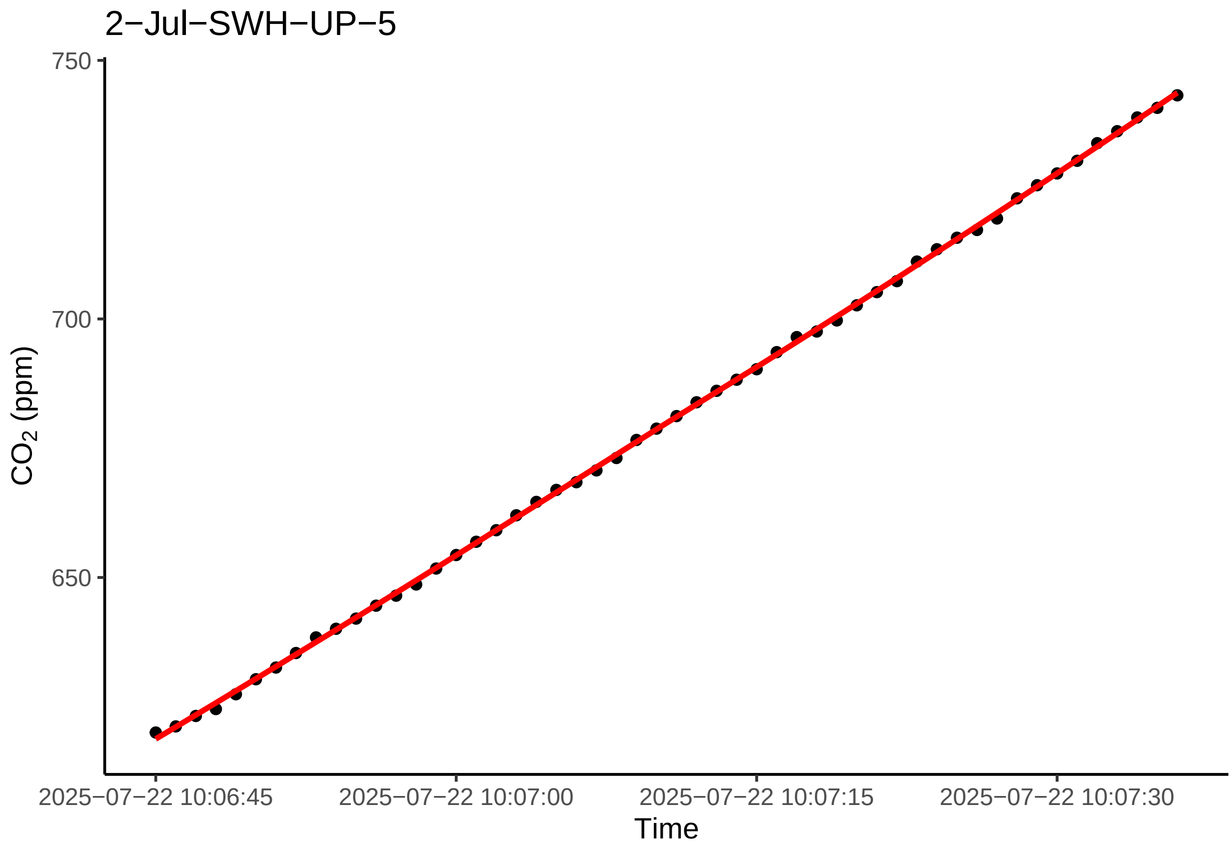

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



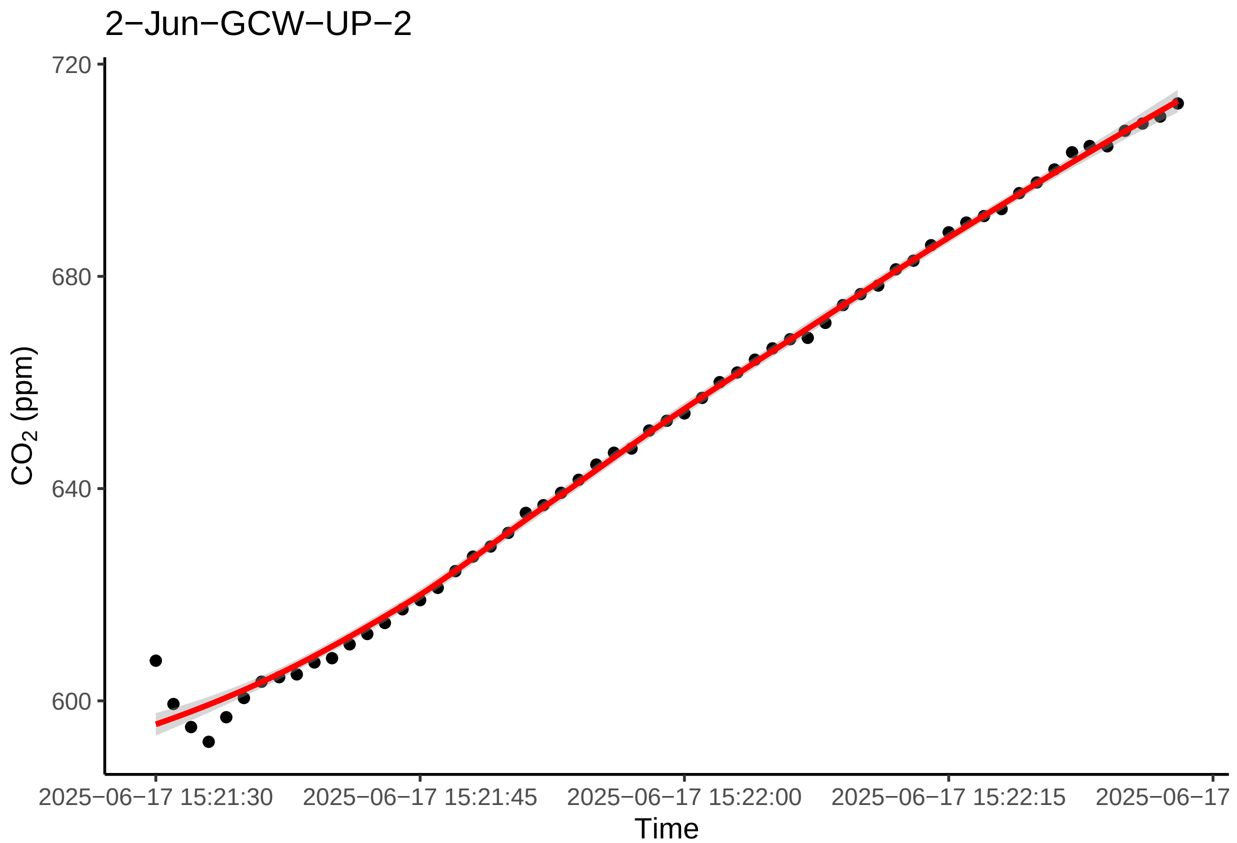

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



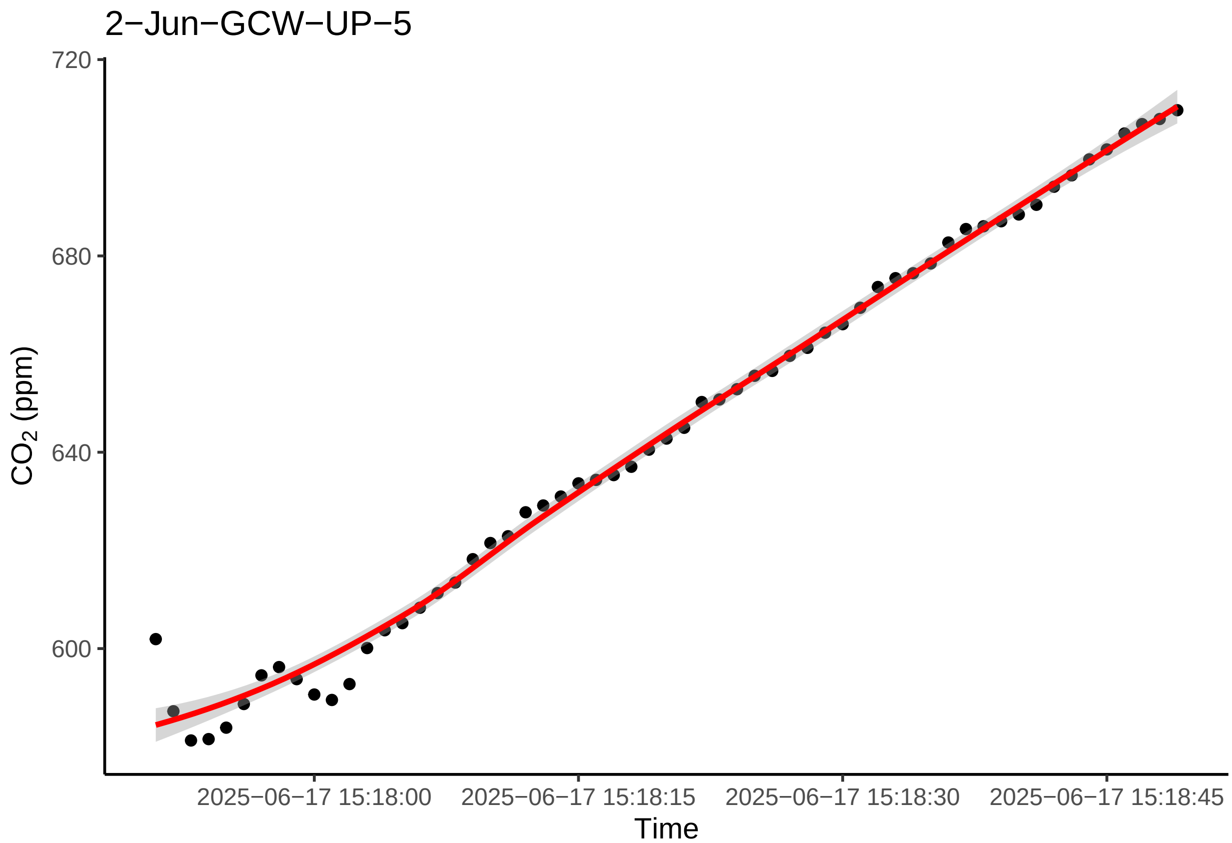

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



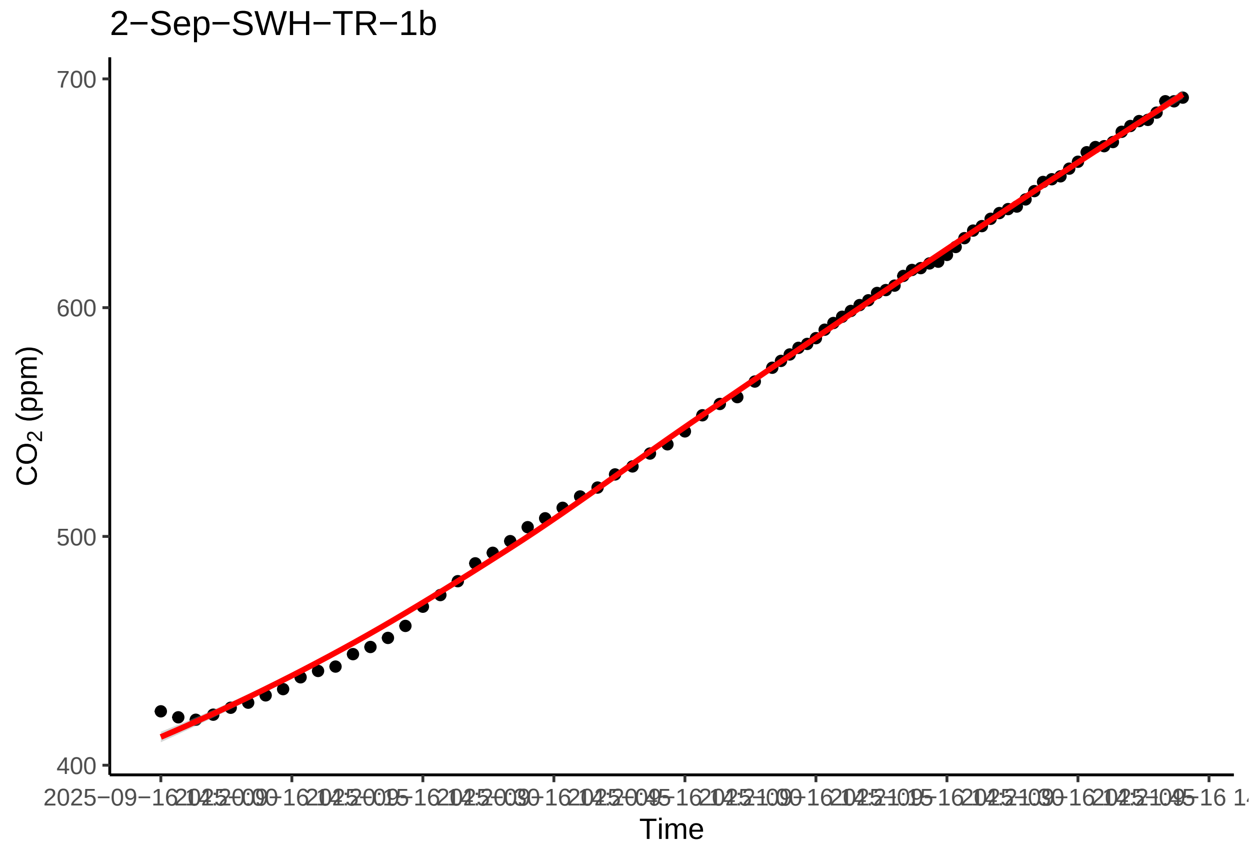

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```




```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



0.16 Check all fluxes

"'{#r, echo=FALSE}

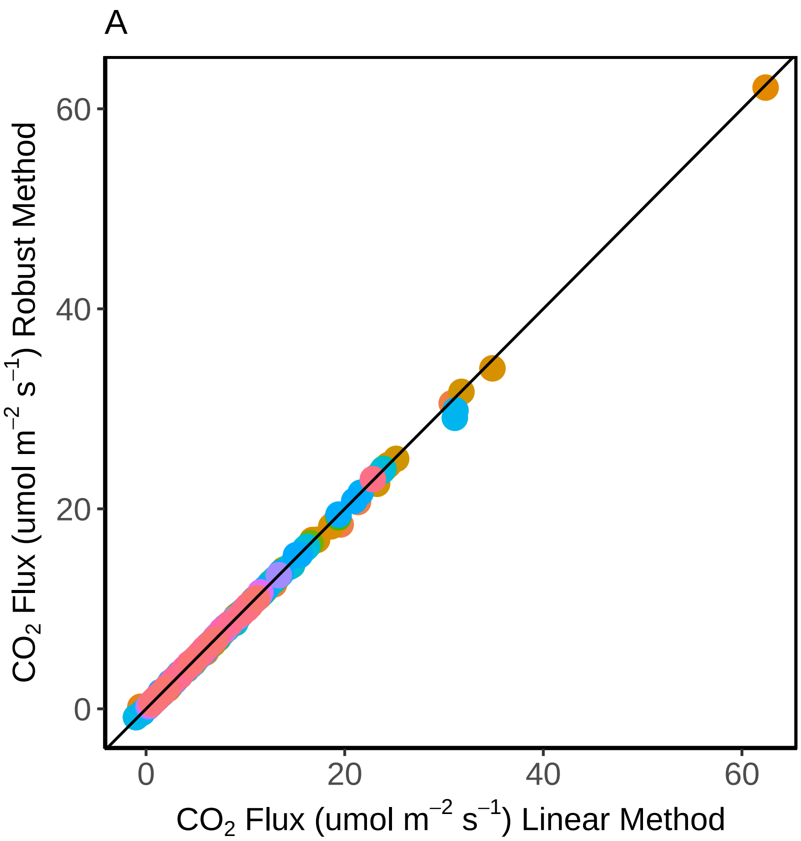
##Put all fluxes together fluxes_all <- as.data.frame(subset(allflux, select = Plot:TIMESTAMP_max)) all_plots <- unique(fluxes_all$Plot)

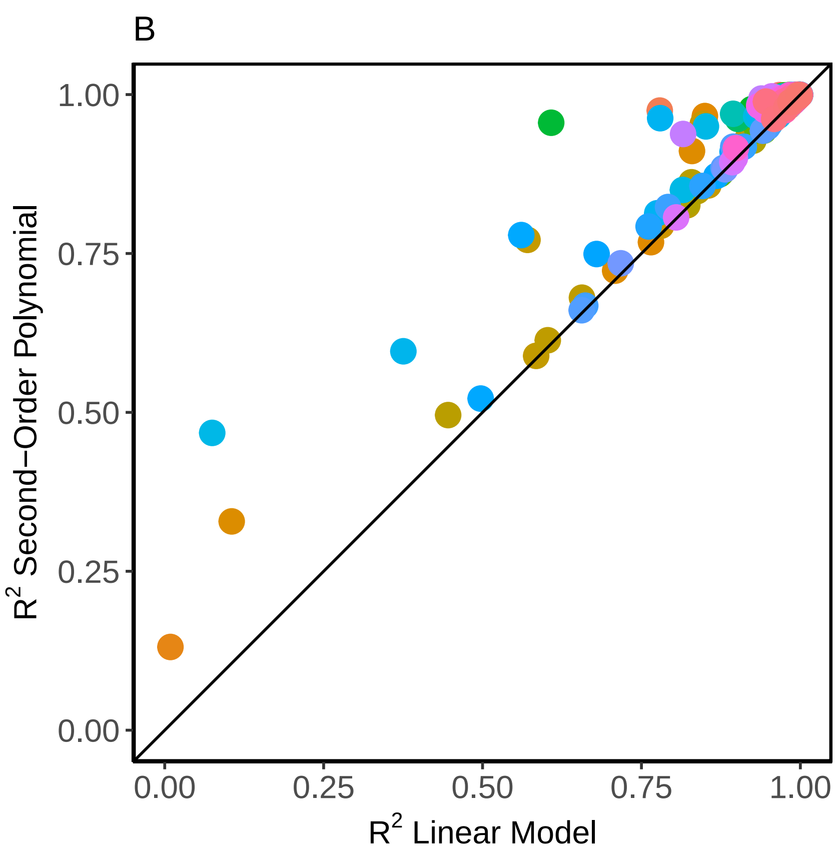
#Loop to run five random plots subset above for(P in all_plots){ r1 <- subset(alldat, Plot == P) p1 <- ggplot(r1, aes(TIMESTAMP, co2)) + geom_point() + theme_classic() + scale_x_datetime(breaks = "15 sec") + ylab(expression(paste(CO [2], " (ppm)"))) + xlab("Time") + labs(title = P) + geom_smooth(color="red") print(p1) }

```
## Flag any fluxes that do not fit criteria

''' r
# previously removed rows that were present in fluxes_weird which have a low R2 and a flux higher than 
# flagged fluxes with low R squared and low flux
# flagged high fluxes that are negative

#allflux_filtered <- allflux[!(allflux$Plot %in% fluxes_weird$Plot), ]

allflux_filtered <- allflux

#Any others that are not in the ebullition or negative category that need to be removed:
  #also flagging ones that are crazy off but were not in the fluxes_weird dataframe (did this by listin
```

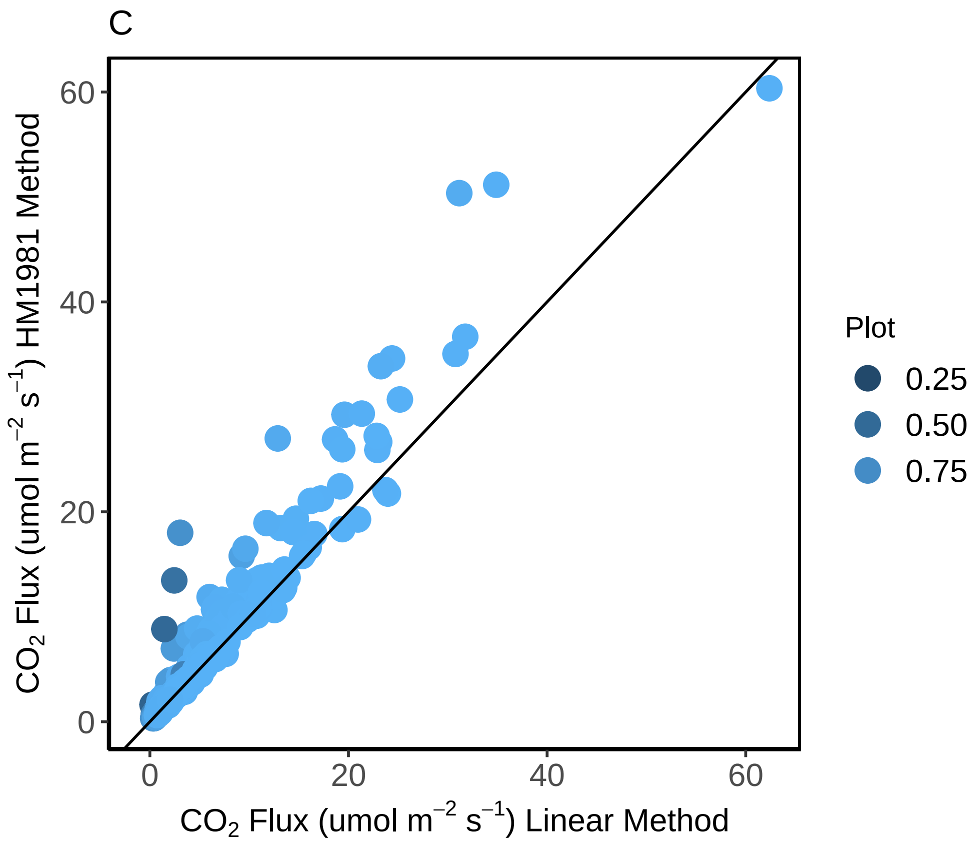
```
# View the result
head(allflux_filtered)
```

0.17 Do some visualization of QAQC plots to assess fluxes





```
## Warning: Removed 171 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```



0.18 Read out final dataframe and export for use elsewhere